School of Computing and Information Technology

CSIT321 - Project

Preliminary Technical Documentation

Project Topic: CSIT-25-S3-16

(Interactive Website for Real Estate Price Prediction in Singapore)

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1. Introduction

1.1 Problem Statement

Property markets in Singapore are influenced by factors such as location, size, tenure, and proximity to amenities, as well as market conditions. However, it is difficult for many people, especially new investors and new buyers, to interpret this data. Property websites are not useful in helping people to see future expectations on property prices for certain property characteristics even though they display listed prices or transaction history. Thus, a gap between actionable insights and data. There is a need for a useful yet usable instrument where people are able to calculate property prices based on actual data. That enables people to base selling or purchasing decisions on facts.

1.2 Purpose

The purpose of this document is to outline the requirements for building an interactive website that predicts property prices in Singapore. It describes the system's key features, functional requirements, non-functional requirements, and overall design direction.

1.3 Our Vision

We aim to develop a user-friendly website that predicts property prices in Singapore. The idea is to make it easy for anyone to get an estimated price by entering details of a property, such as location, floor level, size, and property type. The website will make use of machine learning to make smart predictions, and users will also be able to view maps, trends, and nearby amenities. This way, people can understand not just the price, but also why the price is what it is.

1.4 Project Objectives

This project has several key objectives:

- ➢ To collect and clean real estate data from reliable public sources.
- ➢ To build machine learning models that can predict property prices based on features like location, size, and tenure.
- ➢ To design a simple, responsive interface where users can easily input property information and view predictions.
- ➢ To include geospatial data that highlights factors such as the proximity to amenities, schools, and MRT stations.
- ➢ To present data visually through maps and charts so users can understand trends and comparisons easily.
- ➢ To ensure the website works smoothly across different devices and screen sizes.

Together, these goals aim to deliver a useful and reliable tool for anyone interested in the Singapore property market.

2. Overview

2.1 Competitor Market Research

PropertyGuru

PropertyGuru offers instant property valuation using its proprietary estimation tool, along with neighborhood insights based on market trends. Users can also track their mortgage for better refinancing opportunities and get matched with top-performing property agents in their area.

Note:

Focused mainly on residential properties; no tools for commercial or industrial units.

Features:

- ➢ Accurate property valuation: Proprietary estimate based on recent transaction data
- ➢ Mortgage tracking: Monitor your home loan and receive refinancing alerts
- ➢ Agent matching: Connect with agents ranked by transaction volume in your area
- ➢ Market trends: View historical price trends across towns and property types
- ➢ Login required: Users must log in to access full valuation tools and alerts

SRX X-Value (Singapore)

SRX X-Value is an AI-powered property valuation tool in Singapore that provides highly accurate estimates for HDB and private homes. It uses Comparable Market Analysis (CMA), past sales, and data. Users can enter their address and get valuation results right away. The platform also includes tools for market trends, rental yield, and capital gain projections.

Note:

Does not support commercial or industrial properties.

Features:

- Accurate property valuation: AI-generated estimates based on current and historical data
- Comparable sales: Shows similar properties recently sold nearby
- Market trends: View up to 10 years of property price history
- Rental yield insights: Estimates expected rental income
- Capital gain projections: Helps sellers forecast sales proceeds
- Property tracking: Monitor up to 3 properties with email alerts
- Login required: Needed to check property value
- Updated quarterly: Regular updates and backtesting for accuracy

Edge Fair Value (Singapore)

Edge Fair Value (EFV) is EdgeProp's automated tool that gives instant sale and rental estimates. It uses location, unit size, layout, floor level, and past transactions. After entering details, users get a quick online report with estimated sale/rental values and recent transactions.

Note:

Supports commercial and industrial properties, but the industrial and commercial prediction tool is not functioning.

Features:

- ➢ Instant valuation: Sale and rental price estimate within seconds
- ➢ Supports all property types: HDB, condo, landed, commercial, and industrial
- ➢ Supporting data included: Shows nearby past transactions for transparency
- ➢ Historical trend comparison (Upcoming): Price of your unit vs similar nearby units
- ➢ Assigned agent support: Connects you with a local property agent for advice
- ➢ Login required

Propseller Home Insights (Singapore)

Propseller Home Insights is a free valuation tool that estimates your property's value using over 500,000 past transactions. Users enter basic details like postal code, size, and floor level and get a report via email or WhatsApp.

Note:

Focuses on residential properties only, mainly HDB and condos.

Features:

- ➢ Includes nearby transactions: Shows recent sale prices of similar units
- ➢ Historical trend tracking: View price per sqft over time
- ➢ Popular blocks: Highlights top-performing blocks in your area
- ➢ Free consultation with property agent
- ➢ Fast process: Report delivered quickly after form submission
- ➢ Sent via email & WhatsApp: Easy access and reference later
- ➢ Free to use: No hidden charges or commitments

HomerAI (Singapore)

Ohmyhome is a property platform that offers listings, agent services, mortgage tools, and smart digital tools via HomerAI. Users can get instant valuations, calculate cash proceeds, and check mortgage affordability.

Note:

Only supports HDB and condos. Landed, commercial, and industrial support is not available yet.

Features:

- ➢ Instant home valuation
- ➢ Covers HDB and condos: Landed support in development
- ➢ Cash proceeds & affordability tools: Helps plan for selling or buying
- ➢ Buyer demand tracker: Shows how many people are looking at your unit
- ➢ Smart insights dashboard: Tracks market movements and home value
- ➢ Agent integration: Connect with Ohmyhome's in-house agents

Bluenest MyHomeValue™ (Singapore)

Bluenest's MyHomeValue™ is a free property valuation tool built for HDB and private homeowners. It uses real-time sales and listing data to suggest a fair price.

Note:

Targets residential units. May include some mixed-use listings, but no dedicated support for industrial/commercial properties.

Features:

- ➢ Detailed user inputs: Includes floor level, size, renovation date, occupancy duration
- ➢ Real-time market data: Uses recent sales, current listings, and withdrawn listings
- ➢ Tips for selling: Advice to help owners position their unit better
- ➢ Comparable analysis: Shows similar units for pricing context
- ➢ No instant result on website: Report sent via email
- ➢ Free to use: No payment or registration needed

2.2 Product Comparison Matrix

This matrix compares existing property valuation platforms in Singapore against key features such as real-time price prediction, customizable user inputs, geospatial insights, and rental estimation. The evaluation highlights core functionalities that directly affect the user experience for both unregistered and registered users. While backend technical setups and agent workflow tools are valuable, this comparison focuses on user-facing features that support price transparency, decision-making, and ease of access within the limited 20-weeks development scope of the FYP.

Feature	PropertyGuru	SRX X-Value	EdgeProp (EFV)	Propseller Home Insights	Ohmyhome (HomerAI)	Bluenest MyHomeValue	Our product	
Basic Functions								
Free to use	☑	☑	☑	☑	☑	☑	☑	must have
Real-time price predicton	☑	☑	☑	☐	☑	☐	☑	must have
Machine Learning based	☑	☑	☑	☑	☑	☑	☑	must have
Basic property details (e.g., floor,size)	☑	☑	☑	☑	☑	☑	☑	must have
Geospatial insights (e.g., map, amenities, MRT)	☐	☐	☑	☑	☐	☐	☑	must have
Historical price trends	☑	☐	☐	☐	☑	☐	☑	must have
Data visualisation & user-friendly UI	☑	☑	☑	☑	☑	☑	☑	must have
Enhanced Functions								
Additional property detail (e.g., reno)	☐	☑	☐	☐	☐	☑	☑	nice to have
Rental estimate included	☑	☑	☑	☐	☑	☐	☑	nice to have
Comparable sales display	☐	☑	☑	☑	☑	☑	☑	must have
Registration/login required	☑	☑	☑	☑	☑	☑	☑	must have
Required the filtering of property type	☐	☑	☑	☑	☐	☐	☑	must have
Mobile responsive design	☐	☐	☑	☐	☑	☐	☑	must have
Advanced User Features								
Allow saving address	☑	☑	☑	☐	☐	☐	☑	must have
Allow saving searches	☐	☐	☑	☑	☐	☐	☑	must have
Allow saving property comparison	☐	☑	☑	☐	☑	☐	☑	must have
Supports agent access	☑	☑	☑	☑	☑	☑	☑	must have
Agent contact/integration	☐	☑	☑	☑	☑	☑	☑	must have
Financial calculators (stamp duty, loan, etc.)	☑	☑	☑	☐	☑	☐	☑	nice to have
Referral reward	☑	☐	☑	☐	☑	☑	☑	nice to have
Types of Valuation								
Commercial Valuation	☐	☐	☐	☐	☐	☐	☑	must have
Industrial Valuation	☐	☐	☐	☐	☐	☐	☑	must have

2.3 Conceptualization of Ideas

This project idea was driven by the increasing demand for quick, accurate, and accessible property price checks, especially for *commercial* and *industrial* real estate in Singapore. While many platforms today, such as SRX, EdgeProp, and Ohmyhome offer valuation services, they tend to prioritize residential housing and often have limitations. Some require users to go through agents, others provide delayed results via email or WhatsApp, and few provide immediate predictions with user-controlled inputs.

Our platform was conceptualized to fill this *niche gap*. Unlike existing tools, it focuses specifically on industrial and commercial property segments, categories that are underserved in terms of open, real-time valuation tools.

To achieve this, the system uses trusted and publicly available datasets such as those from URA, along with custom machine learning models trained on historical transaction data. Users are able to input specific property attributes such as property type, floor level, unit size, and location to enable the system to return instant, personalized price predictions for the property.

The platform also incorporates location-based insights, showing how factors like proximity to expressways, MRT stations, or commercial hubs impact pricing, giving users a fuller understanding of what affects property value.

The idea was carefully shaped around feasibility: focusing first on a clean interface, the core ML prediction feature, and simple visual tools like price trend charts.

This concept was validated through user persona analysis and market research, ensuring the platform is useful, achievable within the 20-week development cycle, and uniquely positioned to serve a niche market that is currently overlooked.

2.4 SWOT Analysis of the Product

Strengths

- Multi-Sector AI-Powered Valuation: Our platform uses machine learning for industrial and commercial property predictions. This makes it more comprehensive than most competitors, which focus only on residential.
- User-Friendly Interface: Unlike platforms that depend on agent callbacks or delayed emails, users can enter property details with ease and get instant results, making the experience more seamless.
- Open Data Integration: Data is retrieved from URA and geographic sources.
- Accessible for All User Types: Designed for unregistered users, free and premium members, agents, and system admins, making the platform flexible for different needs.

Weakness

- Short Timeline: It might be challenging to finish advanced features in the allotted 20 weeks.
- Data Quality Issues: Limited datasets and resources for industrial property, it will depend on publicly available data, which could be inconsistent or outdated
- Low Initial Trust: New platform may struggle to gain trust compared to big names like SRX or EdgeProp.

Opportunities

- ➢ Growing PropTech Use: Businesses and investors are increasingly looking for online solutions not only for homes but also for warehouses, retail spaces, and offices.
- ➢ Government Support: Initiatives like Smart Nation and open government datasets support innovative tools like ours, especially those using URA's commercial property data.
- ➢ Premium Add-ons: Can expand later with paid features like comparisons, AI advice, and agent services.

Threat

- ➢ Strong Competition: Competing against established platforms like SRX, EdgeProp, Propseller, etc., which already have valuation tools (although limited to residential).
- ➢ Privacy Concern: Users may not wish to offer personal or property details.
- ➢ Model Misinterpretation: Users might misinterpret the results of predictions or use them unpromptedly, creating liability or dissatisfaction if predictions are wrong.

2.5 Unique Selling Point

This platform supports industrial and commercial properties, such as warehouses, offices, and retail spaces.

Most existing platforms provide general estimates or rely on agent contact to share price details. In contrast, this system is built to be open, fast, and data-driven.

- ➢ Instant Price Estimates: Users can enter property details and receive an immediate prediction, without waiting for agent callbacks or emails.
- ➢ Smart Use of Data: The system applies machine learning to real transaction records, allowing it to provide accurate price predictions tailored to each property.
- ➢ Based on Trusted Public Sources: Predictions are generated using official data from URA and other government datasets, ensuring transparency and reliability.

With its focus on industrial and commercial properties, the platform makes it easier for buyers, investors, and agents to access clear and timely pricing information.

2.5.1 Use of Open Data

The system's approach is based on publicly sourced data from URA and other government authorities for reliability and transparency. In such a manner, the system is backed by actual verified historical transaction records, hence making the predictions more trustworthy and accurate.

2.5.2 Machine Learning-Based Price Prediction

The system uses machine learning models trained on property attributes like location, floor level, area, and transaction record, as opposed to traditional valuation tools that offer static or narrow-range prices. As a result, it can provide more accurate and contextually relevant dynamic and customized price estimates for every user input.

2.5.3 Price Prediction Report in Readable Format

The estimated price range, median price per square foot, and relevant historical comparisons are all included in the clear, easily comprehensible format in which the predicted results are displayed. Without requiring technical expertise, this user-focused reporting format enables users to make informed decisions.

2.5.4 User-Friendly Comparison Tool

The platform has an online interface where users can compare projected prices between various properties. The comparison feature is beneficial for premium users who could be investors in terms of comparing various options side by side according to key characteristics.

2.6 Target Users

Our platform is constructed to support different categories of users with distinct access privileges and intended interactions with the platform. The categories of users determine the scope of functionality, access level, and user interface for successful product implementation.

2.6.1 Unregistered User

An individual who accesses the platform without creating an account.

Typical Behaviors and Goals:

- ➢ Exploring the homepage to understand the website's offerings.
- ➢ Reading user reviews to assess credibility and experience of others.
- ➢ Accessing publicly available pages such as pricing plans, FAQs, and contact information.
- ➢ Encountering prompts to sign up when attempting to access personalized tools or premium services.
- ➢ Registering for an account to unlock additional features.

System Access:

- ➢ Read-only access to selected public content.
- ➢ Cannot save data or receive personalized predictions.

Functional Requirements Reference:

- ➢ Public-facing homepage and navigation
- ➢ View-only access to trial predictions
- ➢ Prompt mechanisms for sign-up
- ➢ Access to registration, FAQ, and contact forms

2.6.2 Registered Free User

Users who have successfully created a free account on the platform.

Typical Behaviors and Goals:

- ➢ Inputting detailed property attributes to generate price predictions.
- ➢ Filtering predictions based on property type.
- ➢ Accessing historical queries and managing saved results.
- ➢ Logging in and out securely.
- ➢ Exploring subscription options and upgrading to a premium plan.

System Access:

- ➢ Access to limited predictive features.
- ➢ Data storage and personalized property dashboards.
- ➢ Upgrade pathway to premium account.

Functional Requirements Reference:

- ➢ User authentication and profile management
- ➢ Property prediction interface (limited access)
- ➢ Subscription upgrade prompt when premium features are clicked

2.6.3 Registered Premium User

Subscribed users with access to full platform capabilities and personalized features.

Typical Behaviors and Goals:

- Accessing detailed price predictions and analytics for properties.
- Viewing historical price trends based on selected property addresses.
- Receiving personalized recommendations based on user preferences.
- Managing subscription plans (e.g., upgrade, downgrade, billing).
- Downloading reports in PDF/Excel formats for decision-making or professional use.
- Inviting others to the platform and participating in referral reward programs.
- Logging in and out securely.

System Access:

- Full access to all core features
- Exclusive access to premium-only reports and tools.

Functional Requirements Reference:

- Advanced analytics and reporting engine
- User preference engine and content filtering
- File export functionality
- Subscription and referral management

2.6.4 Property Agent

Licensed real estate professionals with access to agent-oriented features for supporting their listings, outreach, and client engagement.

Typical Behaviors and Goals:

- Logging in and out securely to manage session access.
- Viewing and editing personal profile details to maintain up-to-date contact information.
- Viewing auto-assigned regional coverage and requesting changes if necessary.
- Accessing FAQs and disclaimers to understand platform capabilities and limitations.
- Providing feedback on the platform to help improve services.
- Sending referral codes to potential users and tracking referral-based rewards.
- Contacting support for queries not resolved via help documentation.

System Access:

- Additional access to agent-exclusive tools (if developed in future scope).
- Referral and feedback integration capabilities.

Functional Requirements Reference:

- Secure login/logout system
- User profile management module and account deactivation mechanism
- Region assignment view and override system
- Contact and support integration
- Feedback submission engine
- Referral system integration
- FAQ and legal disclaimer display

2.6.5 System Admin

Internal platform operator responsible for content, user, and system management.

Typical Behaviors and Goals:

- Monitoring user activity through the user management dashboard.
- Suspending or deactivating accounts.
- Reviewing and moderating user-submitted reviews and content.
- Updating marketing media such as promotional videos on the platform.

System Access:

- Full backend control over user database, content moderation, and system configurations.

Functional Requirements Reference:

- Admin dashboard and user management module
- Role-based access control
- Review moderation and content editing capabilities
- Marketing asset management tools

2.7 Business Model

Our project uses a subscription-based model that starts with a free plan to engage and establish trust with prospective users. This enables them to utilize and interact with the application with restricted features. Upon realizing the advantages of our platform, they have the option to upgrade to premium subscriptions for complete access to all features.

In the future, we aim to maintain the platform by motivating users to opt for premium plans and by offering reliable customer support to guarantee satisfaction and involvement. This method allows us to achieve growth and maintain quality, providing each user with a seamless and satisfying experience.

2.7.1 Target Audience

Our system primarily targets individuals who are directly engaged in Singapore's property market. This comprises:

- **Property buyers and sellers** who want quick and accurate price predictions before making decisions.
- **Real estate developers and investors** who require in-depth analytics, like historical price per square foot or rental trends, to strategize upcoming projects.
- **Property agents** who can utilize the platform to highlight their involvement in specific regions so as to engage with prospective buyers and sellers, and also, present property listings to draw in users who are keen on buying or renting the properties.

By serving these groups, we make the platform valuable for both first-time users (such as someone buying their first office space) and professionals who need deeper insights.

2.7.2 Online Marketing

To reach our target audience, we plan to implement affordable yet effective online strategies:

- Post market updates and content (e.g., price index snapshots, comparison articles) on social media and blogs to highlight the benefits of our system.
- Promote on LinkedIn, focusing on users who are most likely to gain from our platform, including property agents and real estate investors in Singapore.
- Manage our marketing website or landing page to display our platform's features, emphasize its advantages, and lead prospective users to register or investigate the system.

These approaches collectively foster awareness, draw in users, and transform them into engaged subscribers

2.7.3 Partnerships

Potential future collaborations could include:

- **Property agencies and agents**, where our system may provide them with exposure in their assigned regions and potentially offer qualified leads.
- **Property listing portals,** which could allow our tool to be integrated as a “Check Value” feature on their websites.

These partnerships are considered as possible opportunities, depending on feasibility and mutual interest.

2.7.4 Revenue Streams

We designed our revenue model to keep the system sustainable while still being fair to users:

- **Free tier**: lets new users try the system with limited features (e.g., a few predictions).
- **Premium subscription**: unlocks more advanced features like comparisons, detailed trends, and downloadable reports.
- **Agent/Professional tier**: allows property agents to get more visibility, use lead tools.

3. Stakeholders

The success of this project relies on the involvement of the stakeholders, each with an important role in the progress. The following outlines their specific responsibilities:

Stakeholder	Responsibility
Assessor	Assesses the project's overall quality, taking into account the report, implementation, and presentation. Based on project outcomes and adherence to academic standards, offers feedback and assessment.
Supervisor	Guides the team during the course of the project. Provides technical and academic advice, and helps refine ideas while keeping the project on course.
Team Leader	Manage the project's overall progress, communicate with the supervisor, coordinate tasks among team members, and make sure deadlines are met.
Developers	In charge of coding, integrating frontend and backend components, putting machine learning models into practice, and making sure the system works.
Designers	Designs the UI and UX. Ensures the platform is responsive on all devices, easy to use, and aesthetically pleasing.
Testers	Conducts functional and usability testing. Ensures features work as intended, checks for bugs, and gathers user feedback for improvements.
Target Audience (The end users)	Provide real-world usability feedback. Validate if predictions align with market expectations and provide feedback to improve the system usability and features.

4. Data Collection

This section outlines the various data sources used for property price prediction, the types of data collected, methods used for data retrieval, preprocessing techniques applied, and considerations regarding data privacy and compliance. Accurate and reliable data forms the foundation of the prediction engine and overall platform functionality.

4.1 Data Sources

Data for this project is collected from publicly accessible and government-approved datasets to ensure reliability, transparency, and compliance. The following sources are used, with a specific focus on industrial and commercial property markets in Singapore:

Dataset Type	Description	Source URL
Geospatial Data (mapping)	Land maps, region zoning, and distances to commercial hubs, logistics areas, MRTs, and industrial clusters	OneMap.sg or SLA Geospatial / SLA Geospatial Data / Google Maps API / OneMap API
Commercial Property Transactions	Sale records for office, retail and shophouse	teeco.sg commercial
Industrial Property Transactions	Sale records of single-user factory, multiple-user factory, warehouses, and business parks	teeco.sg Industrial

Commercial Rental (Retail)	Rental rates and trend analysis of commercial retail spaces	URA Retail Rental Analysis
Commercial Rental (Office)	Rental rates and trend analysis of office spaces	URA Office Rental Analysis
Commercial Rental Index	Quarterly index tracking rental market performance for commercial spaces	Data.gov.sg - Commercial Rental Index
Industrial Rental Index	Quarterly index tracking rental market trends for industrial properties	Data.gov.sg - Industrial Rental Index

4.2 Types of Data

The data collected includes a mix of structured property records and transactional attributes relevant for price prediction modeling. Key data fields include:

Category	Examples
Property Type	Office, Retail, Shop House, Single-user Factory, Multiple-user Factory, Warehouse, Business Parks
Floor Area/Unit Size	Gross floor area (sqm), net lettable area
Location Attributes	Street name, region, proximity to MRT, business hubs, transport nodes
Storey/Floor Level	Low, mid, or high floors; unit-level floor ranges (where applicable)
Lease Details	Lease start year, tenure (e.g., freehold, 99-year), remaining lease
Transaction Details	Sale/rental price (SGD), transaction quarter/year
Building/Project Info	Property name, development type
Geospatial Data	Latitude/longitude, planning area

4.3 Collection Methods

The following methods are employed for data gathering and preparation:

- CSV Downloads: Datasets from Data.gov.sg (e.g., rental indices) and other open sources are downloaded in .csv format.
- Form-Based Scraping: Data from URA's transaction search page is extracted using custom scraping scripts (Python/Selenium) based on filters (location, date, property type).
- API Requests: Additional geolocation and proximity features (e.g., nearest MRT, schools, malls) are fetched via Google Maps API or OneMap.sg.

Scripts and batch processes are developed to routinely collect and archive updated datasets for consistency across training cycles.

4.4 Data Preprocessing

To ensure the datasets are machine learning–ready, the following preprocessing steps are applied:

- **Data Cleaning**:
  - Removal of duplicates and incomplete records
  - Standardization of date formats, floor levels, and categorical values
- **Feature Engineering**:
  - Calculating remaining lease years
  - Encoding floor levels numerically (e.g., Basement/Low/Mid/High = 0/1/2/3)
  - Adding proximity scores to MRT stations, expressways, cargo hubs, and industrial clusters
- **Integration**:
  - Merging datasets from commercial and industrial transaction records across different sources
  - Mapping location data to planning regions/zones

- **Transformation**:
  - Normalization of price and size values
  - One-hot encoding of categorical features
- **Filtering**:
  - Removing outliers (e.g., abnormal price spikes, typo errors)

4.5 Privacy Considerations

While all data is sourced from open government databases and publicly accessible resources, privacy and ethical considerations are still adhered to:

- No personally identifiable information (PII) is collected or stored.
- Data usage complies with Singapore's Personal Data Protection Act (PDPA).
- All analysis is done at an aggregated and anonymized level.
- Disclaimers are included on the platform to inform users that prediction results are based on historical public records and do not constitute financial advice.

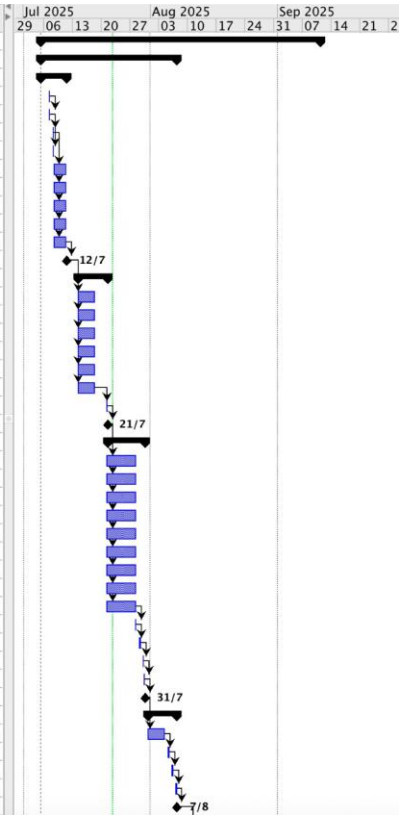
5. Project Timeline

5.1 Project Milestones

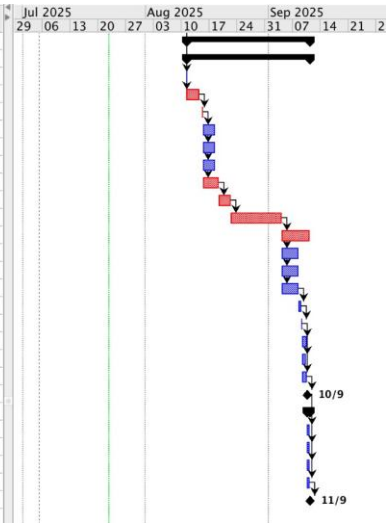
Assessment Item	Responsible	Due Date
1. Project Requirements Documentation	Group	9 Aug 2025
2. Project Progress Report *Preliminary Technical Documentation, Preliminary User Manual, Presentation Slides*	Group	13 Sept 2025
3. Project Progress Presentation *Prototype Demo*	Group	20 Sept 2025
4. Final Product & Documentation *Final Technical Documentation, User Manual, Source Code, Peer Assessment, Project Video, Presentation Slides, Supervisor Review*	Group	15 Nov 2025
5. Reflective Diary	Individual	15 Nov 2025
6. Final Presentation	Group	22 Nov 2025
7. Final Submission of Documents *Final report and artefacts*	Group	29 Nov 2025

5.2 Gantt Chart

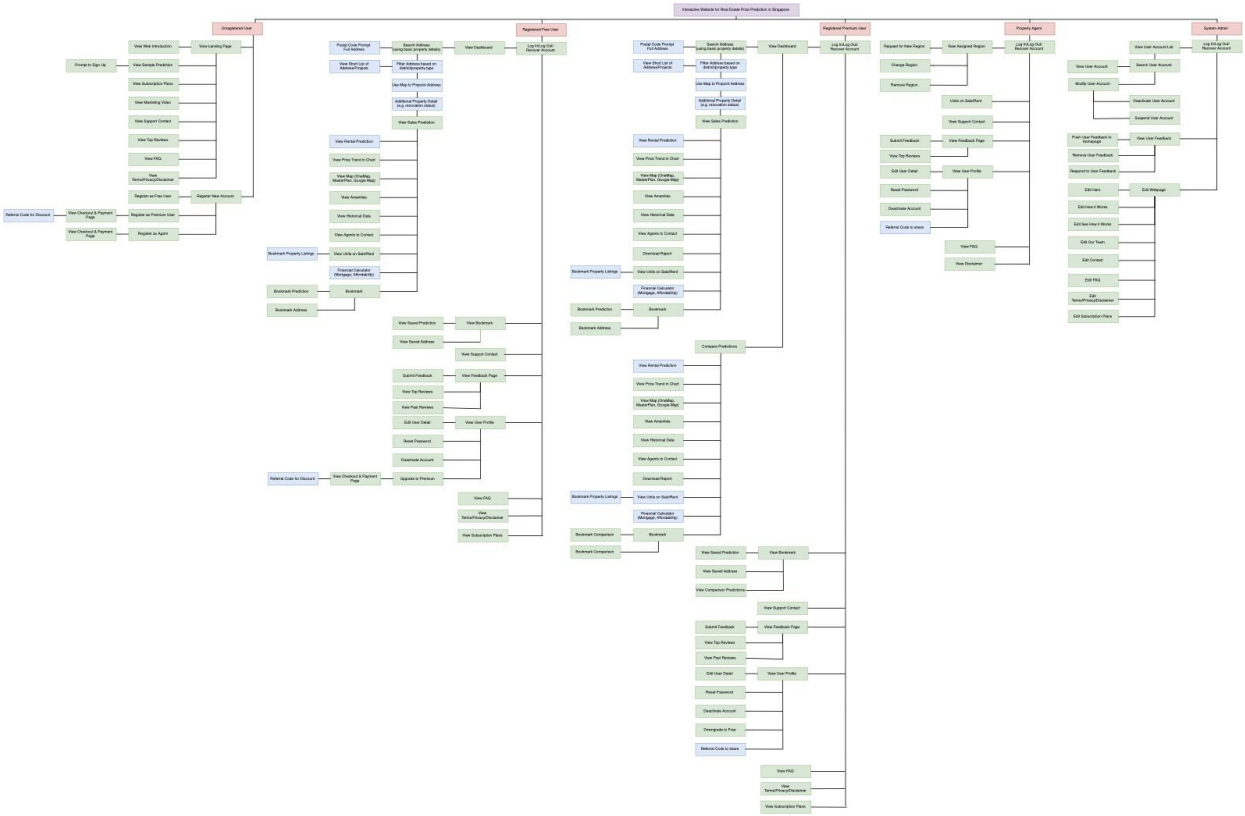
Name	Duration	Start	Finish	Predecessors
Interactive Website for Real Estate Price Prediction	48.5 days?	**5/7/25 8:00 AM**	**11/9/25 1:00 PM**	
First Submission	23.5 days?	**5/7/25 8:00 AM**	**7/8/25 1:00 PM**	
Team Initiation Phase	5 days?	**5/7/25 8:00 AM**	**11/7/25 5:00 PM**	
Kick-off Team Meeting	0.5 days?	5/7/25 8:00 AM	7/7/25 1:00 PM	
Team Meeting with Supervisor	0.5 days	7/7/25 1:00 PM	7/7/25 5:00 PM	4
Team Meeting	0.5 days	8/7/25 8:00 AM	8/7/25 1:00 PM	5
Assign Preliminary Roles	0.5 days	8/7/25 8:00 AM	8/7/25 1:00 PM	5
Draft Project Vision & Problem	3.5 days	8/7/25 1:00 PM	11/7/25 5:00 PM	6
Draft Project Timeline	3.5 days	8/7/25 1:00 PM	11/7/25 5:00 PM	6
Draft Project Charter & Scope	3.5 days	8/7/25 1:00 PM	11/7/25 5:00 PM	6
Stakeholder Identification	3.5 days	8/7/25 1:00 PM	11/7/25 5:00 PM	6
Draft Communication Plan	3.5 days	8/7/25 1:00 PM	11/7/25 5:00 PM	6
Complete Initiation Phase	0 days	11/7/25 5:00 PM	11/7/25 5:00 PM	12
Exploration & Research Phase	5.5 days	**14/7/25 8:00 AM**	**21/7/25 1:00 PM**	
Competitor Product Study	5 days	14/7/25 8:00 AM	18/7/25 5:00 PM	13
Comparison Matrix	5 days	14/7/25 8:00 AM	18/7/25 5:00 PM	13
Define Scope and Users	5 days	14/7/25 8:00 AM	18/7/25 5:00 PM	13
Create WBS	5 days	14/7/25 8:00 AM	18/7/25 5:00 PM	13
Define Technical Exploration	5 days	14/7/25 8:00 AM	18/7/25 5:00 PM	13
Draft SWOT Analysis	5 days	14/7/25 8:00 AM	18/7/25 5:00 PM	13
Team Meeting	0.5 days	21/7/25 8:00 AM	21/7/25 1:00 PM	20
Complete Exploration & Research Phase	0 days	21/7/25 1:00 PM	21/7/25 1:00 PM	21
Documentation & Design Phase	7.5 days	**21/7/25 1:00 PM**	**30/7/25 5:00 PM**	
Begin PRD Compilation	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Create Introduction of Documentation	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Create Overview of Project	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Create Functional Requirements	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Create Non-Functional Requirements	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Create Other Requirements (e.g. Legal / System Requirement	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Create Use Case Descriptions	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Create Use Case Diagrams	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Start Wireframe Sketch	5 days	21/7/25 1:00 PM	28/7/25 1:00 PM	22
Team Meeting	0.5 days	28/7/25 1:00 PM	28/7/25 5:00 PM	32
Revise Work After Meeting	1 day	29/7/25 8:00 AM	29/7/25 5:00 PM	33
Team Meeting	0.5 days	30/7/25 8:00 AM	30/7/25 1:00 PM	34
Team Meeting with Supervisor	0.5 days	30/7/25 1:00 PM	30/7/25 5:00 PM	35
Complete Documentation & Design Phase	0 days	30/7/25 5:00 PM	30/7/25 5:00 PM	36
Finalisation & Submission Phase	5.5 days?	**31/7/25 8:00 AM**	**7/8/25 1:00 PM**	
Revise Based on Supervisor Feedback	3 days?	31/7/25 8:00 AM	4/8/25 5:00 PM	37
Finalise Glossary & Appendics	1 day	5/8/25 8:00 AM	5/8/25 5:00 PM	39
Submit Reflective Diary	1 day	6/8/25 8:00 AM	6/8/25 5:00 PM	40
Submit PRD	0.5 days	7/8/25 8:00 AM	7/8/25 1:00 PM	41
End of Submission Phase	0 days	7/8/25 1:00 PM	7/8/25 1:00 PM	42

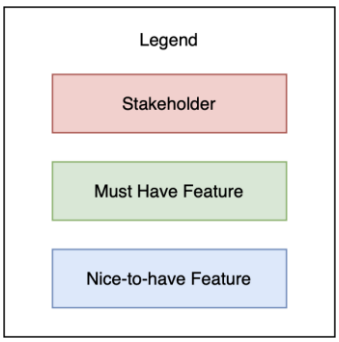


Name	Duration	Start	Finish	Predecessors
Second Submission	23.5 days?	**11/8/25 8:00 AM**	**11/9/25 1:00 PM**	
Documentation & Design Phase	23.5 days?	**11/8/25 8:00 AM**	**11/9/25 1:00 PM**	
Team Meeting	1 day?	11/8/25 8:00 AM	11/8/25 5:00 PM	43
Assigned Tasks	4 days	11/8/25 8:00 AM	14/8/25 5:00 PM	43
Discuss with supervisor	0.5 days	15/8/25 8:00 AM	15/8/25 1:00 PM	47
Draft Prelim Architecture Design	1 day	15/8/25 1:00 PM	18/8/25 1:00 PM	48
Draft Prelim Database Design	1 day	15/8/25 1:00 PM	18/8/25 1:00 PM	48
Draft Prelim Wireframe	1 day	15/8/25 1:00 PM	18/8/25 1:00 PM	48
Finalise functionality for Prototype development work	2 days	15/8/25 1:00 PM	19/8/25 1:00 PM	48
Start development work for Prototype (Frontend)	3 days?	19/8/25 1:00 PM	22/8/25 1:00 PM	52
Study + Exam Week	9 days	22/8/25 1:00 PM	4/9/25 1:00 PM	53
Continue development work for Prototype (Backend)	5 days	4/9/25 1:00 PM	11/9/25 1:00 PM	54
First draft Preliminary Technical Document	2 days	4/9/25 1:00 PM	8/9/25 1:00 PM	54
First draft Preliminary User Manual	2 days	4/9/25 1:00 PM	8/9/25 1:00 PM	54
First draft Presentation Slide	2 days	4/9/25 1:00 PM	8/9/25 1:00 PM	54
Discuss with supervisor	0.5 days	8/9/25 3:00 PM	9/9/25 10:00 AM	58
Revise Based on Supervisor Feedback	0.5 days	9/9/25 10:00 AM	9/9/25 3:00 PM	59
2nd draft Preliminary Technical Document	1 day?	9/9/25 3:00 PM	10/9/25 3:00 PM	60
2nd draft Preliminary User Manual	0.5 days	9/9/25 3:00 PM	10/9/25 10:00 AM	60
2nd draft Presentation Slide	1 day?	9/9/25 3:00 PM	10/9/25 3:00 PM	60
Complete Documentation & Design Phase	0 days	10/9/25 3:00 PM	10/9/25 3:00 PM	63
Finalisation & Submission Phase	0.5 days	**10/9/25 3:00 PM**	**11/9/25 10:00 AM**	
Submit Preliminary Technical Document	0.5 days	10/9/25 3:00 PM	11/9/25 10:00 AM	64
Submit Preliminary User Manual	0.5 days	10/9/25 3:00 PM	11/9/25 10:00 AM	64
Submit Peer Assessment Form	0.5 days	10/9/25 3:00 PM	11/9/25 10:00 AM	64
Submit Presentation Slide	0.5 days	10/9/25 3:00 PM	11/9/25 10:00 AM	64
End of Submission Phase	0 days	11/9/25 10:00 AM	11/9/25 10:00 AM	69



5.3 Work Breakdown Structure (WBS)



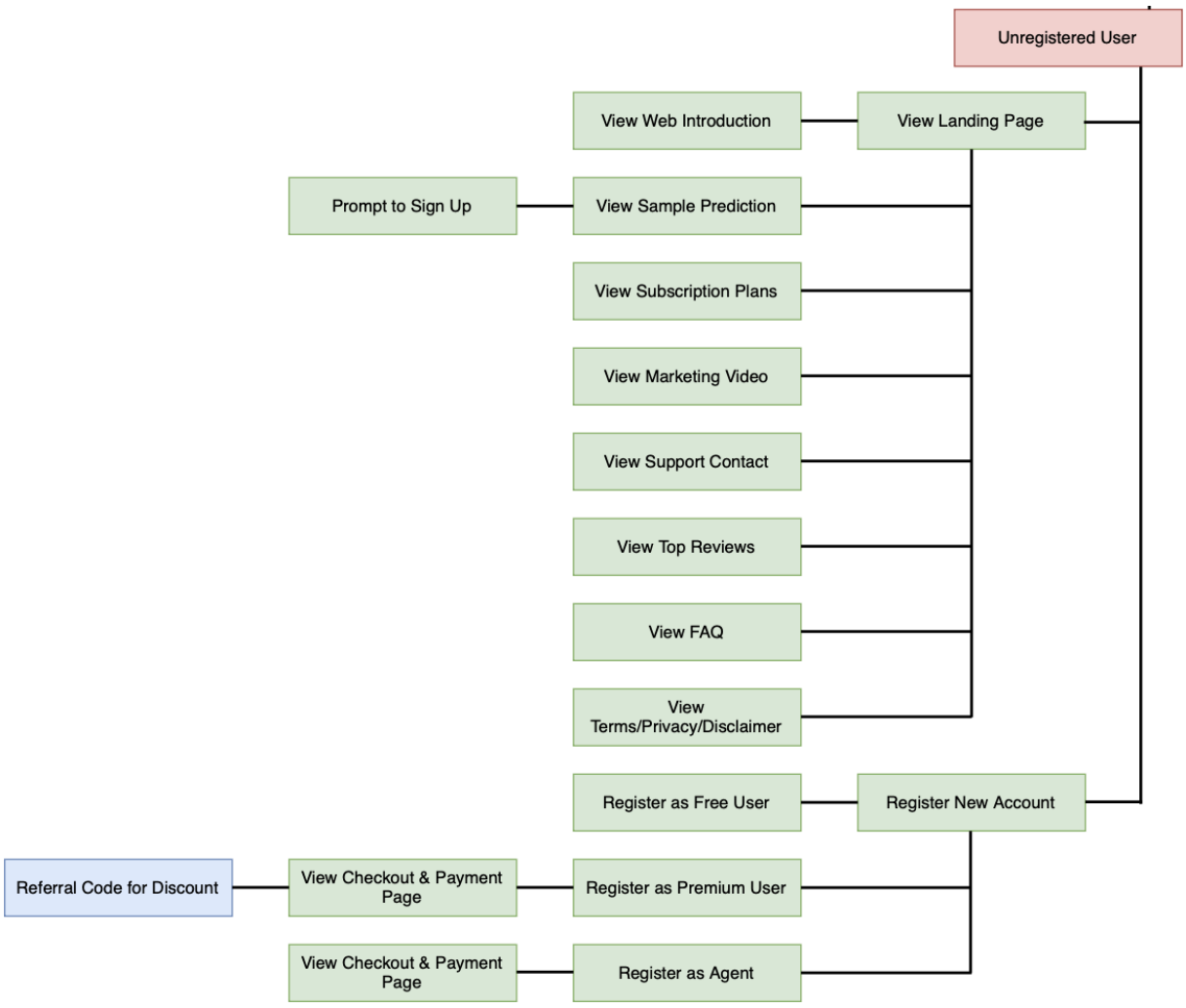


Role	Feature	Description
Unregistered Users	View Landing Page	Unregistered Users will be able to see the introduction of the website, sample prediction, subscriptions plans, marketing video, support contact, top reviews, FAQ, the team behind the web, disclaimer, terms and privacy of the website.
	Register New Account	Unregistered Users will be able to register for an account after verifying their email and submitting the required details.
Registered Free/ Premium Users, Property Agent, System Admin	Login	Users will be able to login to their accounts and be redirected to their respective dashboard.
	Logout	Users will be able to log out from their accounts and be redirected to the landing page.
	Recover Account	Users will be able to recover their account by verifying the email registered, and changing the password.
Registered Free/ Premium Users, Property Agent	View Support Contact	Users will be able to see the contact details for support.
	View Feedback Page	Users will be able to enter their reviews for the website, view the top reviews for the website and see their past reviews for the website.
	View User Profile	Users will be able to edit their user details, reset password and deactivate their account. For Registered Free/Premium Users, users can choose to upgrade or downgrade their account membership.
	View FAQ	Users will be able to see the FAQ of the website so if they have any queries, they can check out the FAQ

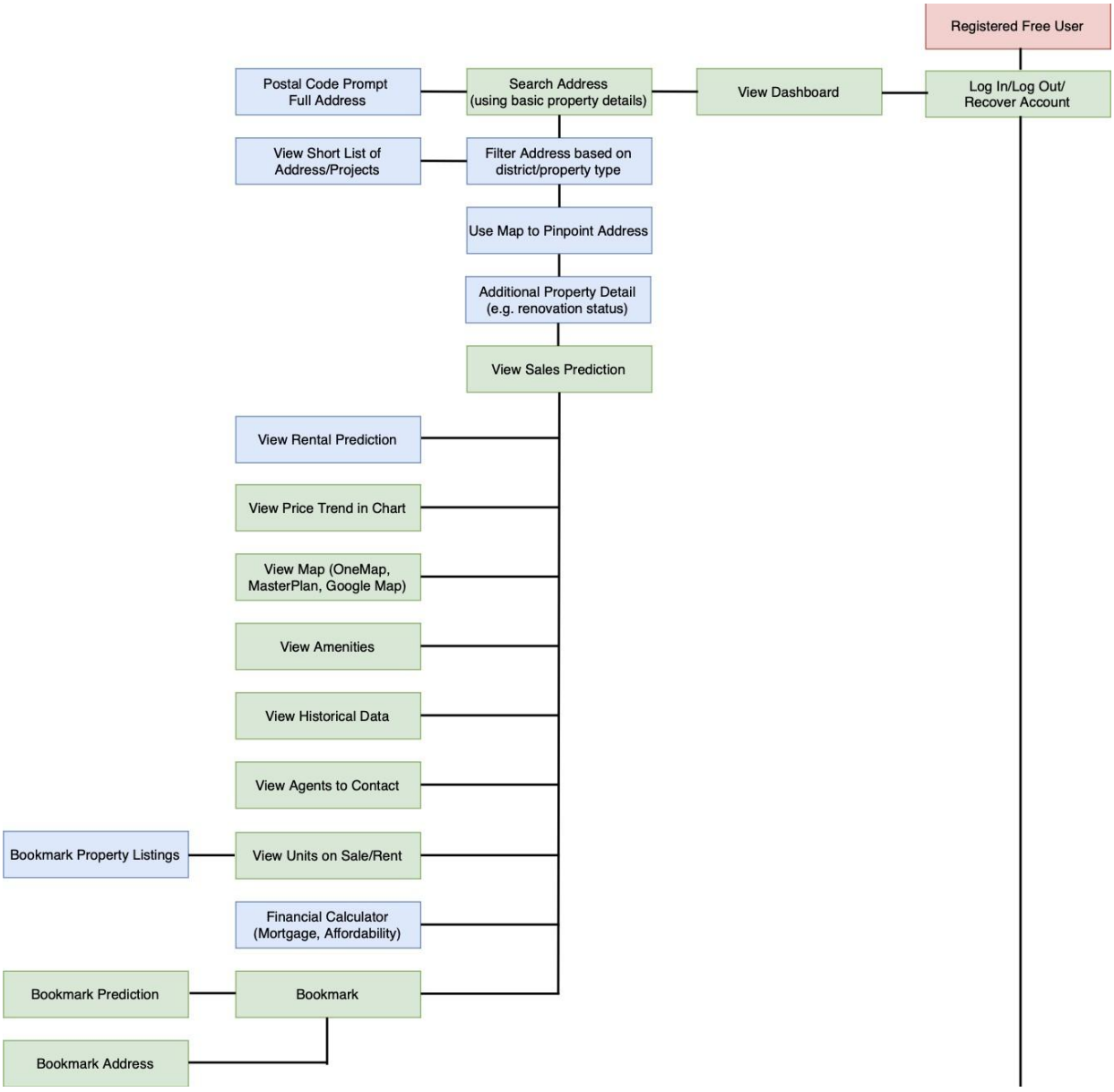
		page to see if their queries have been answered there.
	View Privacy Policy	Users will be able to see the privacy policy for the website.
	View Terms	Users will be able to see the terms for the website.
	View Disclaimer	Users will be able to see the disclaimer for the website.
Registered Free/ Premium Users	View Dashboard	Users will be able to search for an address and view the sales prediction for the property. The sales prediction would include price trend, map, amenities, historical data, units on sale/rent and agents to contact if the users want to get an agent to sell their property. Users can also bookmark the address or the prediction for easy viewing. For Registered Premium Users, they can search for two addresses and compare the sales prediction for those two properties. The comparison would also include price trend, map, amenities, historical data and agents to contact if they want to get an agent to sell their property. Registered Premium Users can also bookmark the address or the comparisons for easy viewing.
	View Bookmarks	Users will be able to see the addresses or predictions that they have bookmarked. For Registered Premium Users, they will be able to see the comparisons that they have bookmarked.
	View Units on Sale/Rent	Users will be able to check out other properties that are on sale or rent.

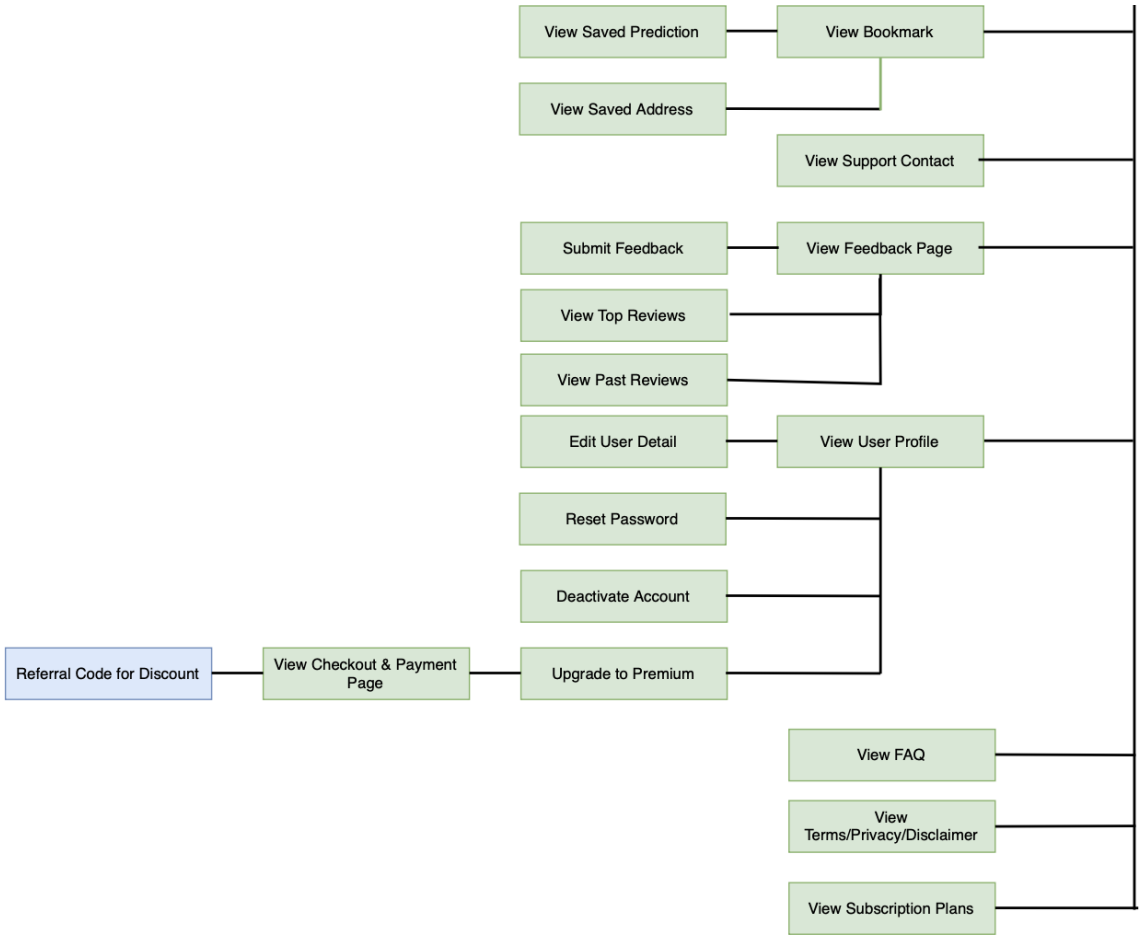
Property Agent	View Assigned Regions	Property Agents will be able to select, change and remove the regions that they will be in charge of, the regions selected will then allow the agents to appear in the price prediction or comparison prediction for Registered Free/Premium Users.
	View Units on Sale/Rent	Property Agents will be able to add, edit and remove listings that they want to market out.
System Admin	View User Account List	System Admin can search for user account, view user account and modify user account (deactivate and suspend user account).
	View User Feedback	System Admin can view user feedback, push user feedback for other users to view, remove user feedback and respond to user feedback.
	Edit Webpage	System Admin can update the hero section, how it works section, see how it works section, and team members section of the landing page. Also, System Admin can update the subscription plan, FAQ, contact, terms, privacy and disclaimer of the website.

5.3.1 Unregistered User

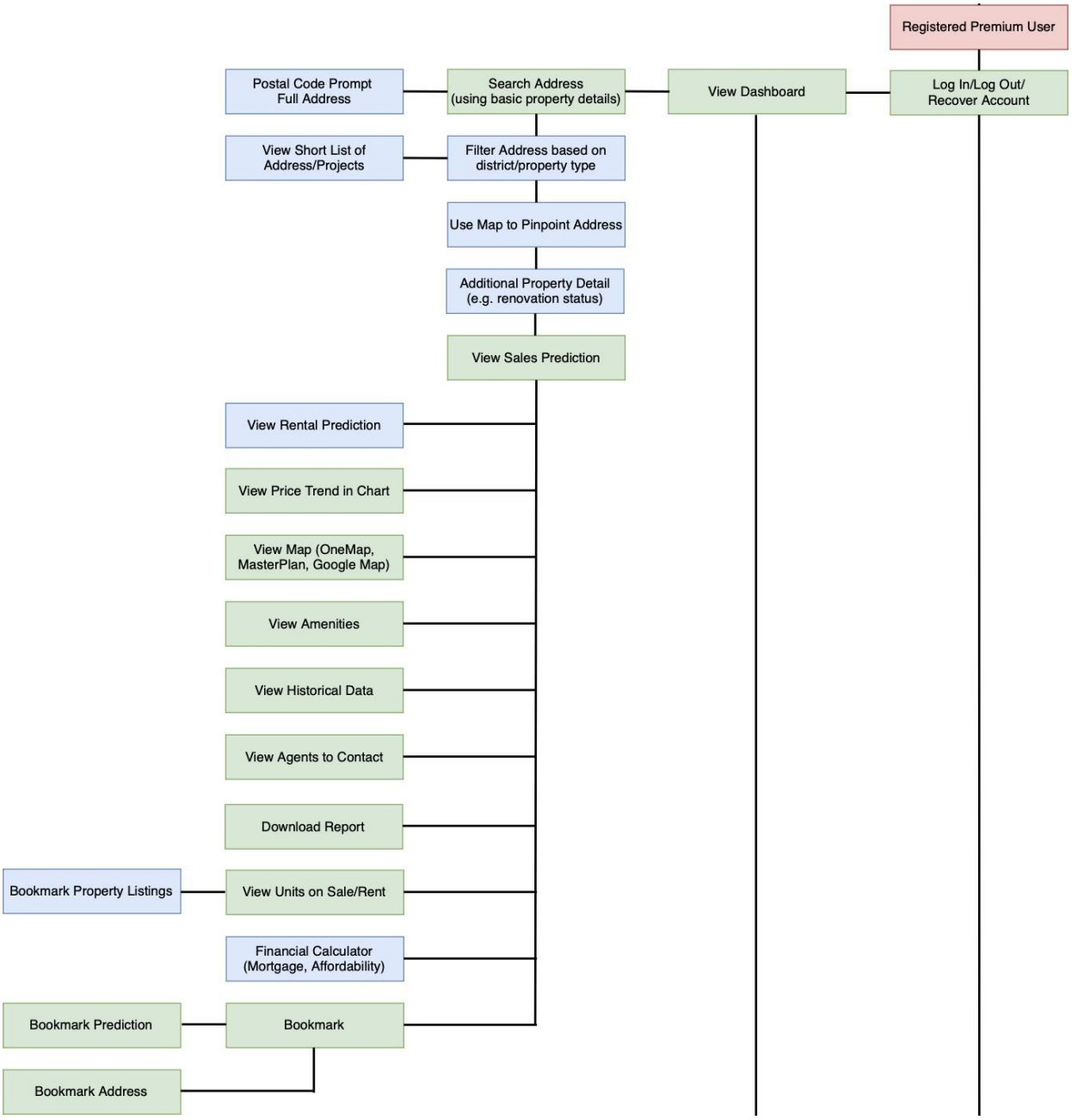


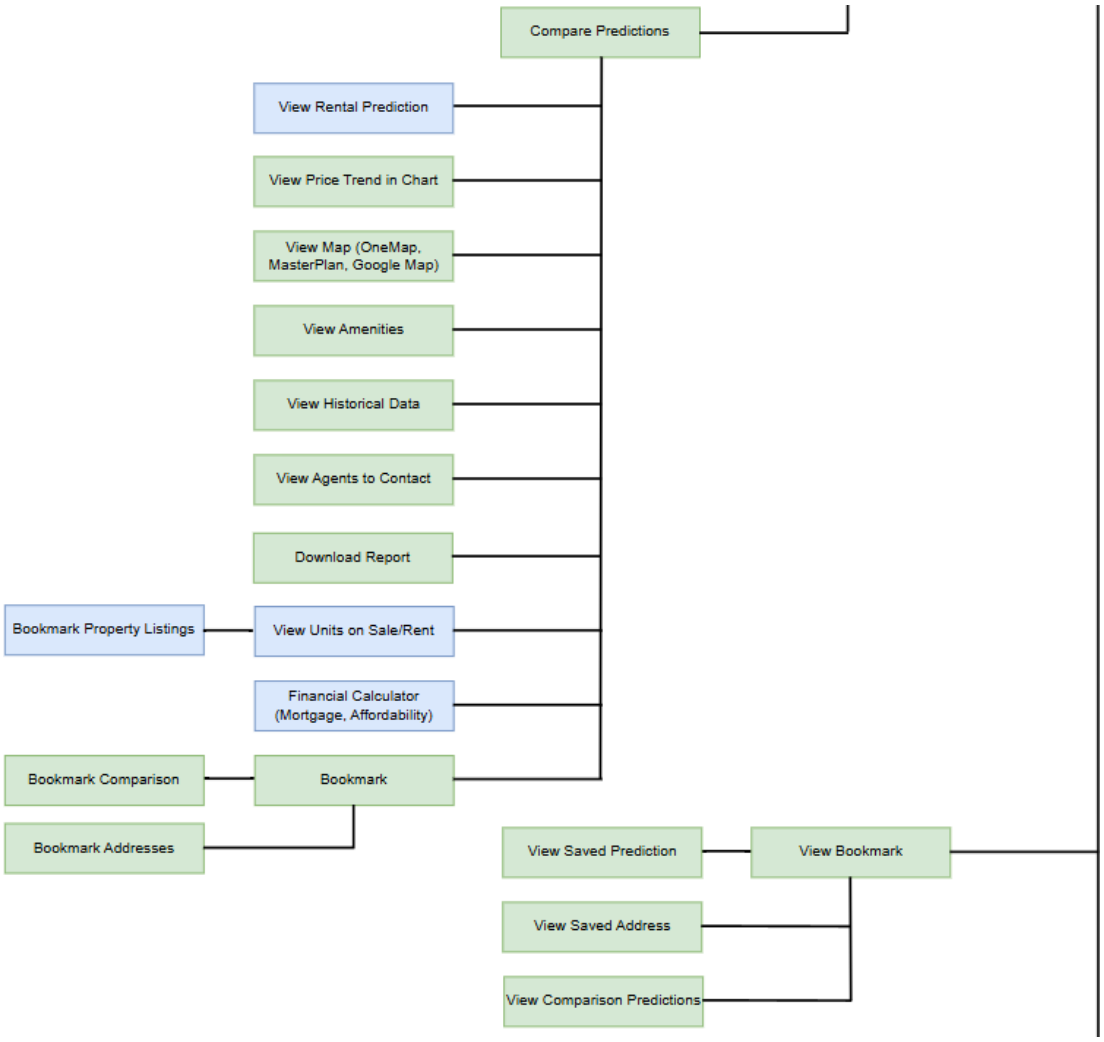
5.3.2 Registered Free User

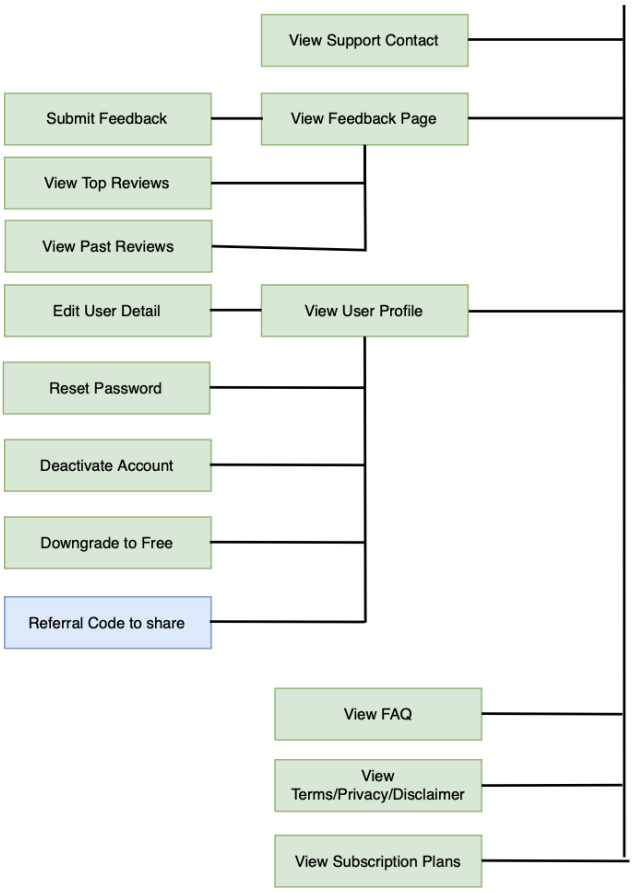



View Saved Prediction
View Bookmark
View Saved Address
View Support Contact
Submit Feedback
View Feedback Page
View Top Reviews
View Past Reviews
Edit User Detail
View User Profile
Reset Password
Deactivate Account
Referral Code for Discount
View Checkout & Payment Page
Upgrade to Premium
View FAQ
View Terms/Privacy/Disclaimer
View Subscription Plans

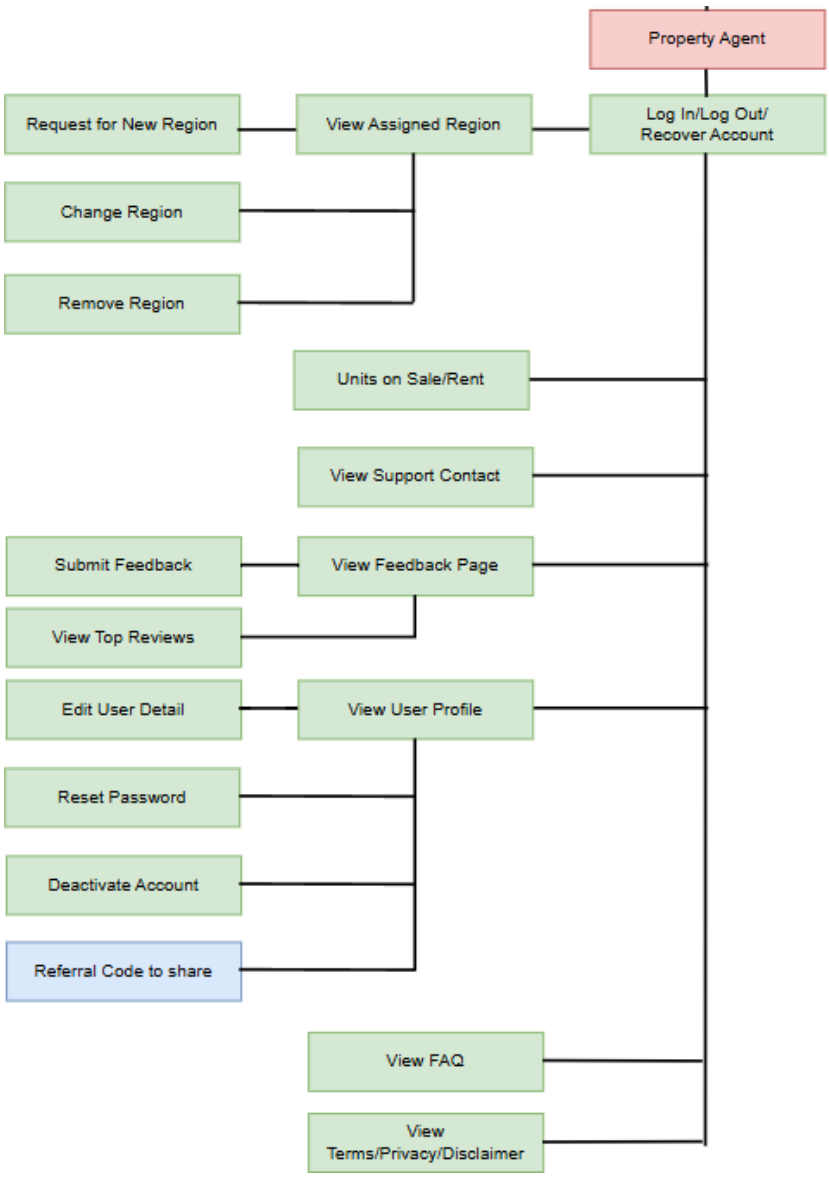
5.3.3 Registered Premium User



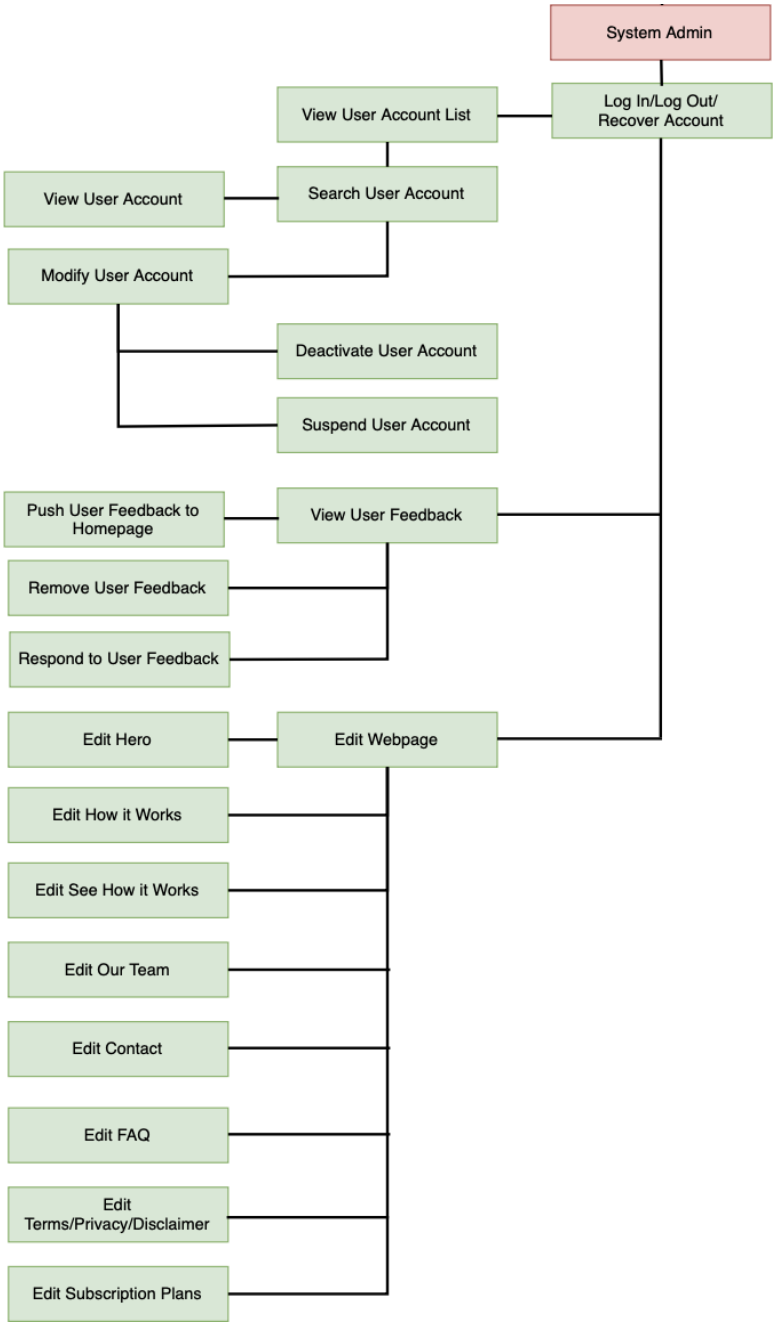

Compare Predictions
View Rental Prediction
View Price Trend in Chart
View Map (OneMap, MasterPlan, Google Map)
View Amenities
View Historical Data
View Agents to Contact
Download Report
Bookmark Property Listings
View Units on Sale/Rent
Financial Calculator (Mortgage, Affordability)
Bookmark Comparison
Bookmark
Bookmark Addresses
View Saved Prediction
View Bookmark
View Saved Address
View Comparison Predictions


View Support Contact
Submit Feedback
View Feedback Page
View Top Reviews
View Past Reviews
Edit User Detail
View User Profile
Reset Password
Deactivate Account
Downgrade to Free
Referral Code to share
View FAQ
View Terms/Privacy/Disclaimer
View Subscription Plans

5.3.4 Property Agent



5.3.5 System Admin



5.4 Project Charter

Project Title: Interactive Website for Real Estate Price Prediction in Singapore
Project Start Date: 5 July 2025 **Projected Finish Date:** 29 November 2025
Project Manager: Mandy, +65 9232 3526, mxlng001@mymail.sim.edu.sg
Project Objectives: The project objective is to design a user-friendly platform to predict real estate prices in Singapore where users can interact with the website to input property details, and receive instant price estimates based on historical data and property attributes.
Success Criteria: Adherence to scope, schedule, and quality expectations
Approach: The Scrum framework will be used to divide our work into targeted sprints. Every sprint will begin with a planning session where we will establish specific objectives. It will then end with a review to reflect on the work that has been done and identify areas that require improvement. Frequent stand-up meetings will also be held to keep everyone informed about the project and possible roadblocks in order to maintain alignment. After that, tasks will be delegated according to each team member's strengths, maintaining flexibility to allow us to assist one another as needed. Being able to work cooperatively and adaptably enables us to overcome obstacles without slowing down our progress.
Roles and Responsibilities

Name	***Role/Position***	***Contact Information***
Mandy Ng Xue Ling	Team Leader, Frontend & Backend Developer, Documentation	mxlng001@mymail.sim.edu.sg
Thet Myat Noe	Documentation, Frontend & Backend Developer	thetmn001@mymail.sim.edu.sg
Angel Lim	Documentation, Frontend & Backend Developer	alim013@mymail.sim.edu.sg
Zafeerah Akbar	Documentation, ML Developer	zafeerah001@mymail.sim.edu.sg
Aung Myint Myat	Documentation, ML Developer	aungmm002@mymail.sim.edu.sg

5.5 Communication Management Plan

5.5.1 Stakeholder Communication Requirements

Stakeholder Group	Preferred Communication Channels
Project Team	Discord, Zoom, Microsoft Teams, Telegram, Taiga, Email, Physical Meetings, Google Drive, Google Docs
Supervisor	Email, Zoom
Assessor	Physical Meetings

5.5.2 Communications Summary

Stakeholder	Communication Type	Delivery Channel	Responsible Party	Period
Project Team	Sprint Review & Planning	Discord, Teams, Zoom, Taiga	Scrum Master / Team Leader	Weekly
Project Team	Task Assignment / Updates	Taiga, Telegram	Team Leader	Continuous (as needed)
Supervisor	Progress Updates / Feedback	Zoom, Email	Team Leader	Biweekly
Assessor	Mid-Project Prototype Demo	Physical Meeting	Full Team	20 Sept 2025
Assessor	Final Presentation	Physical Meeting	Full Team	22 Nov 2025

5.5.3 Communication Guidelines

- **Microsoft Teams** will be the main platform for team meetings, especially for real-time discussions related to development progress and UI/UX implementation.
- **Telegram** will be used for daily communication and reminders such as meeting arrangements.
- **Taiga** will manage the product backlog, sprint planning, and task tracking across the frontend and backend development workflows.
- **Email** will be reserved for official communication with the supervisor, particularly for sending queries which the team has.
- **Google Docs** will be used for collaborative drafting of documents.
- **Google Drive** acts as the central file repository for wireframes, screenshots, and final deliverables.
- In-person discussions may be scheduled during key milestones, such as major design reviews or before submission deadlines.

5.5.4 Conflict Escalation Procedure

- If any team member encounters a project-related issue, they will first notify the Team Leader via Telegram or during scheduled Discord discussions.
- The Team Leader will address the matter within one working day. If the issue cannot be solved or affects critical tasks, it will be escalated to the Supervisor via email along with the necessary information.
- Any conflicts that risk delaying project milestones will also prompt an internal impact analysis and action plan, which may include reassigning tasks, extending buffer time, or simplifying deliverables to ensure continuity.

5.5.5 Document Revision Protocol

- This communication plan will be reviewed at the end of each sprint (every one week) during sprint retrospectives.
- Updates will be made when:
  - Communication tools or platforms are replaced or added.
  - A stakeholder's preferred communication method changes.
  - The frequency or format of meetings is adjusted.
- All changes to the document will be versioned and stored in Google Docs, and updates will be announced in the team Discord.

5.6 Scope Statement

Project Title: Interactive Website for Real Estate Price Prediction in Singapore

Product Characteristics and Requirements:

1. Data Gathering and Processing:
   - ➢ Collect comprehensive real estate data such as property characteristics, transaction history and location-based features.
   - ➢ Clean and pre-process data for consistency, quality, and usability.
2. Model Development:
   - ➢ Select and implement suitable machine learning algorithms.
   - ➢ Train and fine-tune predictive models using historical transaction data.
   - ➢ Evaluate model performance using metrics
3. User Interface Design:
   - ➢ Develop an intuitive and responsive web interface.
   - ➢ Enable users to input property details and view instant predictions.
4. Geospatial Insights:
   - ➢ Integrate geospatial analysis to show location factors such as proximity to MRT, schools and malls.
   - ➢ Assess the impact of amenities, transportation, and neighbourhood attributes on property prices.
5. Data Visualization:
   - ➢ Provide interactive charts and maps showing predicted prices, price trends, and area comparison.
   - ➢ Help users make informed decisions through clear visual feedback.
6. Deployment and Accessibility:
   - ➢ Host the trained model and frontend on a web platform.
   - ➢ Ensure accessibility across major devices and browsers.

Product User Acceptance Criteria:

- ➢ System must provide reasonably accurate price estimates.
- ➢ User inputs must be validated and responded to within a few seconds.
- ➢ Interface must be mobile- and user-friendly.
- ➢ Deliverables must be completed within defined schedule.
- ➢ System passes supervisor review and testing walkthrough.

Summary of Project Deliverables:

Project management-related deliverables: Project charter, scope statement, WBS, Gantt Chart, final presentation, final technical documentation, final user manual, reflective diary, and any other documents required to manage the project.

Product-related deliverables: wireframe, use cases, source code and website

6. Requirement Definition

6.1 Stakeholder Identification

Primary Stakeholders:

- **Property Buyers:** Require accurate and accessible property price predictions for informed decisions.
- **Property Sellers:** Need realistic valuations for competitive pricing.
- **Property Agents:** Require tools for managing listings and client support
- **Investors & Developers:** Seek in-depth financial insights and high-volume analysis.

Secondary Stakeholders:

- **System Admins:** Maintain platform functionality, user account oversight, and data security.
- **Regulatory Bodies:** Ensure compliance with Singapore's PDPA and industry standards (e.g., URA, MAS).
- **External Data Providers:** Supply transactional, geospatial, and market datasets (e.g., URA, teeco).

User Types and Functional Scope:

User Type	Access Scope
Unregistered Users	View landing pages, sample predictions, FAQs, subscription plans, and register
Registered Free Users	View rental/sale info, bookmark properties, feedback
Registered Premium Users	Report downloads, comparison tools
Property Agents	Manage regional listings, view feedback, profile tools
System Admins	Manage users, control feedback, monitor subscriptions, and enforce compliance

The platform for predicting real estate prices involves multiple stakeholders: buyers who need reliable forecasts, sellers looking for market assessments and investors requiring thorough insights for informed decisions, each having distinct viewpoints and needs.

Secondary stakeholders are crucial for platform success. System admins maintain operations and security. Regulatory bodies, like Singapore's Urban Redevelopment Authority and Monetary Authority of Singapore, set compliance frameworks. External data providers deliver essential data for prediction algorithms.

The user classification system divides stakeholders into access levels: unregistered users access limited features, registered free users use basic services, and registered premium users obtain advanced features through subscriptions.

6.2 Requirement Gathering Methods

The process of gathering requirements employed a diverse strategy to identify stakeholder needs and market possibilities. Primary research involved comprehensive market analysis of platforms such as PropertyGuru, 99.co, and SRX to uncover feature deficiencies and user challenges. This competitive analysis provided insights into industry benchmarks, emphasizing differentiation possibilities via improved functionality and user experience.

User story development was pivotal, creating detailed scenarios across stakeholder categories to reflect realistic usage patterns and ensure the final system addressed actual needs. Regulatory research was crucial due to Singapore's stringent data protection laws, ensuring compliance with the Personal Data Protection Act. Data collection focused on analyzing public datasets, including Urban Redevelopment Authority transactions and teeco.sg information, informing technical and functional requirements for effective property price prediction services.

- ➢ Market Analysis: Platforms like PropertyGuru, 99.co, SRX reviewed.

- User Stories: Captured realistic usage scenarios.
- Regulatory Research: Ensured PDPA compliance.
- Public Data Analysis: URA, teeco.sg datasets used.

6.3 Requirement Analysis Approach

Prioritization Framework:

Priority	Features
Critical	Address-based prediction, registration, secure login, feedback display on homepage, charts, property bookmarks
Desirable	Agent contact access, financial calculators, referral programs

Evaluation Dimensions:

- User Journey Optimization
- Technical Feasibility
- Business Value Impact

6.4 Summary of User and System Needs

Analysis of user needs revealed essential criteria for platform success. Accurate price predictions are vital, achieved through sophisticated machine learning algorithms that leverage vast historical data and current market conditions, ensuring reliability and confidence intervals. A user-friendly experience is essential for all parties involved, requiring an easy-to-use, responsive design compatible with desktop and mobile platforms, accessible without technical knowledge, and featuring in-depth market analysis such as past trends and area characteristics for well-informed choices.

The technical infrastructure needs to facilitate a scalable cloud architecture that can handle different user loads and data processing requirements, ensuring smooth integration of various data sources through a strong data pipeline. Dedicated machine learning infrastructure is essential for immediate predictions, necessitating continuous

model training and assessment. Security protocols must guarantee compliance with the Personal Data Protection Act while allowing efficient data access for predictive services.

User Needs:

- Accurate Predictions: Based on historical trends.
- Interactive Tools: Maps and comparison charts.
- Account Management: Easy registration, edits, and upgrade/downgrade capabilities.
- Mobile-first Design: Responsive across devices.

System Requirements:

- Role-Based Access Control: Tailored to WBS-defined user groups.
- Scalable Infrastructure: Cloud-native and auto-scalable.
- Secure Data Pipeline: Integrating external APIs, ETL processes.
- Real-time ML Predictions: With adjustable input parameters.

6.5 Use of Tools

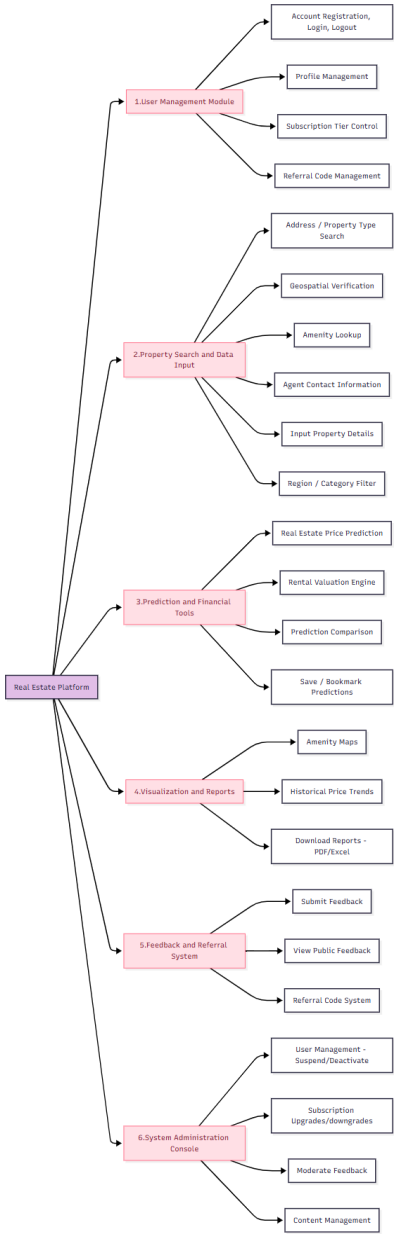
Use Case Diagrams depict user interactions with system features, confirming that stakeholder needs and access control are specified. System Architecture Diagrams illustrate system architecture, highlighting relationships among components, aiding deployment choices, and defining service boundaries for frontend, backend, and machine learning models.

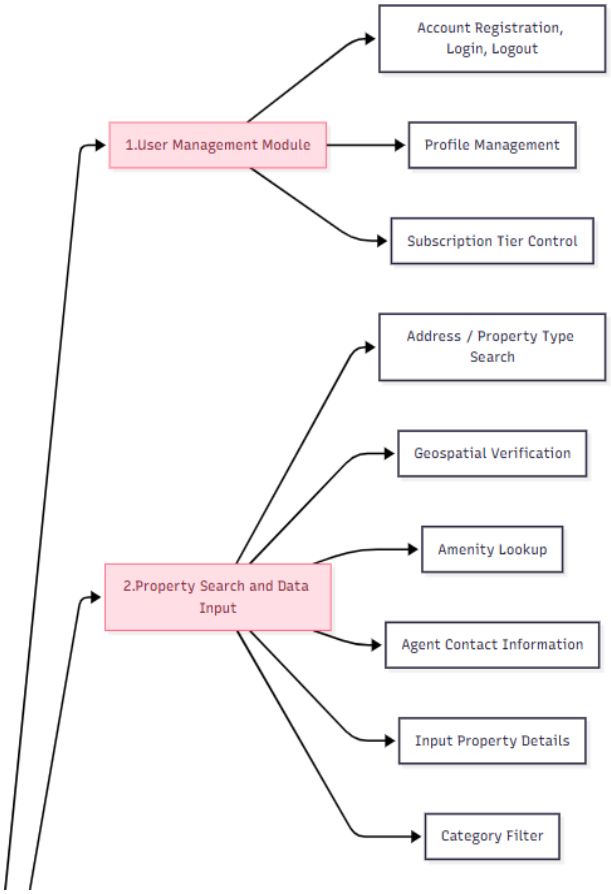
- Use Case Diagrams: Show user-role-based access to platform features.
- System Architecture Diagrams: Architecture showing frontend, backend, ML models, database.

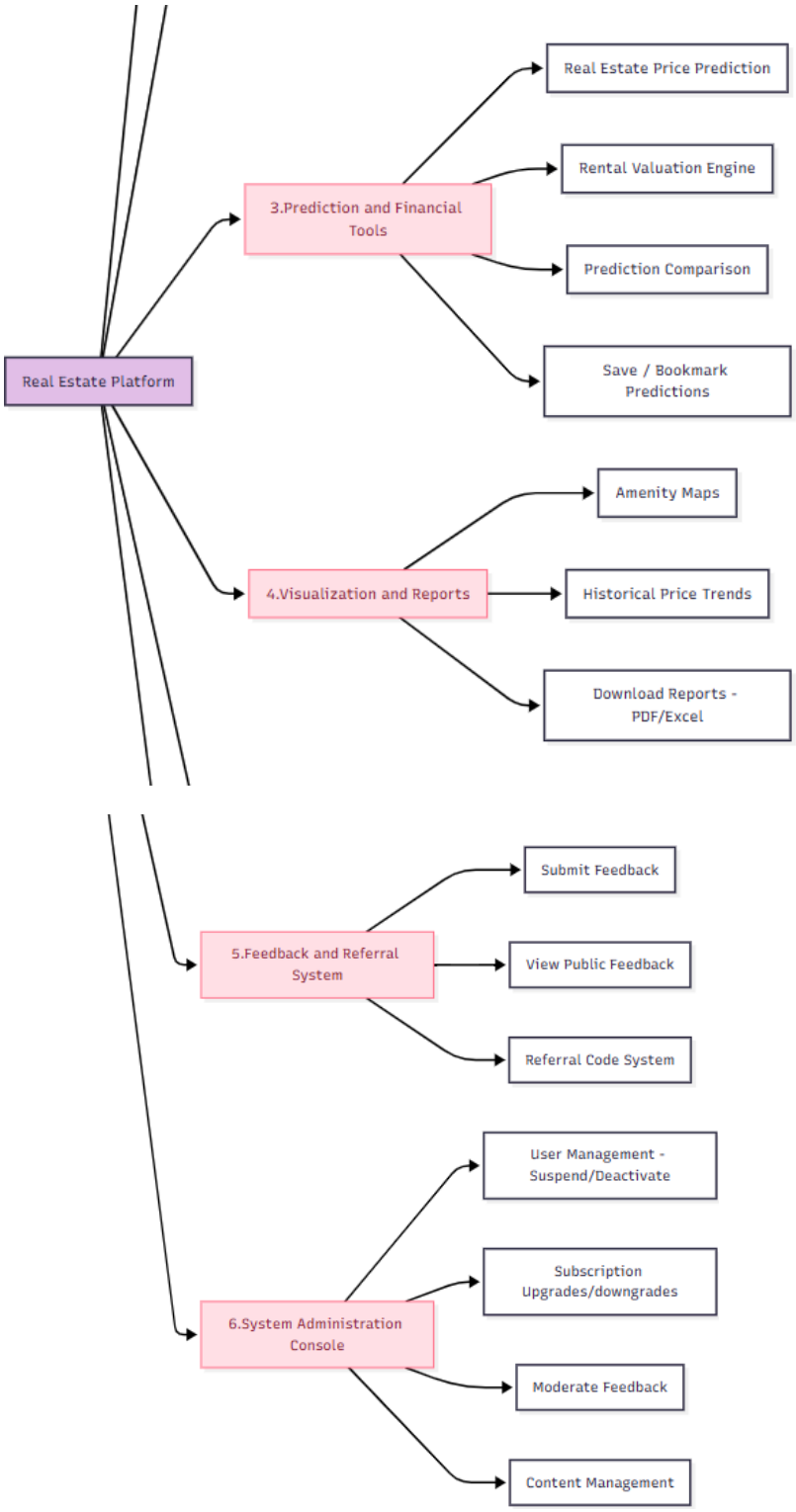
7. Functional Requirements

7.1 Functional Hierarchy

The functional architecture of the real estate price prediction platform is organized hierarchically to provide a clear understanding of system capabilities and component relationships.






Real Estate Platform
3.Prediction and Financial Tools
Real Estate Price Prediction
Rental Valuation Engine
Prediction Comparison
Save / Bookmark Predictions
4.Visualization and Reports
Amenity Maps
Historical Price Trends
Download Reports - PDF/Excel
5.Feedback and Referral System
Submit Feedback
View Public Feedback
Referral Code System
6.System Administration Console
User Management - Suspend/Deactivate
Subscription Upgrades/downgrades
Moderate Feedback
Content Management

This diagram depicts the hierarchical structure of the essential and advanced features of the Real Estate Platform, acting as a guide for organizing development priorities. Features are distinctly labeled as Basic (accessible to all users, even unregistered visitors) or Advanced (reserved for registered or premium-tier users), promoting a balanced rollout approach.

The fundamental features emphasize universal access, including viewing property information, setting up accounts, and conducting regular searches. These guarantee that all users, irrespective of their registration status, have a uniform and engaging experience with the platform.

The Advanced features offer improved tools such as AI-powered forecasts, financial calculators, and high-quality data visualization reports. These features are designed to provide specific value, promote subscription enhancements, and address the requirements of data-informed decision-makers.

Features are categorized into “Must-Have,” “Nice-to-Have,” and “Stretch” (as per the Comparison Matrix) to direct a phased development strategy:

7.1.1. User Management Module

- **Basic:**
    - Account Registration, Login, Logout (**Must-Have**)
    - Profile Management (**Must-Have**)
- **Advanced:**
    - Subscription Tier Control (**Must-Have**)
    - Referral Code Management (**Nice-to-Have**)

7.1.2. Property Search and Data Input

- **Basic:**
    - Address / Property Type Search (**Must-Have**)
    - Region / Category Filter (**Must-Have**)
    - Amenity Lookup (**Must-Have**)
- **Advanced:**
    - Geospatial Verification (**Stretch**)
    - Agent Contact Information (**Nice-to-Have**)
    - Input Property Details (**Must-Have**)

7.1.3. Prediction and Financial Tools

- **Advanced:**
    - Real Estate Price Prediction (**Must-Have**)
    - Rental Valuation Engine (**Nice-to-Have**)
    - Prediction Comparison (**Nice-to-Have**)
    - Save / Bookmark Predictions (**Stretch**)

7.1.4. Visualization and Reports

- **Basic:**
  - Amenity Maps (**Nice-to-Have**)
- **Advanced:**
  - Historical Price Trends (**Must-Have**)
  - Download Reports – PDF/Excel (**Nice-to-Have**)

7.1.5. Feedback and Referral System

- **Basic:**
  - Submit Feedback (**Must-Have**)
  - View Public Feedback (**Nice-to-Have**)
- **Advanced:**
  - Referral Code System (**Nice-to-Have**)

7.1.6. System Administration Console

- User Management – Suspend/Deactivate
- Subscription Upgrades/Downgrades
- Moderate Feedback
- Content Management

7.2 Basic Feature Access Levels

The platform employs a tiered access system that aligns user benefits with revenue creation, offering distinct upgrade motivations throughout the subscription tiers. Unregistered users are granted restricted access meant to showcase platform features while promoting account registration and future subscription.

User Tiers:

Feature	Unregistered User	Registered Free User	Registered Premium User	Property Agent	System Admin
Register	✓				
Login/Log out		✓	✓	✓	✓
Property Search (with location)		✓	✓		
Prediction Download			✓		
View Amenities/Charts/Maps		✓	✓		
Submit Feedback		✓	✓	✓	
Bookmark Properties		✓	✓		
Edit Profile		✓	✓	✓	
Upgrade/downgrade Tier		✓	✓		
Region Assignments				✓	
User Account Management					✓
Push Feedback to Homepage					✓

Unregistered Users:

- ➢ Can explore the homepage for platform features, FAQs, and help resources.
- ➢ Review pricing plans and subscription benefits.
- ➢ Contact customer support for assistance.
- ➢ Registration is simple and user-friendly.

Registered Free Users:

- ➢ Enjoy enhanced features with clear boundaries encouraging premium subscriptions.
- ➢ Users can log into personalized accounts storing prediction histories.
- ➢ Ability to save properties and retrieve detailed price predictions.
- ➢ Prediction record lasts for tracking inquiries and analyzing results.
- ➢ Fundamental search and filtering tools facilitate market navigation.
- ➢ Transparent subscription upgrade details aid premium transitions.

Registered Premium Users:

- ➢ Enjoy full access to all platform features.
- ➢ Receive detailed historical price trend evaluations.
- ➢ Tailored property suggestions based on user preferences.
- ➢ Download prediction reports in PDF and Excel.
- ➢ Participate in referral programs with rewards.
- ➢ Access priority customer support for assistance.

Property Agents:

- ➢ Gain dedicated tools for showcasing property listings with enhanced visibility.
- ➢ Receive leads from interested buyers and sellers through the platform.
- ➢ Manage multiple property portfolios with centralized dashboards.
- ➢ Manage assigned regions for users to contact when needed.

System Admins:

- ➢ Extensive platform management skills for efficient operation and user support.
- ➢ Full user management: view, suspend, and deactivate accounts.hi

- ➢ Oversee user reviews, refresh marketing materials, and manage help documentation.
- ➢ Subscription management allows for account enhancements and support.

7.3 Dependencies

Key Dependency Categories:

Dependency type	Components	Critical Impact
ML framework	Python, Scikit-learn, XGBoost, SHAP, GeoPandas, NumPy, Flask	Prediction accuracy, interpretability, real-time response
Cloud infrastructure	Render (backend & database hosting), Vercel (frontend hosting), CDN, PostgreSQL with PostGIS	Availability, scalability, data integrity, low latency
Data Integration	ETL pipelines sourcing data from URA, CSV imports	Data accuracy, timeliness, prediction reliability
Functional Modules	React.js frontend, Flask backend APIs, PostgreSQL database, authentication & subscription logic, input validation	User experience, security, feature access control

The platform's capabilities depend on an integrated collection of dependencies that includes machine learning libraries, cloud infrastructure, data integration pipelines, and frontend/backend frameworks. Each is essential in guaranteeing the system functions dependably, securely, and with optimal performance.

ML Framework:

- **Components:** Libraries in Python such as Scikit-learn, XGBoost for strong and understandable regression models; SHAP for explaining models; GeoPandas and PostGIS for managing geospatial data; Flask for delivering machine learning models through RESTful APIs.
- **Objective:** To facilitate precise, understandable property price forecasts while accommodating immediate user inquiries with low delay.
- **Critical Impact:**
  - Guarantees that the predictive algorithms work consistently and produce trustworthy results for users.
  - Facilitates the backend API that links the ML models to the frontend interface.
  - Affects general user trust and the efficiency of the platform.

Cloud Infrastructure:

- **Components:**
  - **Render:** Provides the Flask backend and PostgreSQL database, delivering managed services, automated deployments, backup solutions, and scalability.
  - **Vercel:** Provides hosting for the React.js frontend with automated deployments and a worldwide CDN to enhance content delivery speed.
  - **PostgreSQL with PostGIS:** Stores structured real estate data with spatial capabilities for location-based queries.
- **Objective:** To provide a secure, scalable, and reliable environment for application hosting and data storage.
- **Critical Impact:**
  - Facilitates ongoing service accessibility and reactivity throughout user locations.
  - Guarantees safe user verification, session oversight, and data uniformity.

- Enables spatial queries crucial for functionalities such as proximity filtering and map displays.

Data Integration:

- **Components:** ETL processes that extract data from official sources such as the Urban Redevelopment Authority (URA) and CSV imports from verified datasets.
- **Objective:** To ingest and transform real estate and geographic data into a format compatible with the prediction models and user interface.
- **Critical Impact:**
  - Ensures data timeliness and correctness, directly affecting the reliability of predictions.
  - Stops old or inconsistent information from distorting model results or user choices.

Functional Modules:

- **Components:**
  - **Frontend:** React.js framework facilitating dynamic user interfaces, real-time predictions, and data visualization using libraries like Chart.js and Leaflet.
  - **Backend:** Flask API handling authentication, subscription management (free and premium tiers), input validation, and prediction request processing.
  - **Database:** PostgreSQL with PostGIS for relational and geospatial data.
- **Objective:** To ensure smooth user engagement, safe access management, and dependable feature provision.
- **Critical Impact:**
  - Controls user access according to subscription levels, guaranteeing appropriate feature restrictions.
  - Protects the platform from unauthorized entry and harmful inputs.

- Facilitates seamless navigation and boosts user interaction with adaptive UI elements.

7.4 Inputs and Outputs

Inputs:

- User: Email, password, address, property details
- Admin: User ID, feedback content, account actions
- Property Info: Size, type, age, location, amenities

Outputs:

- Price/Rental Predictions
- Comparison charts, amenity maps, historical graphs
- PDF reports, downloadable Excel summaries
- Real-time UI updates (predictions, bookmarks, edits)

The platform efficiently processes various user and external system inputs to provide valuable outputs for property investment and purchasing. User registrations involve capturing account-related information like email addresses and secure passwords, with validations for format, strength, and uniqueness to enhance security. Property price predictions require extensive property details. Such as type, location, size, age, floor level, renovations, and amenities, with strict input checks for compliance, geographic relevance, and market range. Administrative commands for user management, content updates, and configurations necessitate privilege checks and impact assessments to safeguard against unauthorized changes.

Outputs of the platform include actionable insights like estimated property values, comparable listings with transaction histories, per square foot analyses, and rental yield estimates for investors. Data visualizations represent market data in user-friendly formats such as interactive maps, historical price trends, and comparative performance charts, aiding strategic planning. Report capabilities generate professional

documentation like PDF valuation reports and Excel exports. The user interface is responsive and optimized for various devices, featuring interactive forms for data entry and real-time feedback to inform users about processing status and results.

Input Category	Required Fields	Validation Rules
Property Type	Classification	Singapore market categories
Location	Precise coordinates	Geographic boundary verification
Physical Details	Size, floor level	Reasonable range validation

8. Non-functional Requirements

8.1 Performance Requirements

Response Time Standards:

Operation Type	Target Response Time	Success Rate
Price Predictions	< 3 seconds	95%
Page Load Times	< 2 seconds	100%
Interactive Maps	< 5 seconds	95%
Database Queries	< 1 second	99%
User Authentication	< 1 second	99%

Performance requirements establish technical standards for optimal user experiences across platform functions. Property price predictions must be delivered within three seconds for ninety-five percent of requests. Static content page load times should not exceed two seconds, and interactive maps need to load within five seconds.

User interaction database queries should complete promptly, with authentication processes responding almost instantly. Throughput must support many concurrent users and high request volumes during peak times.

Premium report generation must allow multiple simultaneous operations for subscribers. Infrastructure scalability is essential for future growth, requiring horizontal scaling to accommodate a significant user influx.

Database systems need to manage extensive property records, and machine learning should support daily predictions with acceptable response times.

8.2 Security Requirements

Highlights:

- **Role-Based Access Control (RBAC)**
- **Database Encryption** for sensitive user info
- **PDPA Compliance:** Consent

Data Protection Standards:

The platform protects sensitive user information by encrypting the database, guaranteeing that personal data stays secure even if a breach occurs. Password security adheres to industry norms to safeguard against credential breaches.

Role-based access control (RBAC) limits feature access based on user subscription tiers, ensuring usability alongside security.

In accordance with Singapore's Personal Data Protection Act (PDPA), the system oversees user consent for data gathering and fosters transparency regarding data utilization.

8.3 Interface and Usability Requirements

UI Design:

- Responsive UI (Desktop, Tablet, Mobile)
- Support for Chrome and Safari

Usability Metrics:

- Task completion rate: >90%
- Onboarding success: >80%
- User satisfaction: >4.2 / 5
- Support contact rate: <5%

Interface and usability requirements establish standards for user interaction design that ensure the platform is accessible, intuitive, and effective across diverse user populations and usage scenarios. User interface standards mandate responsive design that adapts seamlessly to desktop computers, tablets, and mobile devices, ensuring consistent functionality regardless of the user's chosen access method.

- **Cross-browser compatibility:**
  - Supports major browsers (Chrome and Safari) for universal access.
  - Navigation design allows reaching any feature within three clicks, minimizing friction and promoting efficient task completion.
- **Usability metrics:**
  - Onboarding completion rates should exceed 80%, while task completion rates for core functions should surpass 90%.
- **User satisfaction scores and ratings:**
  - User satisfaction scores need to average above 4.2 out of 5.0, reflecting a positive user experience.
  - Support contact rates must be below 5% of active users, indicating an intuitive interface that reduces support needs.
- **Design specifications:**
  - Emphasize a minimalistic, professional aesthetic for financial decision-making applications.
  - Consistent color palettes and fonts decrease cognitive strain and enhance information processing.
  - Loading indicators for tasks over one second help manage user expectations regarding system responsiveness.

Designing error messages guarantees that users obtain understandable and actionable details when issues arise, aiding them in comprehending what happened and how to fix problems on their own.

8.4 Portability and Scalability Requirements

Portability and scalability requirements ensure the platform adapts to technical changes and user demands while maintaining performance.

Portability:

The platform uses GitHub for code versioning and team collaboration, ensuring easy integration, tracking, and environment consistency through CI/CD practices.

Scalability:

The platform is designed to facilitate growth across various property types like commercial, industrial, and potentially residential, each equipped with customized data and predicting logic. A modular backend structure enables each category to expand independently according to user needs. The infrastructure can be enhanced to accommodate more users and functionalities over time, including future elements such as maps, charts, or feedback tools.

9. Other Requirements

Beyond the functional and non-functional requirements, there are additional considerations that are just as important for the success of the system. These include factors that affect how the system is developed, how it operates, and how it stays compliant with relevant regulations.These subsections below provide detailed explanations of the technical, user, legal and regulatory, system and operational considerations.

9.1 User Requirements

Our system has three types of users such as Unregistered Users, Registered Users, Premium Users, Property Agents and System Admins. All users should be able to as follow:

- Unregistered users must be able to browse basic features such as sample price predictions, subscription plans and FAQs without creating an account.
- Registered users should be able to log in securely, perform property price predictions, bookmark predictions, and view their personal prediction history.
- Premium users must have access to advanced features such as downloadable reports, and property comparison tools.
- Property agents must be able to edit user details, view assigned sales regions and request changes if necessary, and deactivate accounts when no longer using the platform.
- System admin must be able to view, manage, deactivate and suspend all user accounts, including registered free, registered premium, and property agent accounts.

Overall, the system must ensure easy-to-use interfaces, clear navigation, and support on both desktop and mobile devices. The platform must provide sufficient guidance through FAQs and contact support options.

9.2 Legal and Regulatory Requirements

The system must comply with the Personal Data Protection Act (PDPA) of Singapore:

- ➢ Collect only essential user information with user consent.
- ➢ Protect user data through encryption.
- ➢ Include legal disclaimers clarifying that predictions are estimates and not financial advice.
- ➢ Implement clear Terms of Use and Privacy Policy on the platform.

9.3 System and Operation Requirements

The system should be designed to perform reliably, scale as usage grows, and be flexible enough to support future expansions without major redevelopment. It must provide a smooth user experience across devices while ensuring stability, security, and maintainability.

- ➢ The system must support high concurrency with minimal downtime (99% uptime goal).
- ➢ The prediction engine must return results within 3 seconds for standard inputs.
- ➢ The platform must be mobile-responsive and cross-browser compatible.
- ➢ Backend must be scalable to handle more users. Designed to be modular for easy future expansion. Able to add new property categories without major changes.
- ➢ Provide a secure admin panel for monitoring and maintenance.

10. Risk Management

Managing risks is a fundamental part of developing any software system, especially one that relies heavily on data, such as our Interactive Real Estate Price Prediction Platform. The success of this project relies on several important factors, including the reliability of data sources, the accuracy of the machine learning model, system security, and the trust of our users. Since the platform involves integrating publicly available datasets with real-time predictions and user input, we need to consider various risks including technical, legal, operational, and user-related things.

If these risks aren't handled properly, they could cause major consequences, such as inaccurate property price estimates, data security breaches, violating regulations like Singapore's PDPA, or a lack of user engagement. To prevent these issues, we're taking a proactive approach. This includes identifying potential risks early, assessing how likely and how serious they might be, and putting the right measures in place to manage or reduce their impact

This risk management strategy will help us track project health, address problems before they escalate, and ensure that the platform stays reliable, secure, and legally compliant. By staying flexible through agile development methods and being transparent with users, we aim to build a dependable product that fulfills its intended purpose while maintaining a high standard of quality and user satisfaction.

Risk	Description	Impact	Likelihood	Mitigation Plan
Data Reliability Risk	The system depends on open datasets (URA, Data.gov.sg). If data is outdated, inaccurate, or incomplete, predictions will be unreliable.	High	Medium	Use only verified government datasets, perform data cleaning, and validate before training models.
Machine Learning Model Performance Risk	The prediction model may fail to deliver accurate results for certain user inputs or market conditions, affecting user trust.	High	Medium	Conduct iterative model training and testing with diverse datasets, apply cross-validation, and collect user feedback for continuous improvement.
Legal & Regulatory Compliance Risk	Breach of the Personal Data Protection Act (PDPA) could lead to legal consequences and damage to reputation.	Critical	Low	Ensure user data is consent-based, enforce strict data protection policies by encryption, and display clear terms and privacy notices.
System Security Risk	Unauthorized access or hacking could lead to data leaks or service disruption, affecting both user trust and system integrity.	Critical	Low	Implement secure authentication methods and encrypting sensitive data.
Integration Risk	Failure in integrating the frontend, backend, and ML	Medium	Medium	Apply continuous integration practices, conduct unit and

	prediction engine may cause functionality breakdown or user experience issues.			integration testing at every stage, and ensure backend APIs are stable before frontend connection.
Hosting & Server Downtime Risk	Unexpected server failures or poor performance may disrupt user access, leading to dissatisfaction and loss of users.	High	Low	Hosts on reliable cloud platforms(Render) implement load balancing and monitoring, and conduct regular performance and testing.
User Adoption & Usability Risk	Poor UI/UX design may result in low user engagement.	Medium	Medium	Focus on user-centered, conduct usability testing, gather early feedback, and iterate on design based on user input.
Scope Creep & Timeline Risk	Expanding project scope beyond initial planning could lead to delays and resource strain.	High	Medium	Use Agile project management with fixed sprint timelines to monitor progress regularly.
Technical Expertise Risk	The team may face technical challenges, especially in ML deployment, API integration, or backend scaling.	Medium	Medium	Provide team training, encourage collaboration and pair programming, and seek external expert guidance when necessary.

11. Development Methodologies

Our project uses Scrum as the main development approach, with Taiga as the management tool. We chose this combination because it gives us a balance of flexibility, structure, and clear communication, which is important for a project that involves both technical modeling and interface design.

Adaptability and Iteration
Scrum breaks work into short sprints. Each sprint focuses on a small set of tasks, so the team can deliver results step by step instead of waiting until the end. At the end of every sprint, we review our progress in Taiga and decide what to improve for the next cycle. In practice, this means we can quickly refine our prediction model or test new features like neighborhood insights or trend visualization without getting stuck on one long development cycle.

Task Tracking
One of the reasons we chose Taiga is because it makes task management simple to follow. The Kanban board and backlog show exactly what stage each task is at. For example, data collection and preprocessing might be “in progress,” while user interface testing might already be marked as “done.” This overview helps the team avoid confusion and keeps deadlines in sight.

Workflows
The project is divided into clear phases: data gathering, model development, integration, and testing. Taiga lets us customize these workflows so they fit the actual order of our work. If new requirements appear, such as adding premium features or more search options, we can insert them into the workflow without disrupting the rest of the plan.

Collaboration
Since not all members can always meet in person, we rely on Taiga’s collaboration tools to stay connected. Features like comments, file uploads, and notifications make it easier to update one another. This is useful for coordinating between the data team, the developers, and the design group, especially when working remotely.

Prioritisation
Scrum also helps the team focus on tasks that deliver the most value. By using Taiga’s backlog, we can rank features based on importance. For example, improving prediction accuracy is treated as a higher priority compared to smaller interface adjustments. This way, we are always working on items that matter most to users.

User Stories
To make sure development stays connected to user needs, we use user stories. Examples include “As a registered free user, I want to save my past price predictions so that I can revisit or compare them later” and “As a property agent, I want to edit my user profile so that I can update my personal data when needed.” Each story is linked to specific tasks in Taiga so that all work leads back to a real requirement.

Feedback and Improvement
Finally, Scrum encourages continuous feedback. At the end of each sprint, we conduct reviews and retrospectives through Taiga. This allows us to spot problems early, whether they relate to system performance, data accuracy, or the usability of the interface. It also creates room for improvement throughout the project instead of leaving changes until the end.

12. Technical Stack

12.1 Frontend Framework

The system uses React.js as the frontend framework to build a dynamic and interactive user interface for real estate price prediction.

- ➢ React’s component-based architecture and strong state management suit handling user input, real-time predictions, and data visualizations. Its large ecosystem and compatibility with libraries like Chart.js and Leaflet also simplify integrating graphs and maps.

While alternatives like:

- ➢ Vue.js offer simplicity, it has a smaller ecosystem;
- ➢ Vanilla JS lacks scalability; and
- ➢ Next.js focuses on server-side rendering, which isn’t needed here.

Therefore, React.js was selected for its flexibility, rich ecosystem, and ability to support the complex UI interactions required in this project.

12.2 Backend Framework

The system uses Flask as the backend framework due to its simplicity and flexibility in building lightweight RESTful APIs for the real estate price prediction platform.

- ➢ Flask's minimalistic design enables rapid development and easy customization without unnecessary overhead, making it ideal for serving Python-based machine learning models and efficiently handling user requests.

While alternative like:

- ➢ Django offers a full-featured framework but adds unnecessary complexity;
- ➢ FastAPI provides high performance and modern features but has a steeper learning curve and may be more than required; and
- ➢ Node.js + Express is flexible and performant but requires switching to JavaScript for backend development, which adds overhead since our ML components are Python-based.

Therefore, Flask was selected for its lightweight nature, ease of integration with Python machine learning components, and straightforward API development aligned with the project's scope.

12.3 Database

PostgreSQL is a powerful, open-source relational database management system (RDBMS) recognized for its dependability, extensive features, and performance. When integrated with PostGIS (a spatial extension), it transforms into a robust engine for managing geospatial data, essential for real estate applications

- ➢ **Relational Structure**: Real estate data (properties, prices, locations, amenities) is best modeled in structured tables with relations (e.g., property -> district).
- ➢ **Geospatial Support (PostGIS)**: Enables spatial queries like:

- “Find properties within 500m of an MRT station.”
- “Sort listings by distance to a park.”

➢ **Scalability and Maturity**: Used in production by many enterprise-grade systems.

➢ **Indexing and Performance**: Supports advanced indexing (B-Tree, GIN, GiST) which speeds up complex queries.

A variety of database alternatives were assessed for the real estate price prediction system, ultimately selecting PostgreSQL with PostGIS. Nevertheless, options like SQLite, MySQL, MongoDB Atlas, and Firebase were also taken into account.

➢ **SQLite** is a compact, file-based database ideal for small or embedded applications but lacks support for simultaneous multi-user access and struggles with larger datasets, making it unsuitable for production-level web applications.

➢ **MySQL** offers a more robust relational database framework but lacks built-in geospatial functionalities, complicating spatial queries for location-based systems.

➢ **MongoDB Atlas** excels in handling unstructured data and horizontal scaling, but does not support relational integrity or complex spatial queries, limiting its utility for structured geospatial property records.

➢ **Firebase** is noted for its real-time synchronization and developer-friendly frontend integration, yet it falls short for data-rich structured datasets like real estate records that require relational joins and spatial indexing, failing to meet the necessary performance and data modeling needs.

These comparisons led to the choice of PostgreSQL with the PostGIS extension due to its robust relational features, built-in geospatial functions, free hosting option through Render, and support for spatial queries crucial for functions such as proximity filtering and district mapping.

12.4 Machine Learning Libraries

This combination offers a modular, scalable, and highly interpretable solution for property price prediction, particularly suited to tabular, structured datasets.

- **Scikit-learn**:
  - Industry-standard for regression and classification on structured data
  - Simple, consistent API for training, testing, and tuning models
- **XGBoost**:
  - High performance for regression tasks, especially with non-linear patterns
  - Regularization and built-in feature importance analysis
  - Excellent handling of missing values, boosting model robustness
- **SHAP**:
  - Provides explainability by showing which features drive predictions
  - Especially important for user trust in pricing models
- **geoPandas**:
  - Seamlessly handles spatial/geographic data in pandas-like syntax
  - Integrates well with PostGIS and helps visualize proximity-based features
- **NumPy**:
  - Core numerical operations, essential for vectorized ML workflows

Comparison with Alternatives:

- **LightGBM** excels in speed and efficiency, especially with large feature sets, making it suitable for pricing prediction despite challenges with categorical data and interpretability.
- **XGBoost** is preferred due to better community support and easier management of categorical features.
- **TensorFlow and PyTorch**, while powerful for deep learning, are not optimal for real estate price prediction, which relies on tabular data where simpler models outperform neural networks. These frameworks introduce unnecessary complexity and longer training times, lacking built-in interpretability.

- **Pandas** is crucial for data analysis, aiding in preprocessing and exploratory data analysis (EDA), but lacks predictive capabilities. It should be used alongside libraries like Scikit-learn or XGBoost for effective machine learning applications.

In the end, the selected combination of Scikit-learn, XGBoost, GeoPandas, and SHAP offers a balance of performance, interpretability, and seamless integration. This integration enables effective preprocessing, spatial feature management, and clear model interpretations, which are all essential for creating a reliable property price prediction system.

12.5 APIs

This project uses three main APIs: OneMap SG, Google Maps (Free Tier), and URA Map API to handle location search, map display, and land zoning data.

- OneMap SG is a map service built for Singapore. It helps find addresses, get coordinates, and locate nearby places like MRT stations or shopping malls. Since it's designed for local use, it gives accurate and reliable results for Singapore properties.
- Google Maps (Free Tier) adds interactive map features like directions, estimated travel time, and route planning. It makes the interface more familiar and easy to use, especially for things like showing where the property is and how far it is from key places.
- URA Map API shows information about land zoning and development plans. For example, it can tell if a piece of land is meant for commercial, residential, or industrial use, or if there are upcoming changes nearby. This helps users understand how the location might affect the future value of the property.

An alternative considered was OpenStreetMap, which is free and open-source. However, it does not provide detailed zoning data or local planning information, and it is not as easy to work with for Singapore-specific features.

12.6 Hosting

For our project, we decided to host the backend and database on **Render** and the frontend on **Vercel**. This combination offers a balance between simplicity, reliability, and cost-effectiveness, which is ideal for a student project like ours.

Backend Hosting:

- **Render** provides an easy way to run both the backend API and a database in the same environment, which saves us the trouble of managing two separate hosting services.
- **Render** works well with Python frameworks such as FastAPI, and setting it up only requires linking our GitHub repository, any new commits are deployed automatically.
- **Rende**r also offers managed database services with automated backups, and its free plan is good enough for our development phase. If the system needs more power later, we can easily upgrade.
- Another advantage is that it scales without too much manual work, so we can start small and grow as needed.

The Alternatives:

- **Netlify** is great for hosting frontends but isn't built to handle a backend that needs to run continuously.
- **Replit** is fine for testing ideas but doesn't perform well for hosting a long-running web service.
- **Railway.app** is quite similar to Render, but its free plan has lower limits and slightly less stability for running both a backend and a database together.
- **GitHub** Pages is for static websites only, so it wouldn't work for a project like ours that needs a live backend and database.

Frontend Hosting:

- For the frontend, we chose **Vercel** because it works perfectly with modern JavaScript frameworks like React.js, which we are using for our interface.
- **Vercel** automatically builds and deploys our site whenever we push updates to GitHub, and it delivers our content quickly through its global CDN.
- The free plan is more than enough for what we need, and we also have the

option to use Vercel's serverless functions if we want to extend the frontend with small backend-like features.

The Alternatives:

- **Netlify** is another solid option, but we went with Vercel since it integrates more naturally with React projects and offers excellent developer tools.
- **Replit** is better for quick prototyping rather than hosting a polished production site.
- **Railway.app** is mainly for backend services rather than static frontends.
- **GitHub** Pages can host static sites, but it doesn't handle modern build tools and frameworks as well as Vercel does.

12.7 Server

We're running our backend and database on Render using a Web Service and Database Instance.

- The main reason is convenience, we can deploy our FastAPI backend and connect it directly to the database without extra setup.
- The server stays running all the time, which is important for a service that has to respond to prediction requests instantly.
- It also comes with SSL encryption by default, which means our API communication is secure.
- If we ever need more capacity, we can simply upgrade the plan instead of moving to a new host.

The Alternatives:

- **Gunicorn + Nginx** is a strong choice for professional production environments, but it requires managing a VPS, server configuration, and security updates. For our group project, this would be unnecessary complexity.
- **Railway** offers similar features to Render but doesn't give us as much flexibility

for running both the backend

- and the database in one place.
- **Google Colab** is great for training our machine learning model but isn't meant to host a live backend.
- **Localhost** works for development and testing, but obviously can't be accessed by real users or our teammates from outside our own computer.

12.8 Deployment

The deployment process is managed using GitHub in combination with Render or Vercel for automatic deployment.

- Once code is pushed to the GitHub repository, Render or Vercel automatically builds and deploys the latest version of the frontend and backend without manual steps.
- This method is efficient for small teams, reduces deployment errors, and allows for rapid updates during testing and development.
- Both platforms support continuous deployment, preview environments and are beginner-friendly, especially when working under time constraints.

While Docker was considered for manual deployment and greater infrastructure control, it involves more complexity and is less suited for fast-paced academic development cycles. GitHub-based auto-deploy tools were chosen for their simplicity, speed, and alignment with the project's timeline.

13. User Stories

13.1 Unregistered User

1. **As an unregistered user**, I want to browse the homepage, so that I can understand what this website offers. #1
2. **As an unregistered user**, I want to view random sample price predictions with limited details, so that I can try the prediction feature before signing up. #2
3. **As an unregistered user**, I want to view the FAQ section, so that I can find answers to common questions. #3
4. **As an unregistered user**, I want to see the contact page, so that I can reach out for queries not answered in the FAQ. #4
5. **As an unregistered user**, I want to see user reviews, so that I can learn about others' experiences. #5
6. **As an unregistered user**, I want to view pricing and subscription plans, so that I can decide whether to register or upgrade. #6
7. **As an unregistered user**, I want to see a prompt to sign up when I try to access restricted features, so that I'm aware of available benefits. #7
8. **As an unregistered user**, I want to register for an account, so that I can access features exclusive for registered users. #8
9. **As an unregistered user**, I want to view the marketing video to know what are the available features, so that I can easily understand the website. #9
10. **As an unregistered user**, I want to view the disclaimer for the website, so that I can understand that no machine-learning based prediction can be 100% accurate. #10
11. **As an unregistered user**, I want to use referral code, so that I can get a discount on the subscription plans available. #11

13.2 Registered Free User

1. **As a registered free user**, I want to log in securely, so that I can access the features in the website. #12
2. **As a registered free user**, I want to input basic commercial/industrial property information (e.g., postal code, property type, floor area), so that I can get a price prediction. #13
3. **As a registered free user**, I want to save my past price predictions, so that I can revisit or compare them later. #14
4. **As a registered free user**, I want to bookmark interesting commercial/industrial property addresses, so that I can access them easily later. #15
5. **As a registered free user**, I want to filter options for properties (e.g., location, type: office, retail, warehouse, factory), so that I can view properties that match my search filter. #16
6. **As a registered free user**, I want to view subscription plans, so that I can decide if I want to upgrade. #17
7. **As a registered free user**, I want to purchase a premium subscription, so that I can access advanced features. #18
8. **As a registered free user**, I want to see the contact page, so that I can reach out for queries not answered in the FAQ. #19
9. **As a registered free user**, I want to view commercial/industrial units on sale or for rent, so that I can identify opportunities for my business. #20
10. **As a registered free user**, I want to write feedback for the website, so that I can view and share my experience on the usage process. #21
11. **As a registered free user**, I want to edit my user profile, so that I can update my personal data when needed. #22
12. **As a registered free user**, I want to view commercial/industrial property agents' contacts, so that I can reach out for queries related to a particular property. #23

13. **As a registered free user**, I want to view historical transaction data for similar commercial/industrial properties, so that I can understand market pricing trends. #24
14. **As a registered free user**, I want to view an interactive map of Singapore showing nearby commercial hubs, logistics facilities, and transport links, so that I can assess location suitability. #25
15. **As a registered free user**, I want to log out securely, so that I can manage my session safely. #26
16. **As a registered free user,** I want to deactivate my account, so that I can stop all activities in that account. #27
17. **As a registered free user**, I want to view the FAQ section, so that I can find answers to common questions.#28
18. **As a registered free user**, I want to view the disclaimer for the website, so that I can understand that no machine-learning based prediction can be 100% accurate. #29
19. **As a registered free user**, I want to use referral code, so that I can get a discount on the subscription plans available. #30
20. **As a registered free user**, I want to see user reviews, so that I can learn about others' experiences. #68
21. **As a registered free user**, I want to view the marketing video to know what are the available features, so that I can easily understand the website. #70

13.3 Registered Premium User

1. **As a registered premium user**, I want to input basic commercial/industrial property information (e.g., postal code, property type, floor area), so that I can get a price prediction. #31
2. **As a registered premium user**, I want to save my past price predictions, so that I can revisit or compare them later. #32
3. **As a registered premium user**, I want to download prediction reports (PDF/Excel), so that I can share the report with others. #33
4. **As a registered premium user**, I want to manage my membership, so that I can have the choice to downgrade my subscription plan. #34
5. **As a registered premium user**, I want to invite others to the platform using referral codes, so that I can get referral rewards. #35
6. **As a registered premium user**, I want to log in securely, so that I can access the features in the website. #36
7. **As a registered premium user**, I want to view commercial/industrial property agents' contacts, so that I can reach out for queries related to a particular property. #37
8. **As a registered premium user**, I want to view commercial/industrial units on sale or for rent, so that I can identify opportunities for my business or investment. #38
9. **As a registered premium user**, I want to bookmark interesting commercial/industrial property addresses, so that I can access them easily later. #39
10. **As a registered premium user**, I want to compare between different price predictions, so that I can easily assess the difference between my prediction interests. #40
11. **As a registered premium user**, I want to write feedback for the website, so that I can view and share my experience on the usage process. #41

12. **As a registered premium user**, I want to edit my user profile, so that I can update my personal data when needed. #42
13. **As a registered premium user**, I want to log out securely, so that I can manage my session safely. #43
14. **As a registered premium user**, I want to filter options for properties (e.g., location, type: office, retail, warehouse, factory), so that I can view properties that match my search filter. #44
15. **As a registered premium user**, I want to see the contact page, so that I can reach out for queries not answered in the FAQ. #45
16. **As a registered premium user**, I want to view an interactive map of Singapore showing commercial hubs, logistics facilities, and transport links, so that I can assess location suitability. #46
17. **As a registered premium user**, I want to view the FAQ section, so that I can find answers to common questions. #47
18. **As a registered premium user**, I want to view the disclaimer for the website, so that I can understand that no machine-learning based prediction can be 100% accurate. #48
19. **As a registered premium user,** I want to deactivate my account, so that I can stop all activities in that account. #49
20. **As a registered premium user**, I want to view historical transaction data for similar commercial/industrial properties, so that I can understand market pricing trends. #66
21. **As a registered premium user**, I want to see user reviews, so that I can learn about others' experiences. #69
22. **As a registered premium user**, I want to view the marketing video to know what are the available features, so that I can easily understand the website. #71

13.4 Property Agent

1. **As a property agent**, I want to log in securely, so that I can access the features in the website. #50
2. **As a property agent**, I want to log out securely, so that I can manage my session safely. #51
3. **As a property agent**, I want to edit my user profile, so that I can update my personal data when needed. #52
4. **As a property agent**, I want to view a prompt to select regions, so that I can choose the regions that I want to be allocated to. #53
5. **As a property agent**, I want to see the contact page, so that I can reach out for queries not answered in the FAQ. #54
6. **As a property agent**, I want to write feedback for the website, so that I can view and share my experience on the usage process. #55
7. **As a property agent**, I want to invite others to the platform using referral codes, so that I can get referral rewards. #56
8. **As a property agent**, I want to view the FAQ section, so that I can find answers to common questions. #57
9. **As a property agent**, I want to view the disclaimer for the website, so that I can understand that no machine-learning based prediction can be 100% accurate. #58
10. **As a property agent**, I want to deactivate my account, so that I can stop all activities in that account. #59
11. **As a property agent**, I want to mark properties for sale or rent, so that I can access more interested parties. #67

13.5 System Admin

1. **As a system admin**, I want to log in securely, so that I can access the features in the website. #60
2. **As a system admin**, I want to log out securely, so that I can manage my session safely. #61
3. **As a system admin**, I want to view all user accounts, so that I can manage platform usage and user roles. #62
4. **As a system admin**, I want to suspend problematic accounts, so that I can enforce platform policies. #63
5. **As a system admin**, I want to moderate reviews, so that user-generated content remains constructive. #64
6. **As a system admin**, I want to update homepage banners, FAQs, or marketing content, so that the website stays fresh and up to date. #65

14. Use Case Descriptions

14.1 Shared Across Users

#3 As an unregistered user, I want to view the FAQ section, so that I can find answers to common questions.

#28 As a registered free user, I want to view the FAQ section, so that I can find answers to common questions.

#47 As a registered premium user, I want to view the FAQ section, so that I can find answers to common questions.

#57 As a property agent, I want to view the FAQ section, so that I can find answers to common questions.

Name:	View FAQ Section	3, 28, 47, 57
Stakeholders and Goals:	**Unregistered Users, Registered Free Users, Registered Premium Users, Property Agents** • Find answers to common questions without needing to contact support or register. • Understand how the website works, what features are available, and how to use them. • Clarify doubts related to pricing, data accuracy, account creation, and predictions. **Platform Owner/Business** • Reduce support workload by addressing common inquiries in a self-service format. • Build trust and credibility by being transparent and informative. • Support user onboarding by resolving concerns before registration.	
Description:	This use case allows unregistered users to freely access the FAQ section. It provides answers to commonly asked questions about the platform's functionality, usage, pricing, and registration to help users feel more confident in using or signing up for the platform.	

Actor:	Unregistered User, Registered Free User, Registered Premium User, Property Agents
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is not logged in or registered.
Trigger:	The user interacts with the FAQ feature.
Normal Flow:	1. The unregistered user navigates to the FAQ page via the homepage or footer link. 2. The system displays a list of categorized frequently asked questions. 3. The user clicks on a question to reveal the answer (expand/collapse or full page view). 4. The user browses through multiple entries until their query is answered. 5. Optional: the system prompts the user to register if they still need help or want access to more features.
Alternative/ Exception Flow:	**1a.** FAQ content fails to load → display fallback message: “Unable to load FAQs. Please try again later.” **5a.** The user doesn’t find the answer and clicks “Contact Support” → redirected to support form (optional).
Post Condition:	1. The user has read one or more FAQ entries. 2. The user may proceed to register or use available features based on the clarification received.

#4 As an unregistered user, I want to see the contact page, so that I can reach out for queries not answered in the FAQ.

#19 As a registered free user, I want to see the contact page, so that I can reach out for queries not answered in the FAQ.

#45 As a registered premium user, I want to see the contact page, so that I can reach out for queries not answered in the FAQ.

#54 As a property agent, I want to see the contact page, so that I can reach out for queries not answered in the FAQ.

Name:	View and Use Contact Page	4, 19, 45, 54
Stakeholders and Goals:	**Unregistered Users, Registered Free Users, Registered Premium Users, Property Agents** ● Reach out to customer support for help with questions not addressed in the FAQ. ● Get personalized support without needing an account. ● Understand the available support channels (e.g., email, phone). **Platform Owner/Business** ● Provide a communication channel to help visitors convert into users. ● Maintain a positive and responsive brand experience. ● Gather feedback or uncover gaps in FAQ or feature clarity.	
Description:	Users can visit the contact page to get in touch with customer support. This enables them to ask specific questions that are not answered elsewhere, enhancing trust and user experience even before account creation.	
Actor:	Unregistered User, Registered Free User, Registered Premium User, Property Agent	
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is not logged in or registered.	
Trigger:	The user interacts with the contact page section.	
Normal Flow:	1. The unregistered user clicks on "Contact Us" from the header, footer, or FAQ redirect. 2. The system displays the contact page with options (e.g., support email, phone number). 3. The user chooses their preferred method. 4. The system confirms message receipt. 5. The support team follows up via the chosen method.	
Alternative/ Exception Flow:	**3a.** Contact Options do not work → display fallback message: “Unable to load contact”	

Post Condition:	1. The user obtains contact details. 2. The system acknowledges receipt (if form is used) or opens communication (e.g., email link).

#5 As an unregistered user, I want to see user reviews, so that I can learn about others' experiences.

#68 As a registered free user, I want to see user reviews, so that I can learn about others' experiences.

#69 As a registered premium user, I want to see user reviews, so that I can learn about others' experiences.

Name:	View User Reviews	5, 68, 69
Stakeholders and Goals:	**Unregistered Users, Registered Free Users, Registered Premium Users** ● Learn what others think about the platform's reliability, usefulness, and overall experience. ● Decide whether the platform is worth using, registering for, or upgrading. ● Evaluate user trust and satisfaction before engaging further. **Platform Owner/Business** ● Increase credibility and transparency to attract new users. ● Encourage registration by showcasing positive user experiences. ● Maintain control over what content is visible to users (e.g., limit review details).	
Description:	Users are allowed to browse a selection of public user reviews on the platform.	
Actor:	Unregistered User, Registered Free User, Registered Premium User	
Pre-conditions:	1. The user has an internet connection to access the website.	
Trigger:	The user navigates to the "Feedback/Reviews" section, which is found on the homepage, about page, or in a dedicated	

	reviews/testimonials tab.
Normal Flow:	1. The user accesses the platform reviews section. 2. The system displays a list of user-submitted reviews about the platform. 3. Each review includes a rating (e.g., stars), a comment, and metadata like submission date and user name/initials. 4. The user browses and reads reviews freely.
Alternative/ Exception Flow:	**2a. System fails to load reviews** → Display an error message and retry option.
Post Condition:	1. The user has freely viewed platform reviews. 2. They may be influenced to register, return, or trust the platform based on the shared feedback.

#9 As an unregistered user, I want to view the marketing video to know what are the available features, so that I can easily understand the website.

#70 As a registered free user, I want to view the marketing video to know what are the available features, so that I can easily understand the website.

#71 As a registered premium user, I want to view the marketing video to know what are the available features, so that I can easily understand the website.

Name:	View Marketing Video on Platform Features	9, 70, 71
Stakeholders and Goals:	**Unregistered Users, Registered Free Users, Registered Premium Users** ◦ Understand how the platform works before using it. ◦ Learn how to navigate and use key features (e.g., predictions, filters, reviews) through a quick, accessible video. ◦ Decide whether to register based on ease of use and clarity. **Platform Owner/Business** ◦ Improve user onboarding by lowering the learning curve.	

	◦ Build user trust and confidence by transparently showcasing the platform's capabilities. ◦ Increase registration conversion by demonstrating value upfront.
Description:	A short marketing video is embedded on the homepage, visible and accessible to all visitors, including unregistered users. The video provides a guided walkthrough of how to use key platform features.
Actor:	Unregistered User, Registered Free User, Registered Premium User
Pre-conditions:	1. The user has internet access. 2. The homepage is accessible. 1. The marketing video is available and properly embedded.
Trigger:	The user lands on the homepage and scrolls to the marketing video section.
Normal Flow:	1. The user visits the homepage. 1. The marketing video section is prominently displayed (e.g., banner, card, or modal). 2. The user clicks the play button to watch the marketing video. 3. The video plays inline or in a modal window. 4. The user watches the video to understand how to use the platform.
Alternative/ Exception Flow:	**3a. Video fails to load** → Display fallback message: "Video unavailable. Please try again later or check our Help section." **4a. User has poor internet connection** → Offer video transcript or a "Read instead" text guide below the video. **5a. User skips video or only watches part** → No action required; user can still explore other parts of the homepage freely.
Post Condition:	1. The user has watched (or had the opportunity to watch) the marketing video. 2. They gain a basic understanding of how to use the platform

	features, even without an account.

#10 As an unregistered user, I want to view the disclaimer for the website, so that I can understand that no machine-learning based prediction can be 100% accurate.

#29 As a registered free user, I want to view the disclaimer for the website, so that I can understand that no machine-learning based prediction can be 100% accurate.

#48 As a registered premium user, I want to view the disclaimer for the website, so that I can understand that no machine-learning based prediction can be 100% accurate.

#58 As a property agent, I want to view the disclaimer for the website, so that I can understand that no machine-learning based prediction can be 100% accurate.

Name:	View Disclaimer About Prediction Accuracy	10, 29, 48, 58
Stakeholders and Goals:	**Unregistered Users, Registered Free Users, Registered Premium Users** ◦ Understand that the platform's predictions are generated using machine learning and are not guaranteed to be 100% accurate. ◦ Be aware of the limitations and intended use of the prediction results before deciding to register or rely on the platform. ◦ Avoid misunderstandings or misplaced expectations about the platform's accuracy. **Platform Owner/Business** ◦ Protect the business legally by clearly communicating that prediction results are estimates. ◦ Build trust through transparency about the strengths and limitations of the technology. ◦ Reduce liability or complaints arising from misinterpreted results.	
Description:	A disclaimer section is made publicly accessible (e.g., linked in the homepage footer or next to the prediction tool) to inform users that all price predictions are generated using statistical or machine learning models and may not reflect actual market outcomes. This	

	disclaimer is viewable by all users, including unregistered ones, before any registration or engagement.
Actor:	Unregistered User, Registered Free User, Registered Premium User
Pre-conditions:	1. The user has internet access. 2. The platform has a publicly accessible disclaimer page or section.
Trigger:	The user navigates to the disclaimer page via a link in the footer or user sees a disclaimer near the price prediction tool.
Normal Flow:	1. The user visits the homepage or prediction section. 2. The user sees a clearly labeled link to the disclaimer (e.g., "Disclaimer: Learn more about prediction accuracy"). 3. The user clicks the link. 4. The system displays a static page explaining the disclaimer. 5. The user reads the disclaimer and proceeds accordingly.
Alternative/ Exception Flow:	**2a. Disclaimer link not visible due to layout error** → Default disclaimer should still be shown near prediction output. **4a. Disclaimer page fails to load** → Display an error message: "Disclaimer is temporarily unavailable. Please try again later."
Post Condition:	1. The user is informed of the prediction model's limitations. 2. The platform has fulfilled its duty of transparency and legal clarity.

#11 As an unregistered user, I want to use referral code, so that I can get a discount on the subscription plans available.

#30 As a registered free user, I want to use referral code, so that I can get a discount on the subscription plans available.

Name:	Use Referral Code for Subscription Discount	11, 30
Stakeholders and Goals:	**Unregistered Users, Registered Free Users** • Enter a valid referral code during sign-up or subscription	

	purchase. • Receive a discount on available subscription plans. • Incentivized to register and become a paying member. **Platform Owner / Business** • Increase new user sign-ups through referral incentives. • Reward existing users for bringing in new subscribers. • Promote subscription upgrades by lowering initial cost barriers.
Description:	An unregistered user can apply a referral code during the sign-up or subscription purchase process. If the referral code is valid, the system applies the corresponding discount to the subscription plan. This feature encourages new users to join the platform while rewarding existing referrers.
Actor:	Unregistered User, Registered Free User
Pre-conditions:	1. The user has a valid referral code from an existing user. 2. The platform offers subscription plans that support referral discounts. 3. The user has an internet connection to access the platform.
Trigger:	The unregistered user begins the sign-up or subscription purchase process and enters a referral code.
Normal Flow:	1. The unregistered user visits the sign-up or subscription page. 2. The user selects a subscription plan. 3. The user enters the referral code in the provided field. 4. The system validates the referral code. 5. The system applies the discount to the subscription price. 6. The user completes the registration and payment process. 7. The system confirms the purchase and applies any linked referral rewards to the referrer's account.
Alternative/ Exception Flow:	**4a**. Invalid referral code → system displays "Invalid code" message and does not apply discount. **4b**. Expired referral code → system displays "Code expired"

	message. **4c**. Referral code already used by this user → system displays "Code already used" message. **5a**. System error during discount application → display error message and retry option.
Post Condition:	1. If successful, the user receives the discounted subscription plan. 2. The referrer's account receives the associated referral reward.

#12 As a registered free user, I want to log in securely, so that I can access the features in the website.

#36 As a registered premium user, I want to log in securely, so that I can access the features in the website.

#50 As a property agent, I want to log in securely, so that I can access the features in the website.

#60 As a system admin, I want to log in securely, so that I can access the features in the website.

Name:	Log In	12, 36, 50, 60
Stakeholders and Goals:	**Registered Users, Registered Premium User, Property Agent, System Admin** • Access personalized features available to logged-in users. • Log in using a secure and reliable process. • Ensure that their credentials and personal data are protected. **Platform Owner / Business** • Authenticate users reliably and securely. • Protect user data and maintain platform integrity. • Provide a smooth login experience to encourage repeat visits and engagement.	
Description:	Registered users log in through a secure login form using their email and password. The system authenticates the credentials	

	and grants access to features tied to the user's account.
Actor:	Registered Free User, Registered Premium User, Property Agent, System Admin
Pre-conditions:	1. The user has previously registered an account. 2. The platform supports secure login methods.
Trigger:	The user enters the login page.
Normal Flow:	1. The user navigates to the login page. 2. The user enters their registered email and password. 3. The system authenticates the credentials securely. 4. Upon success, the user is granted access to their dashboard or previous page.
Alternative/ Exception Flow:	**3a.** Invalid login credentials → system displays an error and prompts the user to try again. **3b.** User forgets password → redirected to "Forgot Password" process.
Post Condition:	1. The user gains access to their account upon successful login.

#13 As a registered free user, I want to input basic commercial/industrial property information (e.g., postal code, property type, floor area), so that I can get a price prediction.

#31 As a registered premium user, I want to input basic commercial/industrial property information (e.g., postal code, property type, floor area), so that I can get a price prediction.

Name:	Input Property Information for Prediction	13, 31
Stakeholders and Goals:	**Registered Users** ● Enter detailed property characteristics (e.g., size, location, number of rooms, age, amenities) ● Receive a more personalized and accurate price prediction based on the data.	

	● Understand how input variations affect predicted value. **Platform Owner/Business** ● Provide accurate predictions to increase user trust in the platform. ● Collect richer property data for improving the prediction model. ● Offer a valuable and interactive tool that encourages ongoing user engagement.
Description:	Registered users can access a form or tool to input detailed property information. The system uses this data to generate a customized price prediction, offering users a better understanding of the estimated market value of their property.
Actor:	Registered User
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is logged in with a valid account. 3. The prediction tool is available and operational.
Trigger:	The user interacts with the price prediction tool.
Normal Flow:	1. The registered user navigates to the price prediction tool. 2. The system presents a form with various property input fields (e.g., address, floor area, number of bedrooms, proximity to amenities). 3. The user fills in the details and submits the form. 4. The system validates the input and processes the prediction. 5. The system displays the predicted price with supporting data or confidence level. 6. The user can save or download the result for reference.
Alternative/ Exception Flow:	**1a.** System failure or prediction model timeout → system displays an error and suggests retrying later. **3a.** The user skips some non-mandatory fields → system still generates a prediction, but with lower accuracy indicated. **4a.** Invalid or incomplete input → system prompts user to correct

	fields before proceeding.
Post Condition:	1. The system processes the entered property details. 2. The user receives a price prediction tailored to the input.

#14 As a registered free user, I want to save my past price predictions, so that I can revisit or compare them later.

#32 As a registered premium user, I want to save my past price predictions, so that I can revisit or compare them later.

Name:	Save Price Predictions	14, 32
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users** • Save their past property price predictions to review or compare later. • Easily access previously predicted data without re-entering property details. • Track multiple predictions for different properties or time periods. **Platform Owner/Business** • Improve user retention through personalized history features. • Encourage deeper engagement with the prediction tool. • Gather insight on what types of properties users are most interested in.	
Description:	Registered users can choose to save the results of their property price predictions. These saved entries are stored in their user account and can be reviewed, compared, or deleted later from their personal dashboard.	
Actor:	Registered Free User, Registered Premium User	
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is logged in with a valid account. 3. A prediction has been successfully generated.	
Trigger:	The user interacts with the save price prediction feature.	

Normal Flow:	1. The user completes a property price prediction. 2. The system displays the result with an option to “Save this Prediction.” 3. The user clicks the save button. 4. The system stores the prediction data linked to the user’s account. 5. The system displays a success message and provides a link to “View Saved Predictions.” 6. The user can later view, compare, or manage saved predictions via their dashboard.
Alternative/ Exception Flow:	**1a.** User session expires → system prompts user to log in again before saving. **3a.** The user tries to save a duplicate prediction → system prompts for overwrite confirmation or skip. **4a.** Save fails due to temporary issue → system displays an error with retry option.
Post Condition:	1. The prediction is saved to the user’s profile. 2. The user can access the saved prediction in the future.

#15 As a registered free user, I want to bookmark interesting commercial/industrial property addresses, so that I can access them easily later.

#39 As a registered premium user, I want to bookmark interesting commercial/industrial property addresses, so that I can access them easily later.

Name:	Bookmark Property Addresses	15, 39
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users** • Save interesting property listings for future reference. • Organize and revisit shortlisted properties easily from their dashboard. • Avoid the need to re-search for listings of interest. **Platform Owner/Business** • Increase user engagement and return visits. • Allow personalization features that encourage long-term usage.	

	• Track user preferences for recommendation or analytics purposes.
Description:	Registered users can bookmark property listings they find interesting. These saved bookmarks will be accessible from their user dashboard or "Saved Listings" page, making it easy to revisit them later.
Actor:	Registered Free User, Registered Premium User
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is logged in with a valid account. 1. Property listings are available and viewable.
Trigger:	The user interacts with the save or bookmark feature.
Normal Flow:	1. The registered user browses the property listings. 2. The user sees a “Bookmark” or “Save” icon/button on each listing. 3. The user clicks the button on a property of interest. 4. The system saves the property to the user’s account. 5. The system confirms the action with a notification (e.g., “Saved to bookmarks”). 6. The user can later view all saved listings via their profile/dashboard.
Alternative/ Exception Flow:	**1a.** User tries to bookmark without being logged in → system prompts user to log in or sign up. **3a.** User clicks “Unsave” on an already-bookmarked property → system removes it from bookmarks. **4a.** If the system fails to save the bookmark → display error message with retry option.
Post Condition:	1. The selected listing is added to the user’s saved bookmarks. 2. The user can access it anytime via their dashboard or saved list.

#16 As a registered free user, I want filtering options for properties (e.g., location, type: office, retail, warehouse, factory), so that I can view properties that match my search filter.

#44 As a registered premium user, I want filtering options for properties (e.g., location, type: office, retail, warehouse, factory), so that I can view properties that match my search filter.

Name:	Filter Property Listings	16, 44
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users** • Narrow down property listings to match their specific needs. • Save time by avoiding irrelevant search results. • Quickly find properties within desired location, type, and budget range. **Platform Owner/Business** • Improve user satisfaction and ease of finding relevant listings. • Increase engagement by offering a customizable search experience. • Collect search preference data for recommendations and targeted marketing.	
Description:	Registered users can use filtering options to refine the property listings shown to them. Filters may include location, property type, price range, and other relevant criteria. This allows users to only see listings that match their preferences, making the search process faster and more efficient.	
Actor:	Registered Free User, Registered Premium User	
Pre-conditions:	1. The user has an internet connection to access the website. 1. The user is logged in with a valid account. 2. Property listings are available in the system's database.	
Trigger:	The user selects one or more filtering criteria while browsing property listings.	
Normal Flow:	1. The registered user navigates to the property listings page.	

	2. The system displays available filter options (e.g., location, property type, price range). 3. The user selects desired filter criteria. 4. The system processes the selected filters and refreshes the listing results. 5. The user views only the listings that meet their selected criteria.
Alternative/ Exception Flow:	**1a.** User enters invalid filter values (e.g., non-numeric price) → system prompts to correct the input. **3a.** No listings match the selected criteria → system displays a "No results found" message with suggestions to adjust filters. **4a.** System error occurs while applying filters → display error message and allow user to retry.
Post Condition:	1. The property listings displayed are limited to those matching the selected filters. 2. The user can modify or clear filters to change search results.

#20 As a registered free user, I want to view commercial/industrial units on sale or for rent, so that I can identify opportunities for my business.

#38 As a registered premium user, I want to view commercial/industrial units on sale or for rent, so that I can identify opportunities for my business.

Name:	View Commercial/Industrial Units on Sale or Rent	20, 38
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users** • Search and view commercial or industrial units available for sale or rent. • Identify potential spaces for business opportunities. • Filter listings based on location, type, price, and availability. **Property Agents** • Mark and manage which commercial or industrial units are for sale or rent. • Promote their listings to reach potential business clients. **Platform Owner/Business**	

	• Improve user satisfaction and ease of finding relevant listings. • Increase engagement by offering a customizable search experience. • Collect search preference data for recommendations and targeted marketing.
Description:	Registered free users can browse commercial and industrial property listings that have been marked by property agents as "For Sale" or "For Rent." These listings will display relevant details such as price, size, location, and type, helping users identify potential business spaces.
Actor:	Registered Free User, Registered Premium User
Pre-conditions:	1. The user is logged in with a valid account. 2. Property agents have marked certain units as "For Sale" or "For Rent." 3. The user has an internet connection to access the platform.
Trigger:	The registered free user navigates to the commercial/industrial listings section or applies relevant filters in the search tool.
Normal Flow:	1. The registered free user logs into their account. 2. The user navigates to the property search or listings page. 3. The user selects the “Commercial” or “Industrial” category. 4. The user can optionally filter by “For Sale” or “For Rent.” 5. The system displays matching listings marked by property agents. 6. The user selects a listing to view detailed information.
Alternative/ Exception Flow:	**3a**. No commercial/industrial units available → system displays “No listings found” message. **5a**. System error when retrieving listings → system displays error message and retry option.
Post Condition:	1. The user can view commercial/industrial units currently on sale or for rent. 2. The user can proceed to contact the property agent for

	more details or arrange a viewing.

#21 As a registered free user, I want to write feedback for the website, so that I can view and share my experience on the usage process.

#41 As a registered premium user, I want to write feedback for the website, so that I can view and share my experience on the usage process.

#55 As a property agent, I want to write feedback for the website, so that I can view and share my experience on the usage process.

Name:	Submit Feedback on Website Experience	21, 41, 55
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users, Property Agents** • Share their experiences or suggestions after using the platform. • Provide constructive input to help improve the website. • Feel heard and engaged by the platform team. **Platform Owner/Business** • Collect user insights and feedback to guide future improvements. • Identify usability issues or feature enhancement opportunities. • Foster user satisfaction and community engagement.	
Description:	Registered users (both free and premium) are able to write and submit feedback about the website's usability, layout, or functionality. This feature allows them to share their experience with the platform and suggest improvements.	
Actor:	Registered Free User, Registered Premium User, Property Agent	
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is logged in with a valid account.	
Trigger:	The user navigates to the feedback section and chooses to submit feedback.	

Normal Flow:	1. The user clicks on “Write Feedback” or a similar call-to-action. 2. The system displays a feedback input form. 3. The user fills in their comments and submits the form. 4. The system stores the feedback and shows a confirmation message. 5. The feedback is displayed in a public review section.
Alternative/ Exception Flow:	**3a.** If the user submits an empty form → The system displays a validation error and prompts them to write something. **4a.** If the system fails to store the feedback → An error message is shown and the user is asked to try again later.
Post Condition:	1. The user’s feedback is recorded.

#22 As a registered free user, I want to edit my user profile, so that I can update my personal data when needed.

#42 As a registered premium user, I want to edit my user profile, so that I can update my personal data when needed.

#52 As a property agent, I want to edit my user profile, so that I can update my personal data when needed.

Name:	Edit User Profile	22, 42, 52
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users, Property Agents** • Keep their account details up to date (e.g., name, email, preferences). • Correct any incorrect information previously submitted. • Manage account-related personal settings conveniently. **Platform Owner/Business** • Ensure accurate user data for communication, personalization, and analytics. • Reduce support issues related to outdated or incorrect user information.	

Description:	Registered users can access and update their profile details through the user settings page. This helps them maintain accurate personal information for better service delivery and future correspondence.
Actor:	Registered Free User, Registered Premium User, Property Agents
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is logged in with a valid account.
Trigger:	The user navigates to their profile and chooses to update their information.
Normal Flow:	1. The user clicks on their profile or account settings. 2. The system displays the editable profile fields (e.g., name, email, password). 3. The user makes changes and clicks "Save". 4. The system validates the input and updates the profile data in the database. 5. A success message is shown to confirm the update.
Alternative/ Exception Flow:	**3a.** If the user submits invalid input (e.g., wrong email format) → The system displays validation errors and prompts correction. **4a.** If the update fails due to a system error → An error message appears, and the user is advised to try again later.
Post Condition:	1. The user's updated profile information is saved and reflected in their account.

#23 As a registered free user, I want to view commercial/industrial property agents' contacts, so that I can reach out for queries related to a particular property.

#37 As a registered premium user, I want to view commercial/industrial property agents' contacts, so that I can reach out for queries related to a particular property.

Name:	View Agents' Contact	23, 37
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users** • Get in touch with property agents for further details,	

	questions, or negotiations. • Quickly find agent contact information tied to a specific property listing. **Property Agents** • Be accessible to potential clients via the platform. • Increase chances of direct engagement and lead generation. **Platform Owner/Business** • Facilitate communication between users and agents. • Provide added value to both sides, increasing engagement and trust.
Description:	Registered users (free and premium) can view the contact details of agents listed on specific property pages. Each district has an assigned agent, and their contact information (e.g., phone number, email, or messaging button) is shown to allow users to reach out directly for queries or negotiations.
Actor:	Registered Free User, Registered Premium User
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is logged in with a valid account. 3. The agent has put in their contact in that specific property.
Trigger:	The user views a property listing and wants to contact the agent.
Normal Flow:	1. The user navigates to a property information page. 2. The system displays agent information (e.g., name, profile photo). 3. The system shows contact options such as phone number, email, or a “message agent” button. 4. The user chooses a contact method to initiate communication.
Alternative/ Exception Flow:	**2a.** If the agent has not provided contact info → Display a message like “Contact information unavailable” or offer a platform-based message form (if applicable). **4a.** If the user attempts to use a contact feature that relies on an

	external app (e.g., phone/email) but it fails → Display an error or fallback option.
Post Condition:	1. The user successfully accesses the agent's contact details and may initiate communication outside the platform.

#24 As a registered free user, I want to view historical transaction data for similar commercial/industrial properties, so that I can understand market pricing trends.

#66 As a registered premium user, I want to view historical transaction data for similar commercial/industrial properties, so that I can understand market pricing trends.

Name:	View Historical Transaction Data for Properties	24, 66
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users** • Access past sales and rental transaction records for commercial/industrial properties. • Compare historical prices to understand market trends. • Use data to make informed decisions before buying or renting. **Platform Owner/Business** • Improve user satisfaction and ease of finding relevant listings. • Increase engagement by offering a customizable search experience. • Collect search preference data for recommendations and targeted marketing.	
Description:	Registered free users can view historical transaction data for commercial and industrial properties similar to their chosen listing. Data includes sale prices, rental rates, and transaction dates, helping users identify pricing trends and assess whether current offers are reasonable.	
Actor:	Registered Free User, Registered Premium User	
Pre-conditions:	1. The user is logged in with a valid account. 2. Historical transaction data is available in the system's	

	database. 3. The user has an internet connection to access the platform.
Trigger:	The user navigates to a commercial/industrial property listing and selects the option to view historical data.
Normal Flow:	1. The registered free user logs into their account. 2. The user browses to a commercial/industrial property listing. 3. The user clicks "View Historical Data." 4. The system searches for similar properties (based on location, size, type, etc.) and retrieves transaction records. 5. The system displays a list or chart showing: Transaction date, Sale price or rental rate, Property details (size, location, type). 6. The user reviews the data to identify market trends.
Alternative/ Exception Flow:	**4a**. No historical data available → system displays "No historical data found for similar properties." **5a**. Data retrieval error → system displays error message and retry option.
Post Condition:	1. The user can see past sales and rental data for similar commercial/industrial properties. 2. The user has better insight into market pricing trends.

#25 As a registered free user, I want to view an interactive map of Singapore showing nearby commercial hubs, logistics facilities, and transport links, so that I can assess location suitability.

#46 As a registered premium user, I want to view an interactive map of Singapore showing nearby commercial hubs, logistics facilities, and transport links, so that I can assess location suitability.

Name:	View Interactive Map	25, 46
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users** • Explore different areas of Singapore interactively.	

	• View nearby amenities such as MRT stations, schools, malls, parks, etc. • Use map navigation to assist in evaluating potential properties. **Platform Owner/Business** • Provide a useful visual tool to enhance user experience. • Increase user engagement with interactive features that add value to listings. • Encourage longer session times and better decision-making support.
Description:	Registered users (both free and premium) can access an interactive map of Singapore. The map allows panning and zooming, and shows layers of amenities (e.g., MRTs, schools, malls) relevant to real estate exploration. Users can scroll around to inspect different regions and analyze locations around properties of interest.
Actor:	Registered Free User, Registered Premium User
Pre-conditions:	1. The user is logged into their account. 2. The map module is successfully loaded on the interface.
Trigger:	The user clicks on the “Map” section or accesses a map-enabled feature.
Normal Flow:	1. The user navigates to the interactive map section. 2. The system loads the map centered on Singapore. 3. The user pans, zooms, or searches areas of interest 4. The system displays amenities (icons or layers) based on the zoom level. 5. The user clicks on map markers for more details.
Alternative/ Exception Flow:	**3a.** Map fails to load → Show fallback error message and suggest refreshing. **4a.** Amenity data fails to load → Display a message indicating partial data may be missing.
Post Condition:	1. The user successfully views and navigates the map.

#26 As a registered free user, I want to log out securely, so that I can manage my session safely.

#43 As a registered premium user, I want to log out securely, so that I can manage my session safely.

#51 As a property agent, I want to log out securely, so that I can manage my session safely.

#61 As a system admin, I want to log out securely, so that I can manage my session safely.

Name:	Log Out	26, 43, 51, 61
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users, Property Agent, System Admin** • End their session to protect personal data and prevent unauthorized access. • Ensure account safety, especially on shared or public devices. **Platform Owner/Business** • Provide a secure and seamless user session management experience. • Comply with security standards and good UX practices.	
Description:	Registered users can securely log out from their accounts to end the current session. The option is accessible from the navigation menu and confirms successful logout before redirecting to the homepage or login screen.	
Actor:	Registered Free User, Registered Premium User, Property Agent, System Admin	
Pre-conditions:	1. The user is logged in with a valid account.	
Trigger:	The user clicks the “Log Out” button or link from the menu or user profile dropdown.	
Normal Flow:	1. The user clicks the "Log Out" button. 2. The system terminates the user session.	

	3. The user is redirected to the homepage or login page. 4. A confirmation message (e.g., "You have been logged out successfully") is displayed.
Alternative/ Exception Flow:	**2a.** If the logout process fails due to server issues → Display an error message and suggest retrying. **4a.** If redirect fails → Keep the user on the current page with a message confirming session termination.
Post Condition:	1. The user is fully logged out and must log in again to access personalized features.

#27 As a registered free user, I want to deactivate my account, so that I can stop all activities in that account.

#49 As a registered premium user, I want to deactivate my account, so that I can stop all activities in that account.

#59 As a property agent, I want to deactivate my account, so that I can stop all activities in that account.

Name:	Deactivate Account	27, 49, 59
Stakeholders and Goals:	**Registered Free Users, Registered Premium Users, Property Agent** • Deactivate their account and personal data from the platform. • Stop receiving notifications, emails, or other platform communications. **Platform Owner/Business** • Comply with privacy and data protection regulations. • Maintain trust by giving users full control over their data. • Ensure proper handling and secure deletion of user-related information.	
Description:	Registered users can deactivate their account from the platform. Upon confirmation, all personal data, saved preferences, and associated content (e.g., saved listings, messages) will be	

	removed. This action cannot be undone once completed.
Actor:	Registered Free User, Registered Premium User, Property Agent
Pre-conditions:	1. The user has an internet connection to access the website. 1. The user is logged in with a valid account. 2. The user understands that account deletion is permanent and irreversible.
Trigger:	The user navigates to account settings and selects the "Deactivate Account" option.
Normal Flow:	1. The registered user accesses their account settings. 2. The user selects “Deactivate Account.” 3. The system displays a warning. 4. The user confirms the deactivation action. 1. The system deactivates the account and all associated data from the database. 2. The system logs the user out and displays a confirmation message.
Alternative/ Exception Flow:	**1a.** User tries to deactivate the account without being logged in → system prompts for login. **3a.** User cancels at the warning step → account remains active. **5a.** System error occurs during deletion → display error message and prompt to retry later.
Post Condition:	1. The user account and personal data are deactivated from the platform. 1. The user no longer has access to the platform unless they create a new account.

#35 As a registered premium user, I want to invite others to the platform using referral codes, so that I can get referral rewards.

#56 As a property agent, I want to invite others to the platform using referral codes, so that I can get referral rewards.

Name:	Invite Others via Referral Code	35, 56

Stakeholders and Goals:	**Registered Premium Users, Property Agents** • Invite friends, family, or colleagues to join the platform. • Earn referral rewards or benefits when invited users register successfully. • Share their positive experience with others. **Platform Owner / Business** • Acquire new users through organic referrals. • Reward loyal users for promoting the platform. • Track referral performance for marketing analytics.
Description:	Registered premium users and property agents (paid subscription users) can invite others to join the platform by sharing a unique referral code or link. When the invited person registers using that code, the referrer may receive rewards such as discounts, credits, or bonus features.
Actor:	Registered Premium User, Property Agent
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is logged in with a valid premium account. 3. The referral program is active and available.
Trigger:	The user chooses to invite others via the “Referral” feature in their account dashboard.
Normal Flow:	1. The user navigates to their account dashboard. 2. The user selects the “Referral” or “Invite Friends” option. 3. The system generates a unique referral code or link. 4. The user copies the referral code or shares it via available options (e.g., email, social media). 5. The invited person registers on the platform using the referral code. 6. The system verifies the referral and awards the applicable rewards to the referrer.
Alternative/ Exception Flow:	**1a.** User is not logged in → system prompts for login. **3a.** Referral code cannot be generated due to system error → display error message and retry option.

	5a. Invited person does not use the referral code → no rewards are granted. **6a.** Referral program terms are not met (e.g., invited user does not complete profile) → rewards are not issued.
Post Condition:	1. The referral is recorded in the system. 2. The user receives referral rewards once conditions are met.

14.2 Unregistered User

#1 As an unregistered user, I want to browse the homepage, so that I can understand what this website offers.

Name:	Browse Homepage	1
Stakeholders and Goals:	**Unregistered Users** • Understand the purpose, features, and value of the website at a glance. • Explore basic content and layout without needing to sign up. • Decide whether the platform is relevant to their needs. **Platform Owner/Business** • Make a strong first impression with clear messaging, branding, and user pathways. • Guide users toward registration or deeper feature exploration. • Communicate the platform's benefits and encourage engagement.	
Description:	Unregistered users can freely browse the homepage of the website, which introduces the platform's purpose, features (such as price predictions, market trends, or user testimonials), and calls-to-action. This helps users quickly assess the value of the site and decide whether to explore further or register.	
Actor:	Unregistered User	
Pre-conditions:	1. The user has an internet connection to access the website.	

	2. The user is not logged in or registered.
Trigger:	The user visits the website.
Normal Flow:	1. The unregistered user visits the website URL. 2. The system displays the homepage with core information: • Tagline or slogan • Overview of main features (e.g., price predictions, market data) • Sample visual elements or testimonials • Buttons or links for registration, login, or demo usage 3. The user scrolls through content and navigates to other public pages or features. 4. Optional prompts or banners encourage the user to register.
Alternative/ Exception Flow:	**4a.** The user clicks on a gated feature → redirected to the sign-up prompt.
Post Condition:	1. The user becomes more informed about what the website offers. 2. The user may be encouraged to explore features or register for an account.

#2 As an unregistered user, I want to view sample price predictions with limited details, so that I can try the prediction feature before signing up.

Name:	View Sample Price Predictions	2
Stakeholders and Goals:	**Unregistered Users** • Try out the price prediction tool before committing to registration. • View sample results with limited data (e.g., partial information, blurred details, or capped predictions) to get a sense of how the feature works. • Decide whether the platform is valuable enough to sign up. **Platform Owner/Business** • Attract new users by allowing limited feature exploration. • Use partial feature exposure to convert unregistered users	

	into registered users. • Maintain premium value of full prediction results behind registration.
Description:	Unregistered users are allowed to view limited versions of property price predictions. The system presents sample outputs (e.g., estimated range, visual chart with blurred exact value, or fewer data points) to demonstrate the feature's usefulness while encouraging sign-up for full access.
Actor:	Unregistered User
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is not logged in or registered.
Trigger:	The user navigates to or interacts with the price prediction feature.
Normal Flow:	1. The unregistered user accesses the price prediction feature page. 2. The system displays a sample prediction using pre-selected data. 3. The sample result includes limited or blurred elements (e.g., rough price range, cropped chart, hidden exact values). 4. The system shows a prompt encouraging the user to sign up for full access. 5. The user may choose to register or continue browsing with limited visibility.
Alternative/ Exception Flow:	**2a.** System fails to generate a sample prediction → show error message and encourage retry or sign-up for assistance. **4a.** The user ignores the sign-up suggestion → can continue viewing limited predictions. **5a.** The user signs up → redirected to the full version of the prediction feature.
Post Condition:	1. The user views a limited or blurred version of the prediction. 2. The user is presented with a call-to-action to sign up for full details.

#6 As an unregistered user, I want to view pricing and subscription plans, so that I can decide whether to register or upgrade.

Name:	View Pricing and Subscription Plans	6
Stakeholders and Goals:	**Unregistered Users** • Understand the available subscription tiers and their pricing. • Compare the benefits of each plan (e.g., free vs. premium). • Decide whether signing up or subscribing is worth it. **Platform Owner/Business** • Clearly communicate the value proposition of each pricing tier. • Encourage conversions by highlighting benefits and pricing transparency. • Use pricing plans to upsell premium features.	
Description:	Unregistered users can access a public pricing page or section that outlines the available subscription plans, their features, and their costs. This helps users evaluate the value of registering or subscribing before taking action.	
Actor:	Unregistered User	
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is not logged in or registered. 3. The platform offers different access tiers (e.g., Free, Premium).	
Trigger:	The user interacts with the view plans feature.	
Normal Flow:	1. The unregistered user navigates to the “Plans & Pricing” page via menu or a call-to-action. 2. The system displays available plans with clear pricing and feature breakdown. 3. The user reviews the benefits of each plan. 4. The system shows prompts like “Sign Up” or “Upgrade Now.” 5. The user clicks a button and is redirected to the sign-up or checkout page.	

Alternative/ Exception Flow:	**1a.** Pricing data fails to load → display message: "We're updating our plans. Please try again later." **2a.** If there are temporary promotions, discounts, or trials → system highlights them as banners or tooltips. **4a.** If the user is unsure, the system may offer a comparison table or FAQ section.
Post Condition:	1. The user views all pricing options and benefits. 2. The user may proceed to sign up for a free or paid plan.

#7 As an unregistered user, I want to see a prompt to sign up when I try to access restricted features, so that I'm aware of available benefits.

Name:	Sign Up Prompt for Unregistered User	7
Stakeholders and Goals:	**Unregistered Users** • Register an account to save preferences and access additional features such as saved properties and full data insights. • Continue using basic features (e.g., quick access to public listings or traffic info) without signing up if they prefer a frictionless experience. **Platform Owner/Business** • Increase user registration and engagement. • Convert casual visitors into repeat users by offering more value through account-based features.	
Description:	When an unregistered user attempts to access certain features (e.g., saving a search, viewing full property details, or accessing premium insights), the system will display a prompt encouraging the user to sign up.	
Actor:	Unregistered User	
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is not logged in or registered.	
Trigger:	The user attempts to use a feature that is marked as	

	"account-only."
Normal Flow:	1. The unregistered user browses the site. 2. They click on a restricted feature (e.g., "Save this property"). 3. The system checks and detects that the user is not signed in. 4. A sign-up prompt is displayed with an explanation of benefits. 5. The user clicks “Sign Up” or “Log In”. 6. The system redirects them to the appropriate page. 7. Upon successful sign-up/login, the user is redirected back and the feature is enabled.
Alternative/ Exception Flow:	**4a.** The user closes the prompt → they remain on the current page with limited access. **7a.** The user does not complete the sign-up → no account is created; access remains restricted.
Post Condition:	1. The user is prompted to sign up or log in. 2. If the user completes the sign-up or login process, they gain access to the requested feature. 3. If the user dismisses the prompt, they return to the current page with access to basic features only.

#8 As an unregistered user, I want to register for an account, so that I can access features exclusive for registered users.

Name:	Register for An Account	8
Stakeholders and Goals:	**Unregistered Users** ◦ Create an account easily and quickly. ◦ Gain access to restricted features such as full price predictions, saved searches, and personalized recommendations. ◦ Have a secure and smooth onboarding experience. **Platform Owner/Business** ◦ Convert visitors into registered users.	

	○ Enable access control for premium features.
Description:	This use case allows unregistered users to create an account using a registration form. Once registered, users can access restricted features, such as complete property price predictions.
Actor:	Unregistered User
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is not logged in or registered.
Trigger:	The user interacts with the sign up feature.
Normal Flow:	1. The user clicks on “Sign Up” from the navigation menu or a prompt. 2. The system displays the registration form (e.g., name, email, password). 3. The user receives a subscription choice. 4. The user fills out and submits the form. 5. The system validates the data and creates the account. 6. The user is redirected to their dashboard or price prediction tool.
Alternative/ Exception Flow:	**1a.** Registration system fails (e.g., server issue) → system displays error message and suggests retry later.. **5a.** If validation fails (e.g., weak password, existing email), the system displays an error and prompts correction.
Post Condition:	1. The user has a valid account. 2. The user is redirected to a page where they can use full price prediction features.

14.3 Registered Free User

#17 As a registered free user, I want to view subscription plans, so that I can decide if I want to upgrade.

Name:	View Plans and Pricing	17

Stakeholders and Goals:	**Registered Free Users** • Understand the available subscription tiers and their pricing. • Compare the benefits of each plan (e.g., free vs. premium). • Decide whether signing up or subscribing is worth it. **Platform Owner/Business** • Clearly communicate the value proposition of each pricing tier. • Encourage conversions by highlighting benefits and pricing transparency. • Use pricing plans to upsell premium features.
Description:	Registered free users can access a public pricing page or section that outlines the available subscription plans, their features, and their costs. This helps users evaluate the value of registering or subscribing before taking action.
Actor:	Registered Free User
Pre-conditions:	1. The user has an internet connection to access the website 2. The user is logged in or registered. 3. The platform offers different access tiers (e.g., Free, Premium).
Trigger:	The user interacts with the view plans feature.
Normal Flow:	1. The registered user navigates to the “Plans & Pricing” page via menu or a call-to-action. 2. The system displays available plans with clear pricing and feature breakdown. 3. The user reviews the benefits of each plan. 4. The system shows prompts like “Upgrade Now.” 5. The user clicks a button and is redirected to the checkout page.
Alternative/ Exception Flow:	**1a.** Pricing data fails to load → display message: “We’re updating our plans. Please try again later.” **2a.** If there are temporary promotions, discounts, or trials → system highlights them as banners or tooltips.

	4a. If the user is unsure, the system may offer a comparison table or FAQ section.
Post Condition:	1. The user views all pricing options and benefits. 2. The user may proceed to sign up for a paid plan.

#18 As a registered free user, I want to purchase a premium subscription, so that I can access advanced features.

Name:	Purchase Subscription Plan	18
Stakeholders and Goals:	**Registered Free Users** • Make an informed decision about purchasing a plan. • Seamlessly complete a secure payment and upgrade process. • Gain access to advanced features (e.g., predictions comparisons, detailed analytics, report downloads). **Platform Owner / Business** • Convert free users to paying subscribers. • Clearly communicate the value of premium features. • Ensure a smooth and secure payment experience. • Boost recurring revenue and user retention.	
Description:	Registered users can view available subscription plans and securely purchase a premium subscription in order to unlock additional features and services on the platform.	
Actor:	Registered Free User	
Pre-conditions:	1. The user is logged in with a valid account. 2. Payment gateway is integrated and operational.	
Trigger:	The user enters the subscription plan page.	
Normal Flow:	1. The system displays available subscription plans (e.g., Free, Premium Monthly, Premium Yearly) along with features and pricing. 2. The user selects a desired subscription plan. 3. The system redirects to a secure payment interface.	

	4. The user enters payment details and confirms the purchase. 5. The system processes the payment and confirms the upgrade. 6. The user receives a confirmation message and premium features are unlocked.
Alternative/ Exception Flow:	**2a.** User wants to compare plans → system displays a comparison table. **4a.** User payment fails → system displays an error and invites user to retry or try a different payment method. **4b.** User cancels payment → user is returned to the subscription selection screen.
Post Condition:	1. The user successfully upgrades to a premium subscription or viewed the details they needed. 2. The user gains access to premium-only features. 3. The subscription details are recorded in the user's account.

14.4 Registered Premium User

#33 As a registered premium user, I want to download prediction reports (PDF/Excel), so that I can share the report with others.

Name:	Download Prediction Reports	33
Stakeholders and Goals:	**Premium Users** • Obtain offline copies of property price predictions. • Share detailed insights with clients, partners, or stakeholders. • Maintain records of predictions for future reference or analysis. **Platform Owner / Business** • Provide valuable premium features that encourage subscription. • Facilitate professional use cases by enabling easy data export.	

	• Increase platform credibility through comprehensive reporting tools.
Description:	Premium users can download detailed price prediction reports in formats such as PDF or Excel for sharing and record-keeping purposes.
Actor:	Registered Premium User
Pre-conditions:	1. The user is logged in with a valid account. 2. The user has an active premium subscription. 3. A prediction report is available for the selected property or query.
Trigger:	The user has successfully done a price prediction search.
Normal Flow:	1. The user navigates to a detailed prediction result page. 2. The user clicks the “Download Report” button. 3. The system presents format options (PDF, Excel). 4. The user selects the preferred format. 5. The system generates the report file and initiates the download. 6. The user saves the report locally.
Alternative/ Exception Flow:	**5a.** Report generation fails → system displays error and suggests retrying later.
Post Condition:	1. The user successfully downloads the report in the chosen format. 2. The downloaded report includes relevant data, charts, and summary insights.

#34 As a registered premium user, I want to manage my membership, so that I can have the choice to downgrade my subscription plan.

Name:	Manage Membership	34
Stakeholders and Goals:	**Registered Premium Users** • View current subscription details.	

	• Downgrade to a free plan when needed. • Maintain control over membership preferences. **Platform Owner / Business** • Provide transparency and flexibility in membership. • Reduce churn by giving users the ability to manage their plans easily. • Possibly offer re-engagement incentives during downgrade.
Description:	Premium users can access a dedicated membership management section where they can view their current plan details and initiate a downgrade to a free plan.
Actor:	Registered Premium User
Pre-conditions:	1. The user is logged in as a premium user. 2. A billing/subscription management module is integrated and functional.
Trigger:	The user clicks “Manage Membership” from the profile or settings menu.
Normal Flow:	1. User navigates to the membership settings page. 2. System displays the current subscription details (plan type, next billing date, etc.). 3. User selects the “Downgrade” option. 4. System shows a confirmation prompt with details about loss of premium access. 5. User confirms the downgrade request. 6. System processes the downgrade and updates the user’s plan to “Free”. 1. System sends a confirmation email/notification.
Alternative/ Exception Flow:	**3a.** Subscription system is temporarily unavailable → Show error and suggest trying again later. **5a.** User cancels confirmation → No changes made; stay on the membership page.
Post Condition:	1. The user’s account is downgraded, and access is limited to free-user features only.

#40 As a registered premium user, I want to compare between different price predictions, so that I can easily assess the difference between my prediction interests.

Name:	Compare Price Predictions	40
Stakeholders and Goals:	**Registered Premium Users** • Compare two or more property price predictions side by side. • Quickly assess differences between properties of interest. • Make more informed buying, selling, or investment decisions. **Platform Owner / Business** • Provide premium-exclusive features to justify subscription value. • Increase platform engagement and usage time. • Encourage data-driven decision-making for users.	
Description:	Registered premium users can compare different price predictions by viewing them side by side. The feature allows the selection of two or more properties and displays their prediction details in a clear comparison layout.	
Actor:	Registered Premium User	
Pre-conditions:	1. The user has an internet connection to access the website. 2. The user is logged in with a valid premium account. 3. Price predictions for the selected properties are available in the system.	
Trigger:	The user selects two or more properties to compare from the price prediction results page or saved listings.	
Normal Flow:	1. The registered premium user navigates to the price prediction page. 2. The user selects multiple properties for comparison. 3. The system retrieves prediction data for each selected property. 4. The system displays the prediction details side by side in a comparison table or layout.	

	5. The user reviews the differences in predicted prices and other relevant data.
Alternative/ Exception Flow:	**1a.** User is not logged in → system prompts for login. **2a.** User selects only one property → system prompts to select at least two. **3a.** Price prediction data for a selected property is unavailable → system displays a notice and removes it from the comparison. **4a.** System error occurs while loading comparison → display error message and retry option.
Post Condition:	1. The user can view and assess the differences between multiple property price predictions. 2. The comparison helps the user decide which properties to prioritize for further research or action.

14.5 Property Agent

#53 As a property agent, I want to view a prompt to select regions, so that I can choose the regions that I want to be allocated to.

Name:	Select Assigned Regions	53
Stakeholders and Goals:	**Property Agents** • Select up to three preferred regions to work in. • Ensure their contact details appear only on property listings within their chosen regions. • Focus their efforts on familiar or strategic market areas. **Platform Owner / Business** • Distribute property agent coverage evenly across regions. • Improve user experience by showing relevant, localized agents. • Enable targeted marketing and lead assignments for agents.	
Description:	When a property agent logs in for the first time (or when their region assignment is reset), the system prompts them to select up to three regions they want to be assigned to. Once confirmed, the agent's contact information will automatically appear on property listings within those regions, making them visible to potential clients browsing those properties.	
Actor:	Property Agent	
Pre-conditions:	1. The agent has an internet connection to access the platform. 2. The agent is logged in with a valid account. 3. A list of available regions is stored in the system.	
Trigger:	The agent logs in and is prompted to select their preferred working regions.	
Normal Flow:	1. The property agent logs in to their account. 2. The system detects that no regions are currently assigned (or assignments have expired).	

	3. The system displays a prompt with the list of available regions. 4. The agent selects up to three preferred regions. 5. The agent confirms their selection. 1. The system saves the selected regions to the agent's profile. 2. The agent's contact details now appear on all property listings within those selected regions.
Alternative/ Exception Flow:	**1a.** Agent already has assigned regions → system does not prompt unless re-assignment is required. **4a.** Agent tries to select more than three regions → system displays an error and prevents selection. **5a.** Agent cancels without making a selection → system uses default unassigned status, and the agent's contact will not appear on any listings until regions are chosen. **6a.** System error occurs while saving selections → display error message and prompt to retry.
Post Condition:	1. The agent's selected regions are stored in the system. 2. The agent's contact details automatically appear on property listings within those regions.

#67 As a property agent, I want to mark properties for sale or rent, so that I can access more interested parties.

Name:	Mark Properties for Sale or Rent	67
Stakeholders and Goals:	**Property Agents** • List properties as available for sale or rent to attract potential buyers or tenants. • Increase visibility of their listings in relevant searches. • Update property availability in real time. **Registered Free Users, Registered Premium Users** • Easily find properties that are actively available for purchase or rental. • View accurate and up-to-date availability information.	

	Platform Owner / Business • Distribute property agent coverage evenly across regions. • Improve user experience by showing relevant, localized agents. • Enable targeted marketing and lead assignments for agents.
Description:	Property agents can set the availability status of their listed properties as “For Sale” or “For Rent.” This availability is displayed in search results and property detail pages, ensuring users see only relevant, currently available listings. Agents can update the status at any time.
Actor:	Property Agent
Pre-conditions:	1. The property agent is logged in with a valid account. 2. The property is listed on the platform under the agent’s profile. 3. The agent has the necessary permissions to update property details.
Trigger:	The property agent navigates to their property management dashboard and chooses to update the status of a property.
Normal Flow:	1. The property agent logs into their account. 2. The agent navigates to the property management section. 3. The agent selects the property to update. 4. The agent chooses the status: “For Sale” or “For Rent.” 5. The system updates the property listing with the new status. 6. The updated status appears on the property’s details page and in search results.
Alternative/ Exception Flow:	**3a.** Property not found in the agent’s listing → system displays “Property not found.” **5a**. System error occurs while updating → display error message and retry option. **4a**. Agent tries to mark a property without required details (e.g., price) → system prompts to complete the missing fields before

	status update.
Post Condition:	1. The property is visible to users as available for sale or rent. 2. Search results and filters correctly reflect the property's availability.

14.6 System Admin

#62 As a system admin, I want to view all user accounts, so that I can manage platform usage and user roles.

Name:	View All User Accounts	62
Stakeholders and Goals:	**System Admin** • Access a complete list of all registered accounts on the platform. • Monitor user activity and manage platform usage. • Assign, update, or revoke user roles and permissions as needed. **Platform Owner / Business** • Maintain oversight and control over platform usage. • Ensure users comply with platform policies. • Support data-driven decisions through account analytics.	
Description:	The system admin can access a dashboard to view all registered user accounts, including their account details, current role, and account status. From this view, the admin can filter, search, and manage user roles or take administrative actions.	
Actor:	System Admin	
Pre-conditions:	1. The admin has an internet connection to access the platform. 2. The admin is logged in with valid admin credentials. 3. User account data is available in the system	
Trigger:	The system admin selects the “User Accounts” or equivalent option from the admin dashboard.	
Normal Flow:	1. The system admin logs into the admin dashboard. 2. The admin selects the “View All User Accounts” option. 3. The system retrieves all user account records from the database. 4. The system displays the list of user accounts with relevant details (e.g., username, email, role, status, last login).	

	5. The admin can use search or filters to locate specific accounts. 6. The admin may choose to view more details or manage a specific account.
Alternative/ Exception Flow:	**1a.** Admin is not logged in → system prompts for login. **3a**. Database error occurs while retrieving accounts → system displays error message and retry option. **5a**. Search query returns no results → system displays “No accounts fo
Post Condition:	1. The system admin can see a complete and up-to-date list of all user accounts. 2. The admin may proceed with further actions such as modifying roles, suspending accounts, or reviewing activity.

#63 As a system admin, I want to suspend problematic accounts, so that I can enforce platform policies.

Name:	Suspend Problematic Accounts	63
Stakeholders and Goals:	**System Admin** • Temporarily or permanently suspend accounts that violate platform rules. • Maintain a safe and compliant platform environment. • Enforce community guidelines and terms of service. **Platform Owner / Business** • Protect the platform’s reputation and integrity. • Reduce harmful or fraudulent activity. • Ensure compliance with legal and policy requirements.	
Description:	The system admin can suspend user accounts that are flagged for violating platform rules or engaging in harmful behavior. Suspension prevents the user from logging in or accessing any platform features until the suspension is lifted. The admin may choose between temporary or permanent suspension depending on the severity of the issue.	

Actor:	System Admin
Pre-conditions:	1. The admin has an internet connection to access the platform. 2. The admin is logged in with valid admin credentials. 3. The user account to be suspended exists in the system.
Trigger:	The system admin selects the “Suspend” option for a specific account from the admin dashboard.
Normal Flow:	1. The system admin logs into the admin dashboard. 2. The admin navigates to the list of user accounts. 3. The admin selects the account to be suspended. 4. The admin clicks the “Suspend Account” option. 5. The system prompts the admin to confirm suspension and choose duration (temporary or permanent). 6. The admin confirms the action. 1. The system updates the account status to “Suspended” and restricts user access. 2. The system may optionally notify the user about the suspension via email.
Alternative/ Exception Flow:	**1a.** Admin is not logged in → system prompts for login. **3a**. Account not found → system displays “Account does not exist.” **5a**. Admin cancels at confirmation step → account remains active. **7a**. System error occurs while updating status → display error message and retry option.
Post Condition:	1. The targeted account is suspended and cannot access the platform. 2. The account remains in suspended status until reactivated by an admin.

#64 As a system admin, I want to moderate reviews, so that user-generated content remains constructive.

Name:	Moderate Reviews	64

Stakeholders and Goals:	**System Admin** • Review and approve or reject user-submitted reviews before they appear publicly. • Ensure all published reviews follow community guidelines. • Remove harmful, offensive, or spam content. **Platform Owner / Business** • Maintain a trustworthy and constructive review environment. • Protect platform reputation by preventing abusive or false reviews. • Encourage fair and helpful feedback for properties and agents.
Description:	The system admin can view all submitted user reviews and decide whether to approve or reject before they appear on the platform. Approved reviews will be visible to all users, while rejected ones will remain hidden. This ensures that only constructive and guideline-compliant reviews are displayed.
Actor:	System Admin
Pre-conditions:	1. The admin has an internet connection to access the platform. 2. The admin is logged in with valid admin credentials. 3. User reviews are stored and awaiting moderation.
Trigger:	The system admin accesses the review moderation panel from the admin dashboard.
Normal Flow:	1. The system admin logs into the admin dashboard. 2. The admin selects the “Moderate Reviews” option. 3. The system displays all pending reviews with relevant details (e.g., reviewer name, date, property, review content). 4. The admin reads the review and checks it against moderation guidelines. 5. The admin chooses one of the following actions: a. Approve → review is published on the property or agent profile.

	b. Reject → review is not published and is stored as rejected. 6. The system updates the review status accordingly.
Alternative/ Exception Flow:	**1a.** Admin is not logged in → system prompts for login. **3a**. No pending reviews → system displays "No reviews awaiting moderation." **5a**. Admin tries to approve but system error occurs → display error message and retry option.
Post Condition:	1. Only approved reviews are displayed publicly. 2. Rejected or unapproved reviews remain hidden from public view.

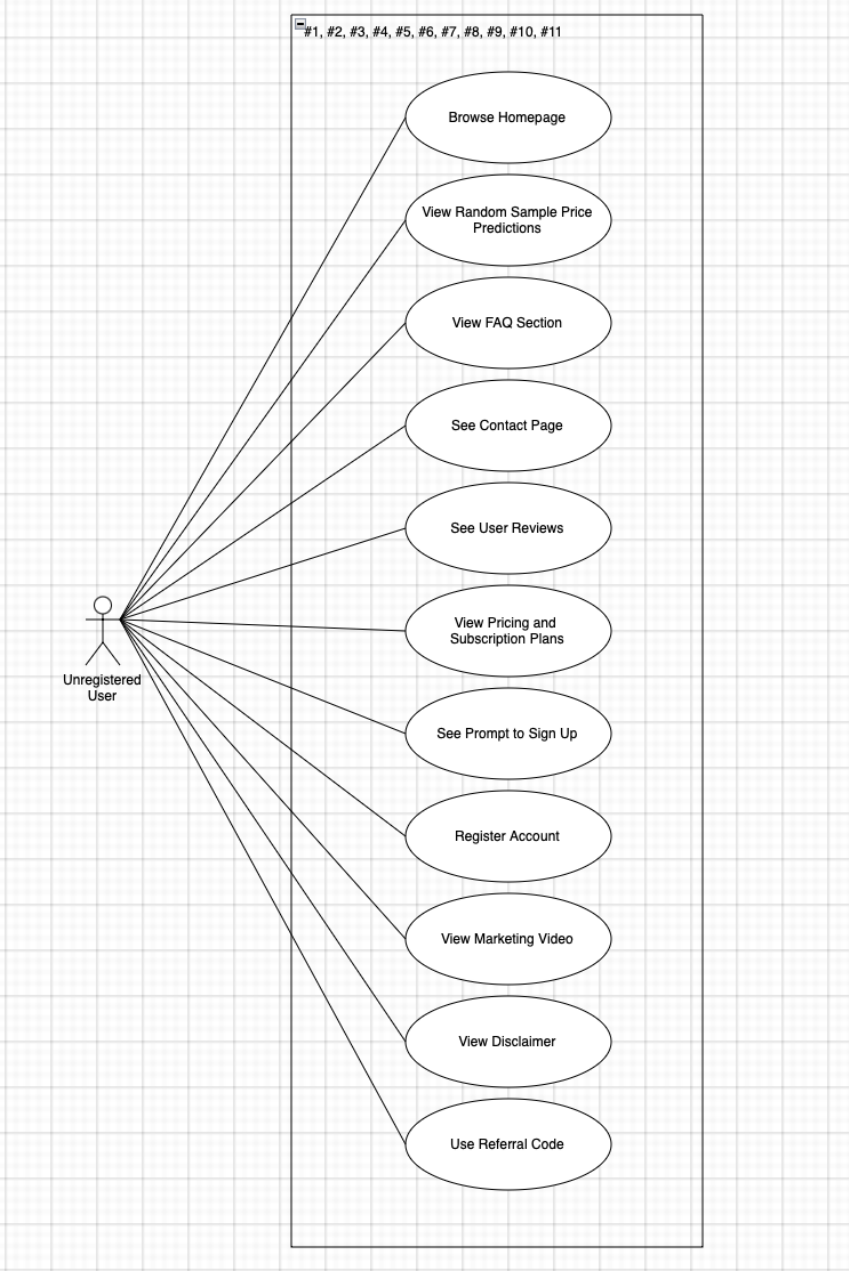
#65 As a system admin, I want to update homepage banners, FAQs, or marketing content, so that the website stays fresh and up to date.

Name:	Update Website Content	65
Stakeholders and Goals:	**System Admin** • Edit and update homepage banners, FAQs, and marketing content. • Keep the website's information accurate and appealing. • Promote new offers, announcements, or important changes. **Platform Owner / Business** • Ensure the platform's content reflects current branding and campaigns. • Improve user engagement with timely and relevant updates. • Maintain trust by providing up-to-date and accurate information.	
Description:	The system admin can access a content management section in the admin dashboard to update homepage banners, FAQs, and marketing content. Changes are reflected on the live website immediately or after admin approval. This keeps the platform visually fresh and ensures that the information presented to users is always relevant.	

Actor:	System Admin
Pre-conditions:	1. The admin has an internet connection to access the platform. 2. The admin is logged in with valid admin credentials. 3. The content management feature is available in the admin dashboard.
Trigger:	The system admin selects the “Content Management” or equivalent option from the admin dashboard.
Normal Flow:	1. The system admin logs into the admin dashboard. 2. The admin navigates to the “Content Management” section. 3. The system displays editable sections for homepage banners, FAQs, and marketing content. 4. The admin selects the section they wish to update. 5. The admin edits or uploads new content. 6. The admin previews the changes. 1. The admin publishes the updated content. 2. The system updates the live website accordingly.
Alternative/ Exception Flow:	**1a.** Admin is not logged in → system prompts for login. **3a**. Selected section not found or unavailable → system displays error message. **5a**. Admin uploads unsupported file format for banners → system displays error and rejects upload. **7a**. Admin cancels before publishing → no changes are made.
Post Condition:	1. The updated homepage banners, FAQs, or marketing content are live on the website. 2. Website visitors see the new and relevant information immediately.

15. Use Case Diagram

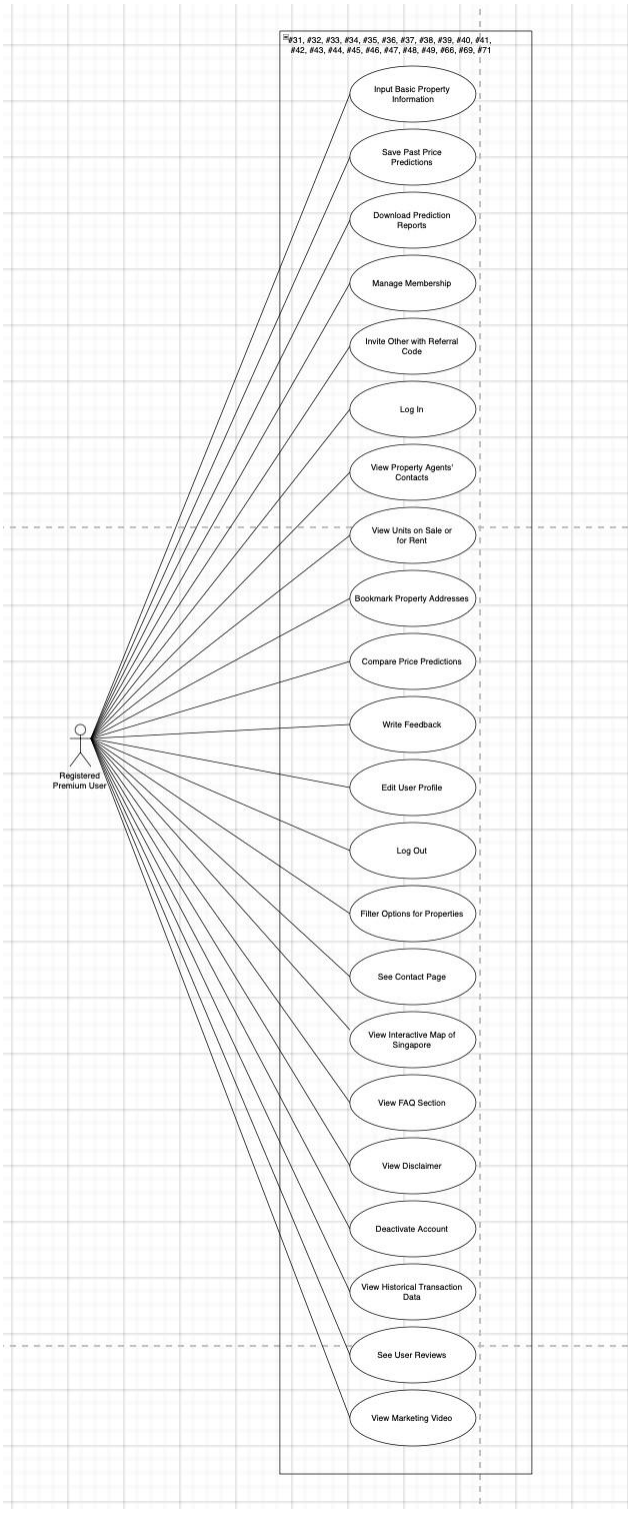
15.1 Unregistered User



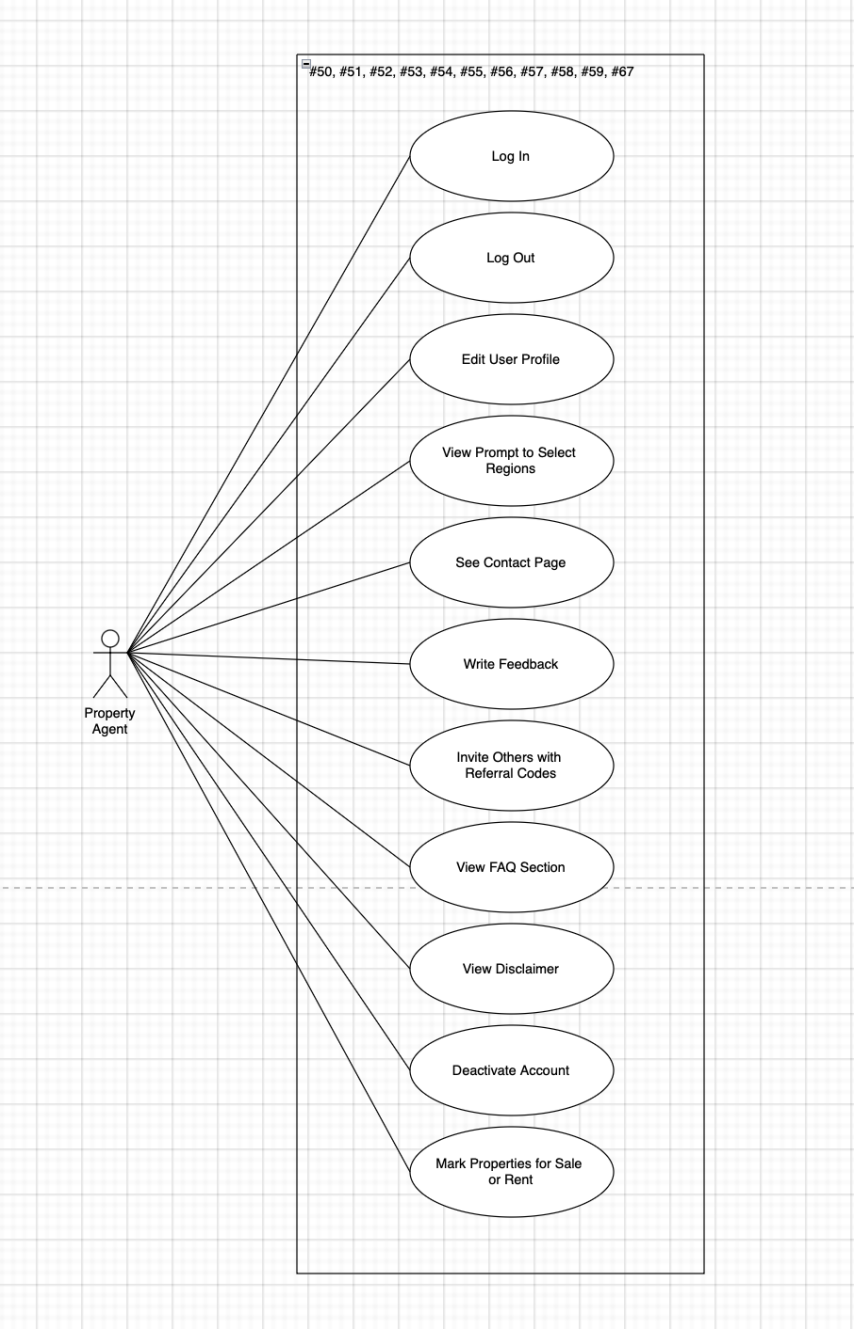
15.2 Registered Free User



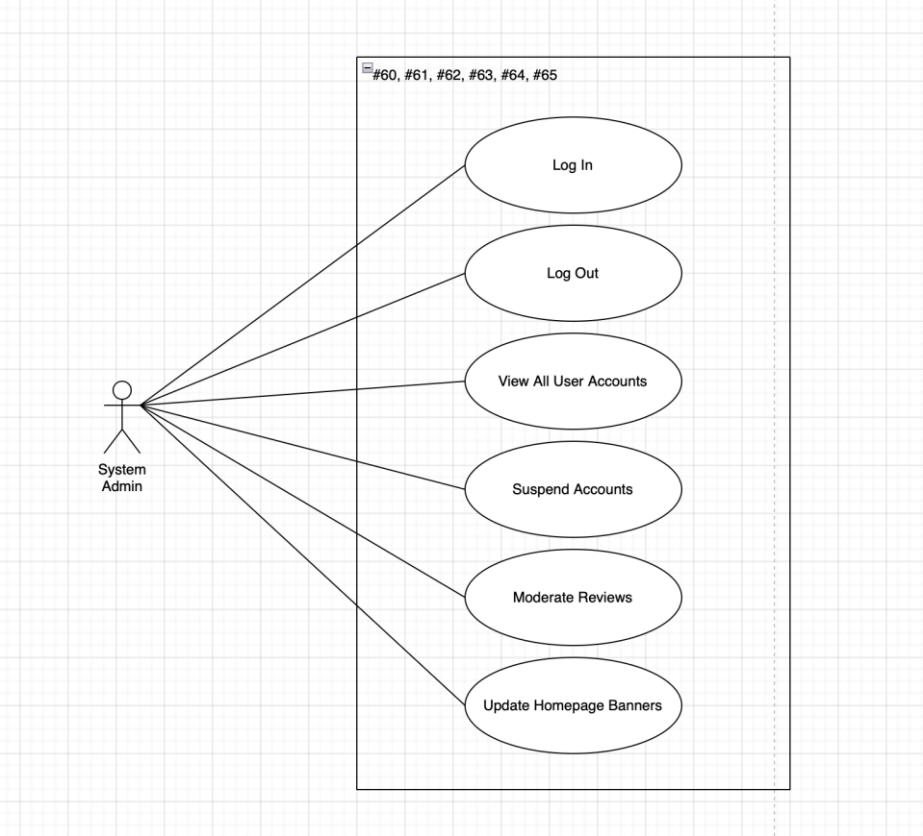
15.3 Registered Premium User



15.4 Property Agent

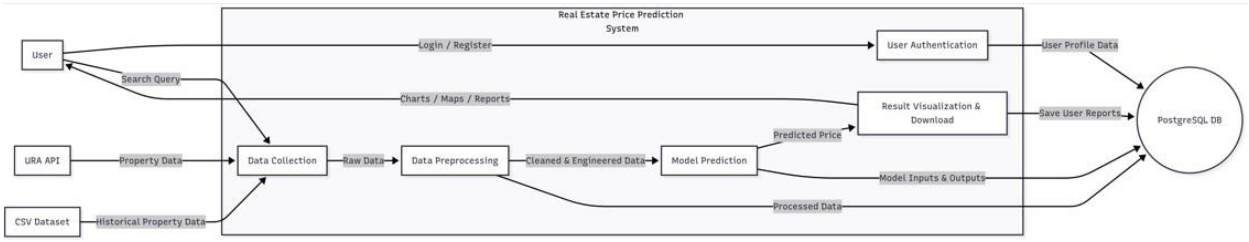


15.5 System Admin

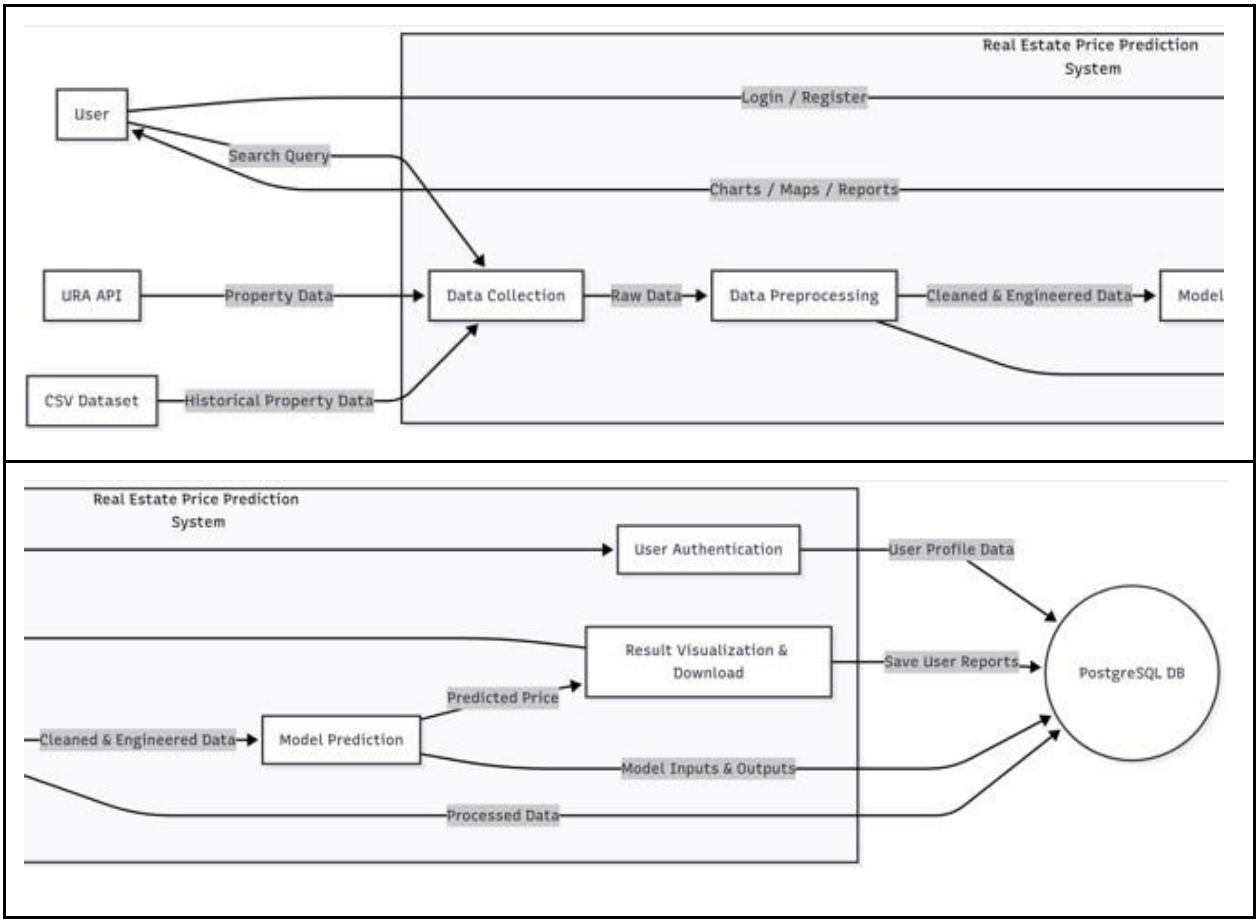


16. System Design

16.1 Data Flow Diagram



Zoomed-in View:



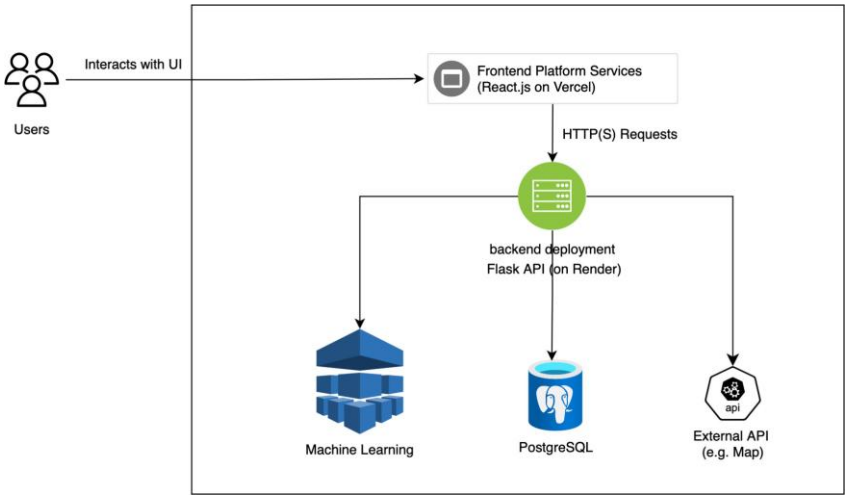
16.2 System Architecture Design

The system uses a modular architecture where the frontend (React.js), deployed on Vercel, interacts with the Flask backend, which is hosted on Render.

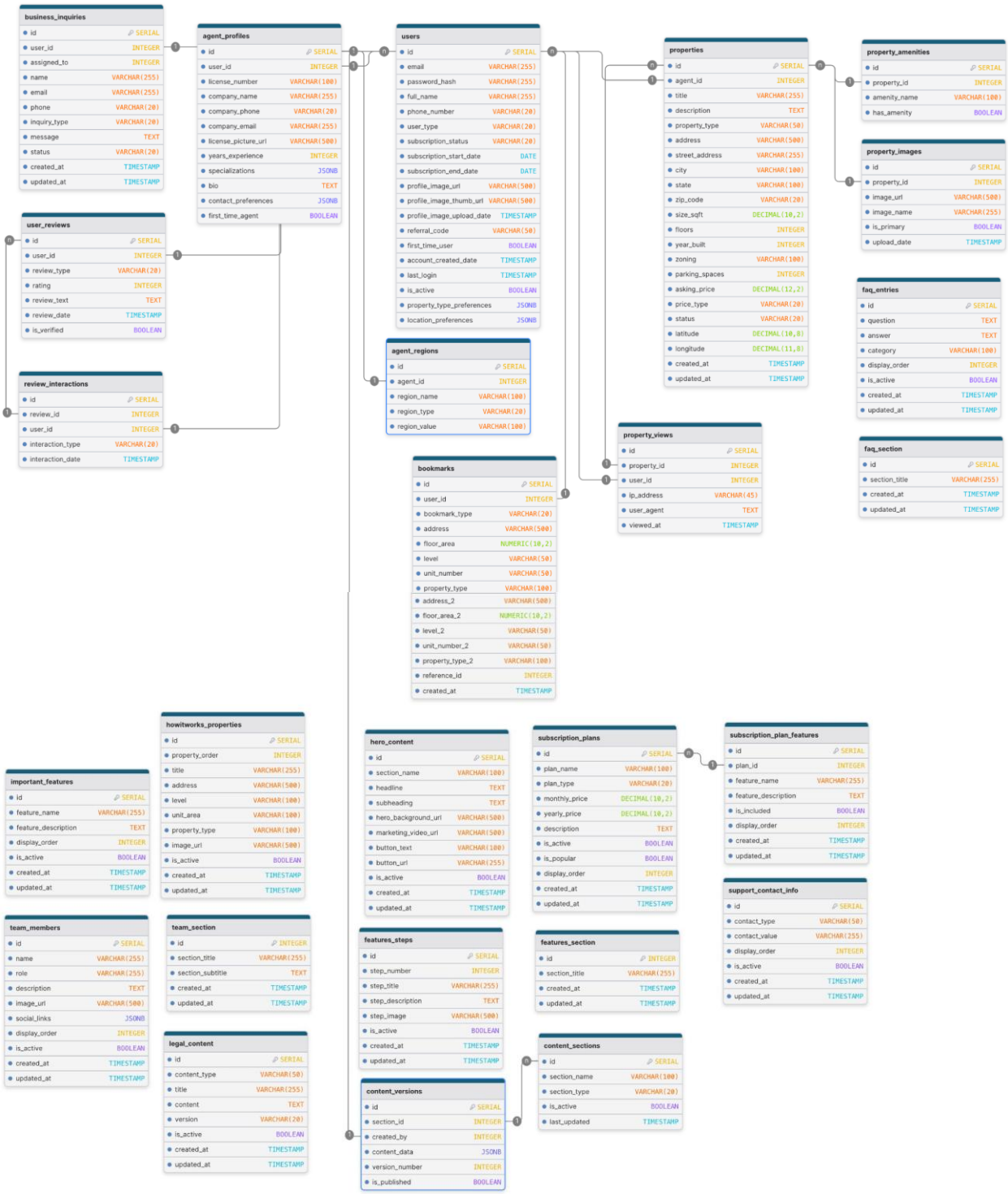
When a user looks for a property or asks for a price forecast, the frontend makes a request to the Flask API. The backend handles the request by:

- ➢ Creating predictions through models developed with machine learning frameworks like Scikit-learn, XGBoost, GeoPandas, and SHAP,
- ➢ Accessing the PostgreSQL + PostGIS database to retrieve spatial and historical information, and
- ➢ Incorporating external APIs such as OneMap SG and Google Maps API to enhance results with mapping, amenities, or planning information.

The reply is subsequently sent back to the frontend, which refreshes the user interface with predictions, interactive graphs, geographic maps, and nearby facilities.

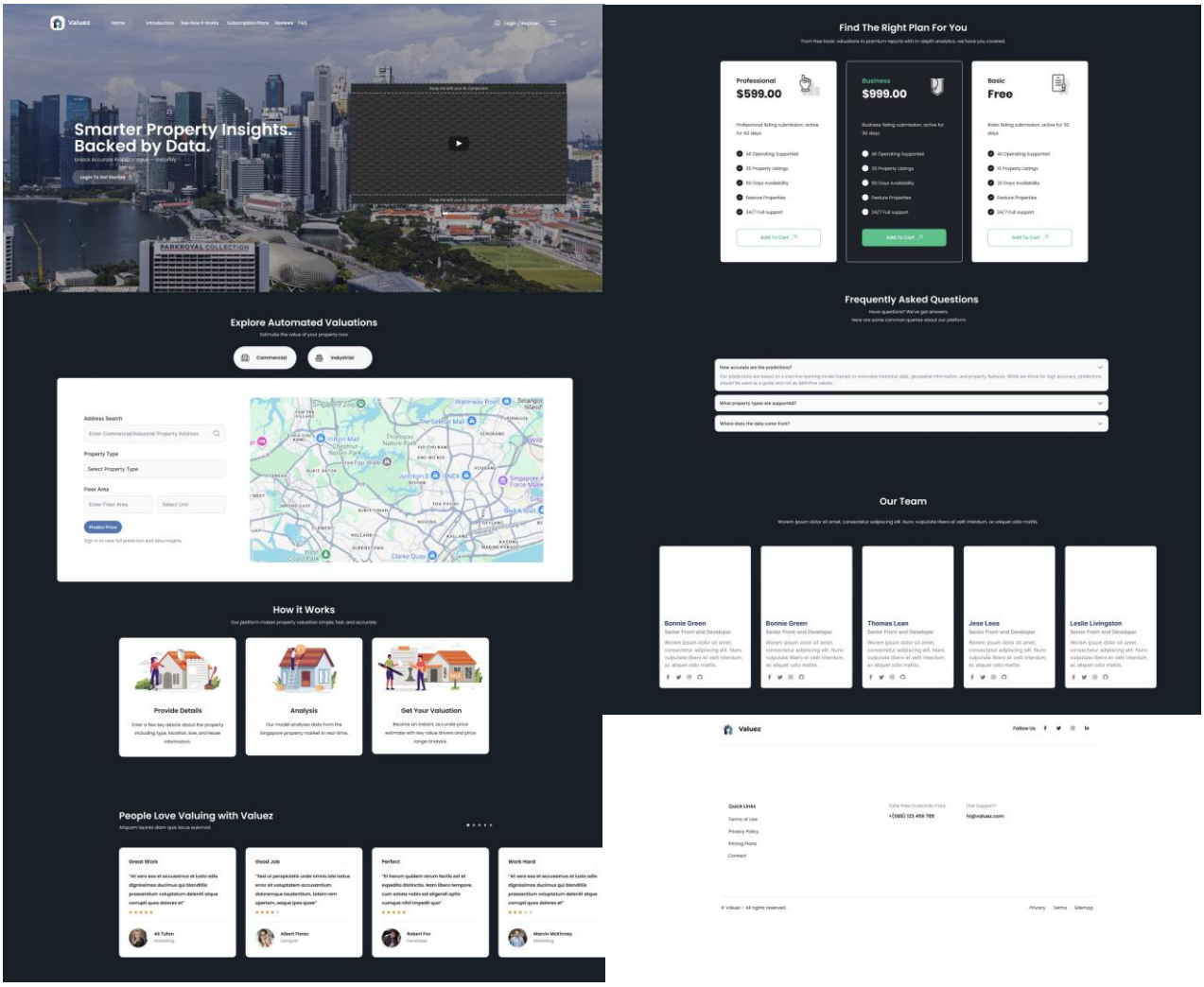


16.3 Database Design

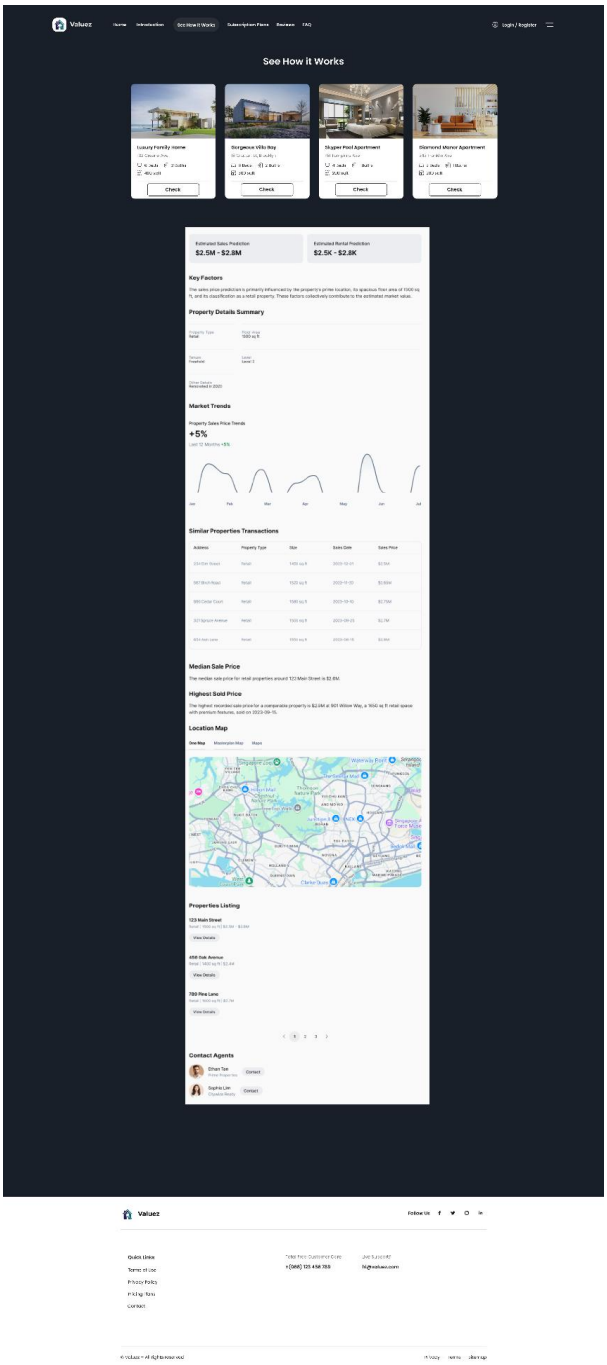


16.4 Wireframe Design

16.4.1 Landing Page (Main)

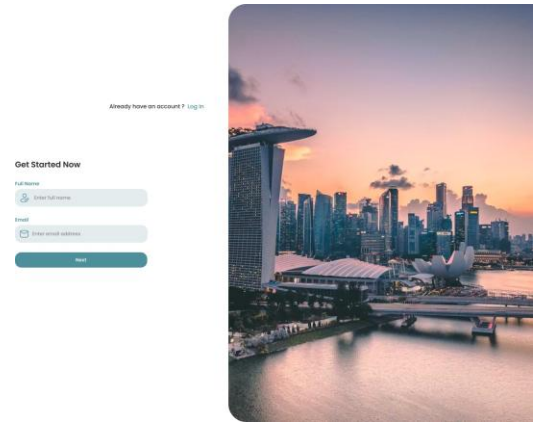


16.4.2 Landing Page (See How it Works)

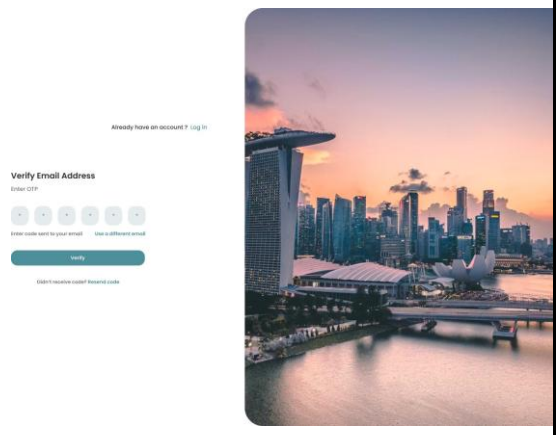


16.4.3 Signup

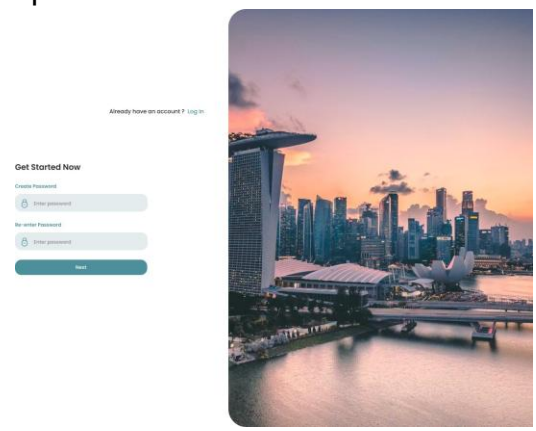
1st Step



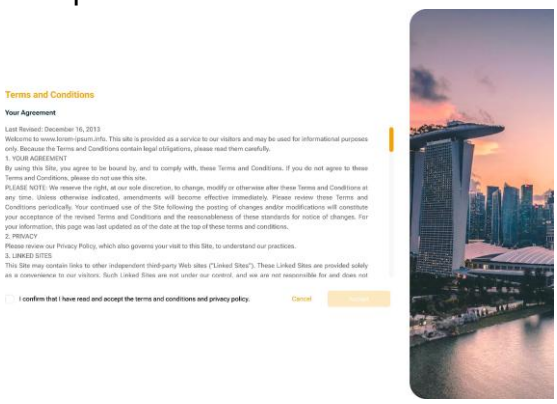
2nd Step



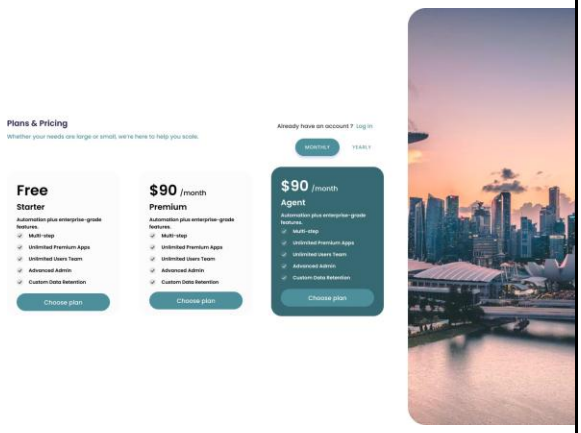
3rd Step



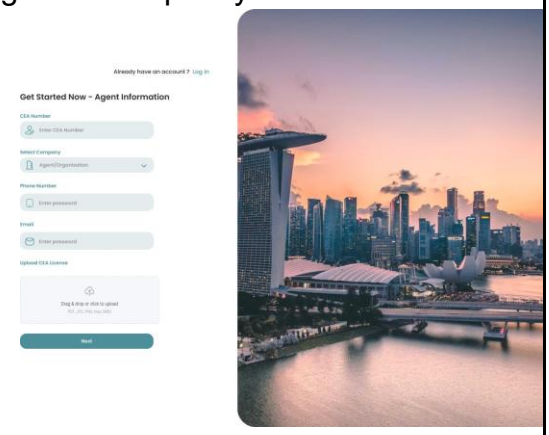
4th Step



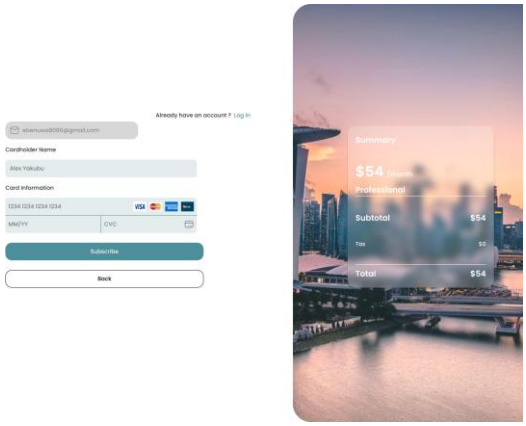
5th Step



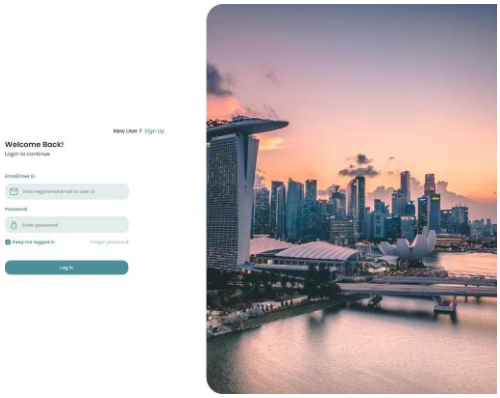
6th Step - additional information details for agent to fill up only



7th Step - payment page for premium user and agent to fill up

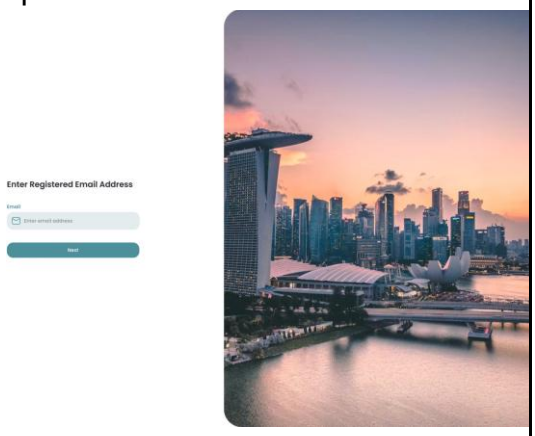


16.4.4 Login

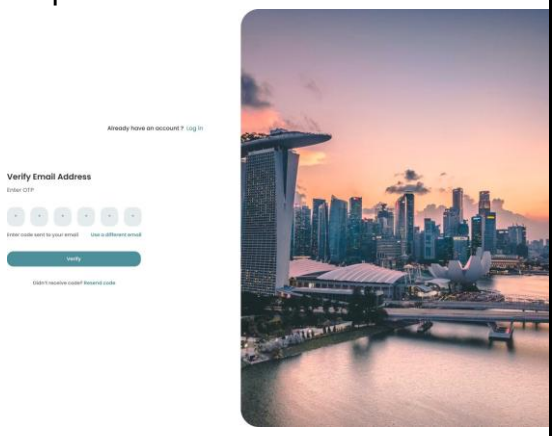


16.4.5 Recovery

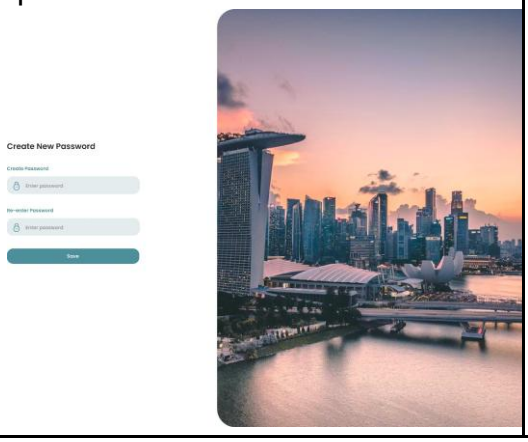
1st Step



2nd Step

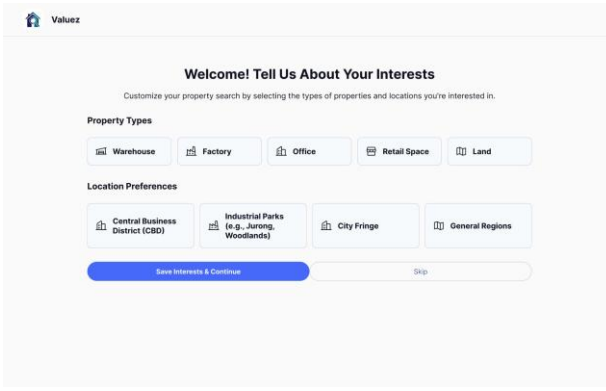


3rd Step

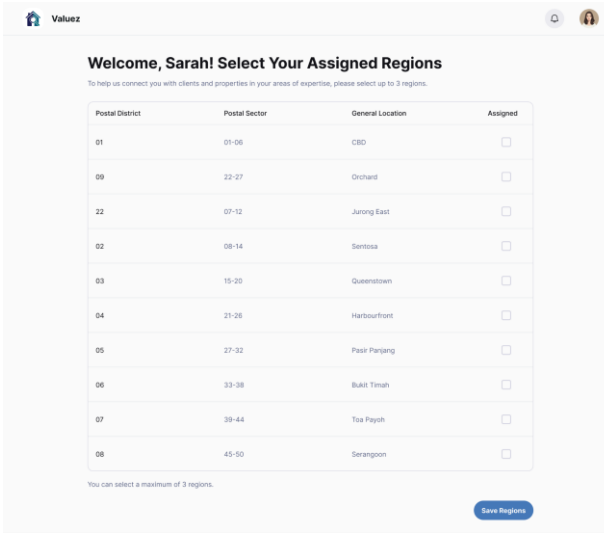


16.4.6 First Time Login (User/Agent)

First Time Login (User)

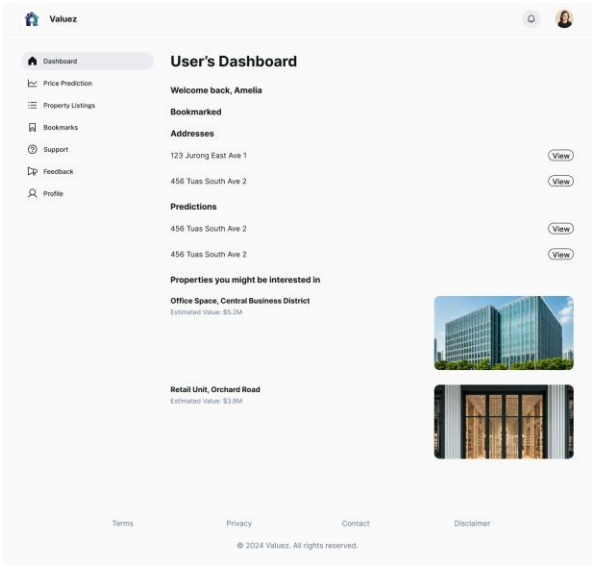


First Time Login (Agent)

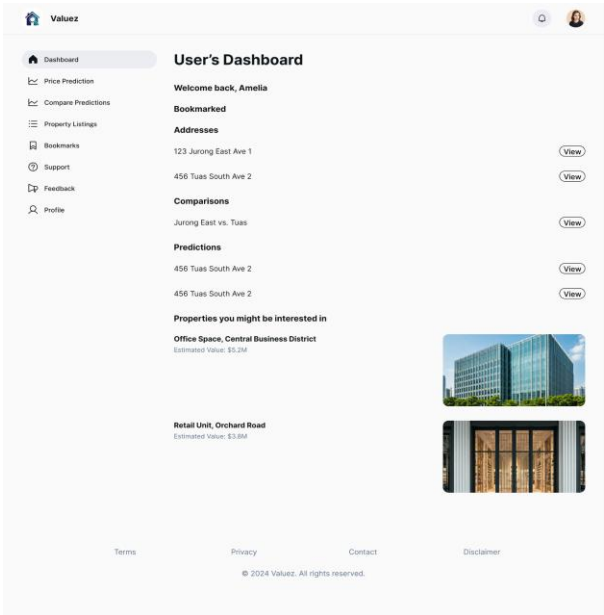


16.4.7 Registered Free/Premium User

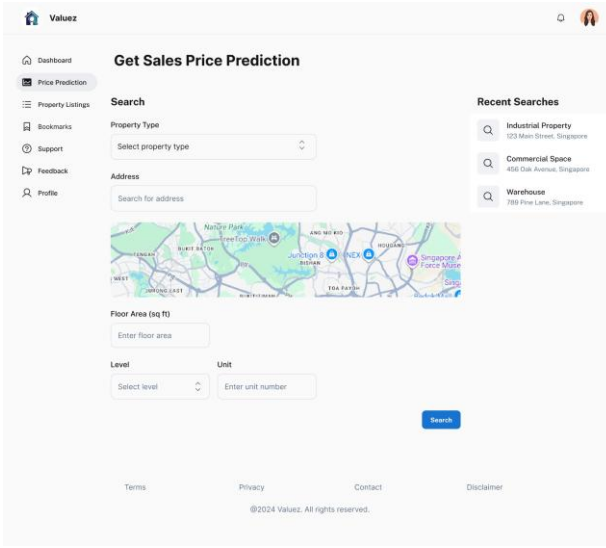
Dashboard - Registered Free User



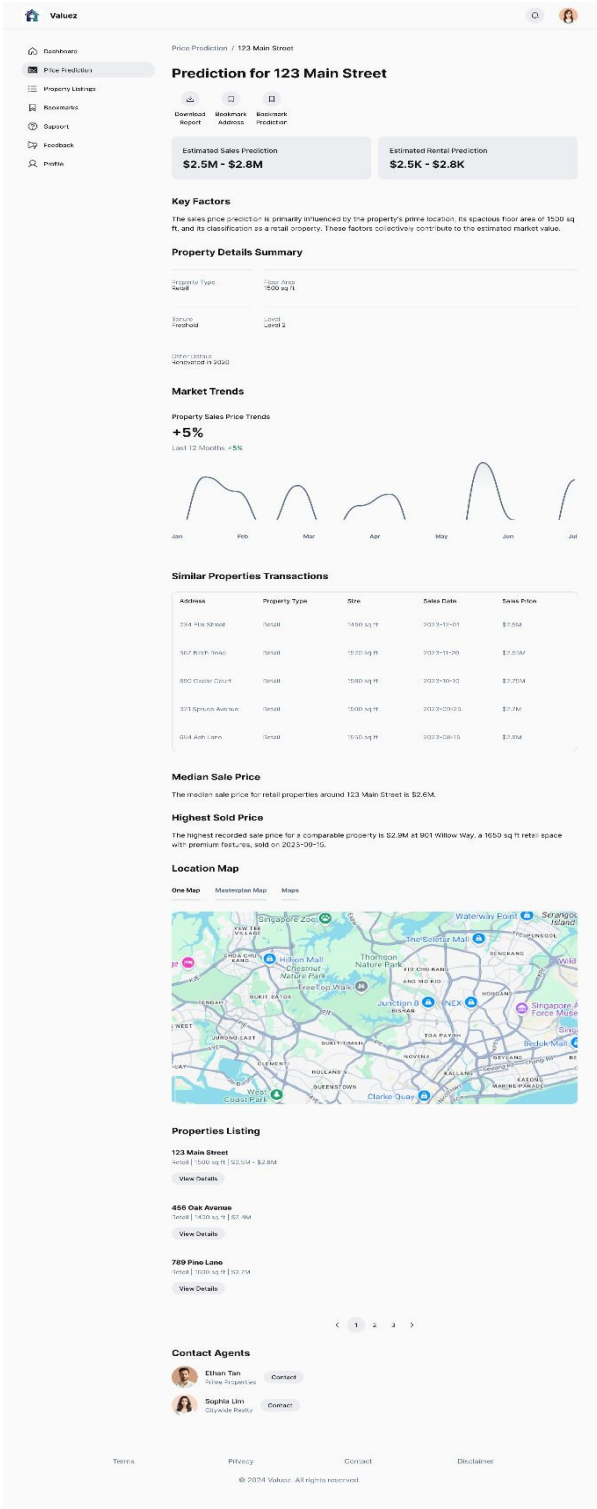
Dashboard - Registered Premium User



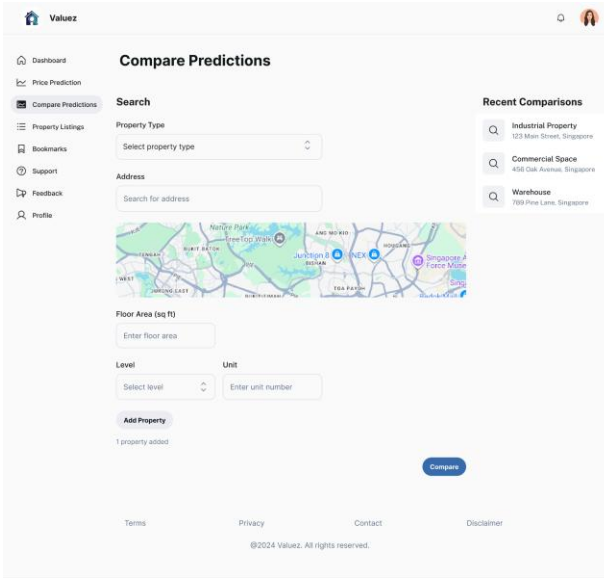
Price Prediction - Search



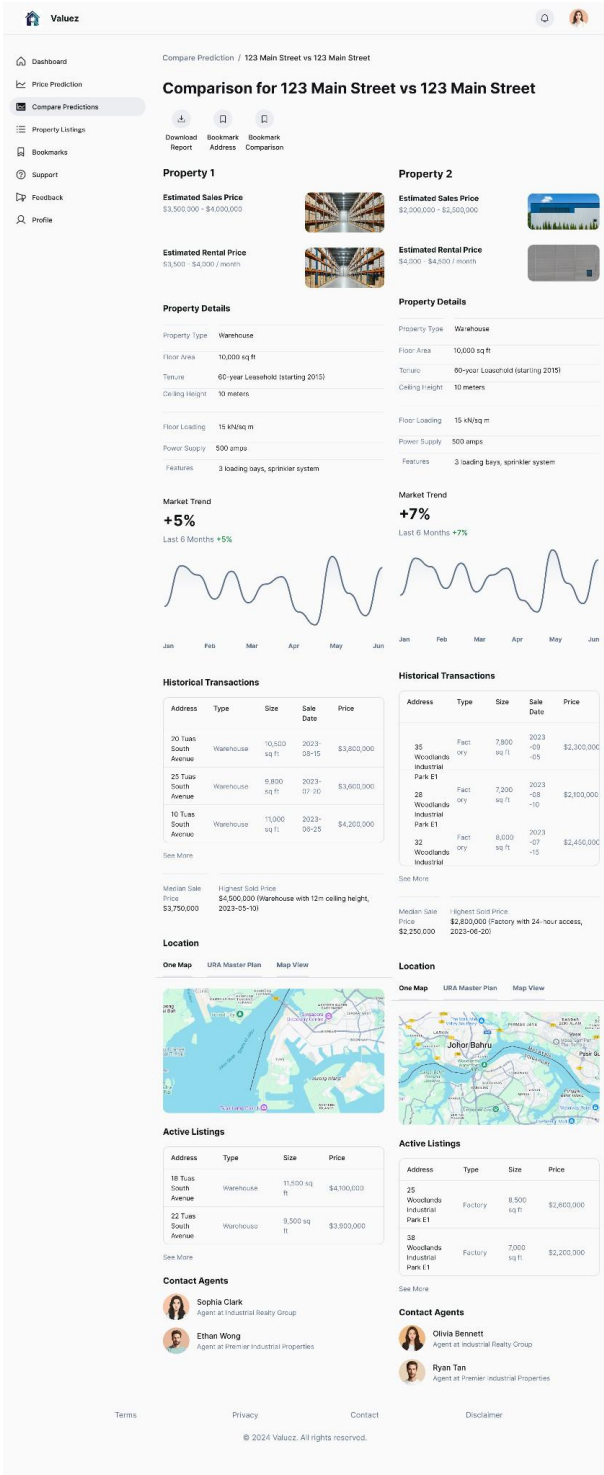
Price Prediction - Result



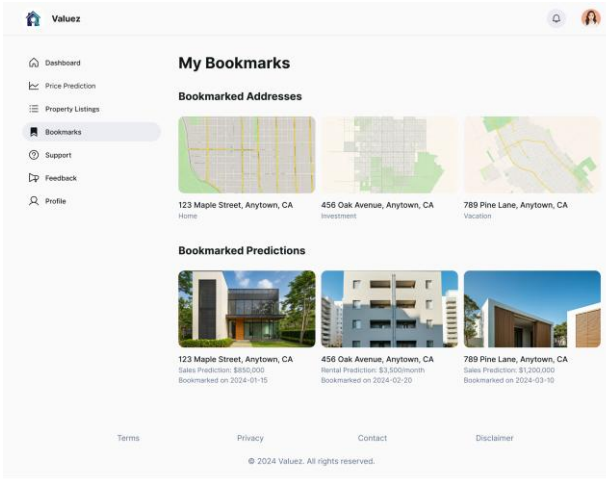
Compare Prediction - Search



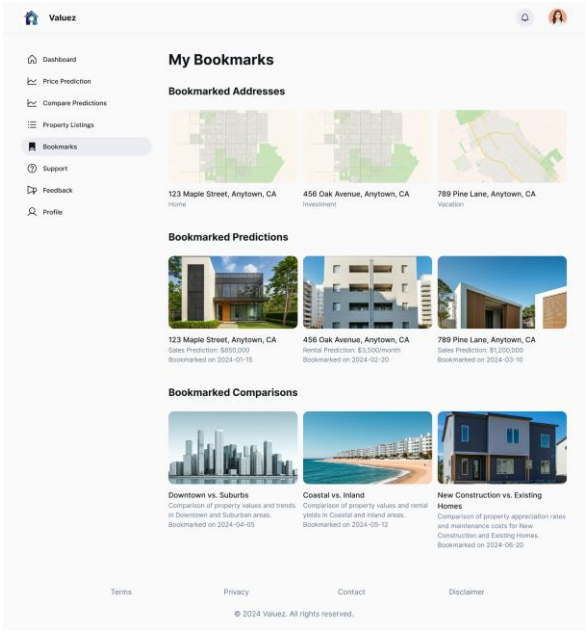
Compare Prediction - Result



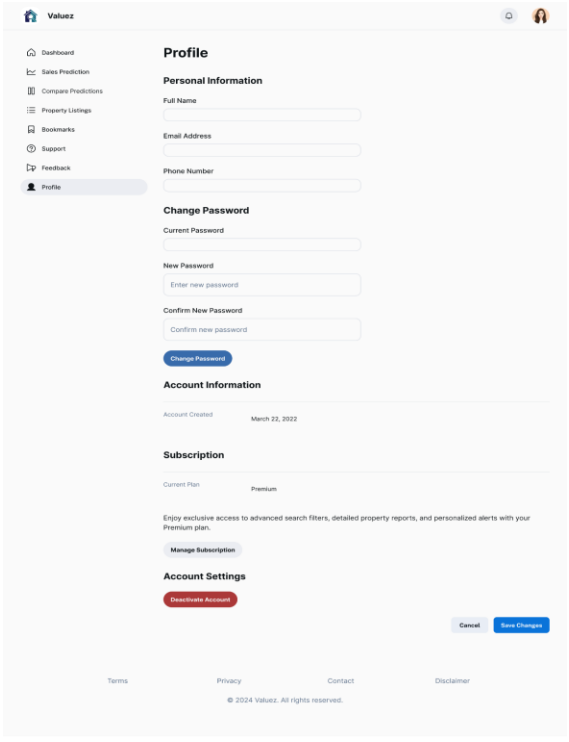
Bookmark Page - Free User



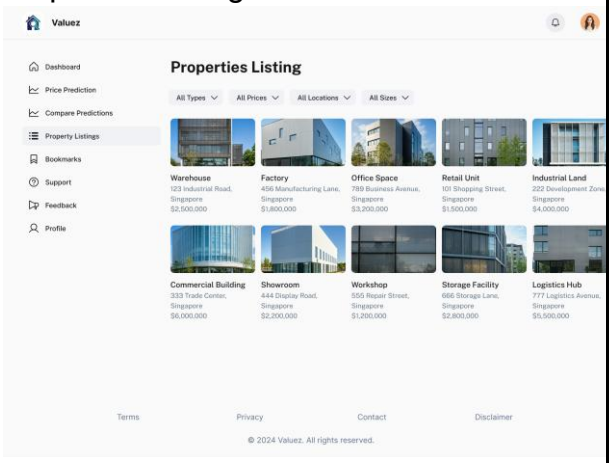
Bookmark Page - Premium User



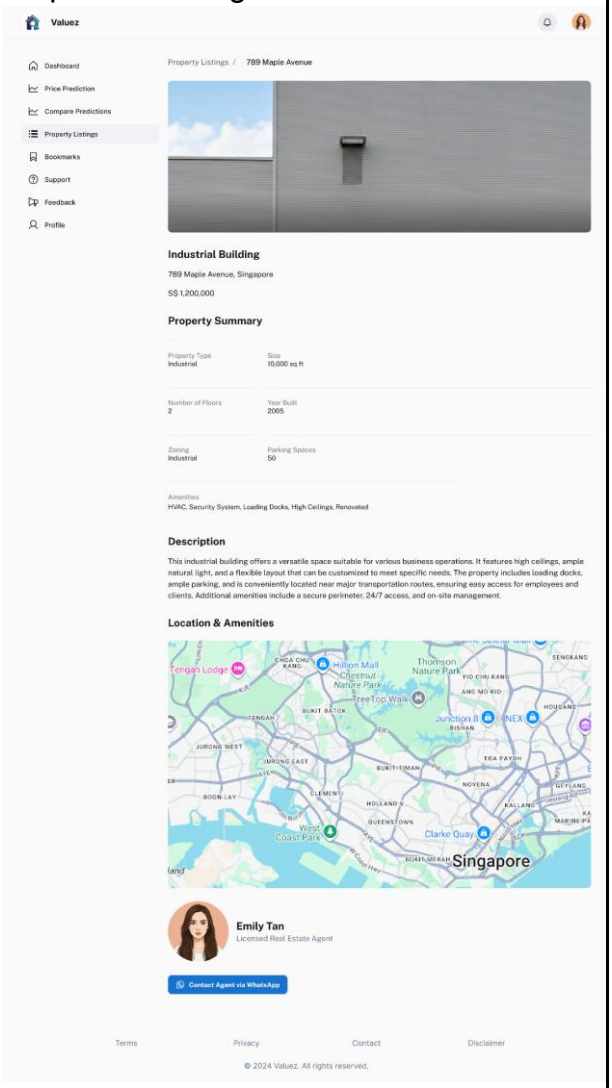
Profile



Properties Listing

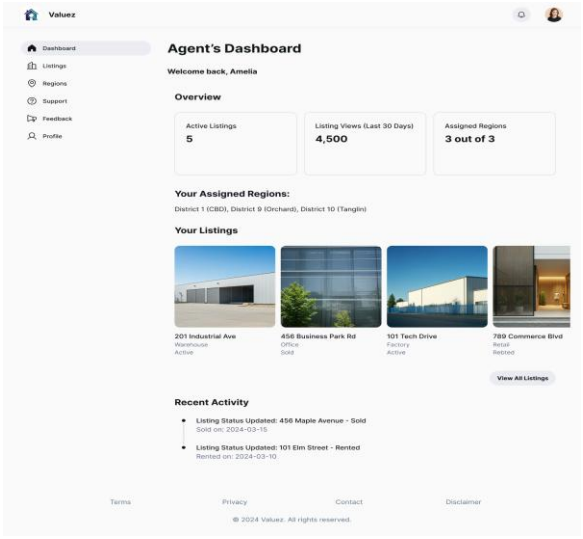


Properties Listing - Individual

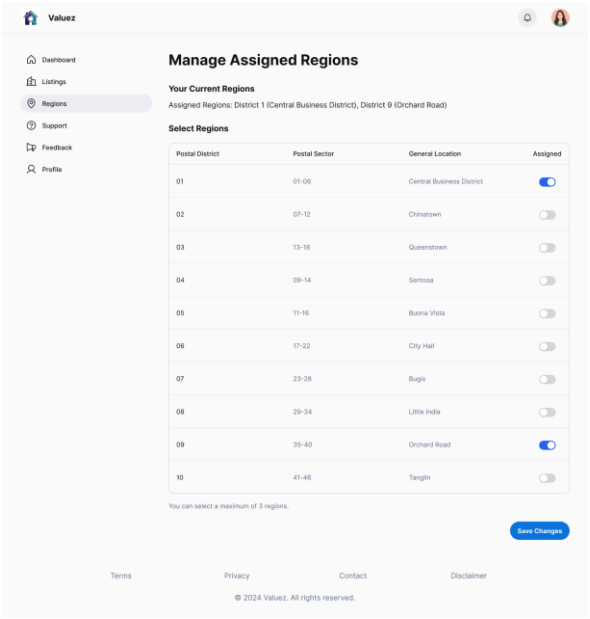


16.4.8 Property Agent

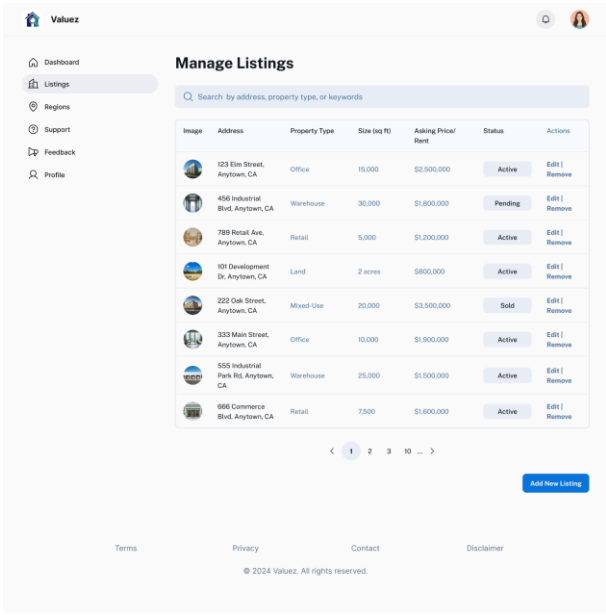
Dashboard



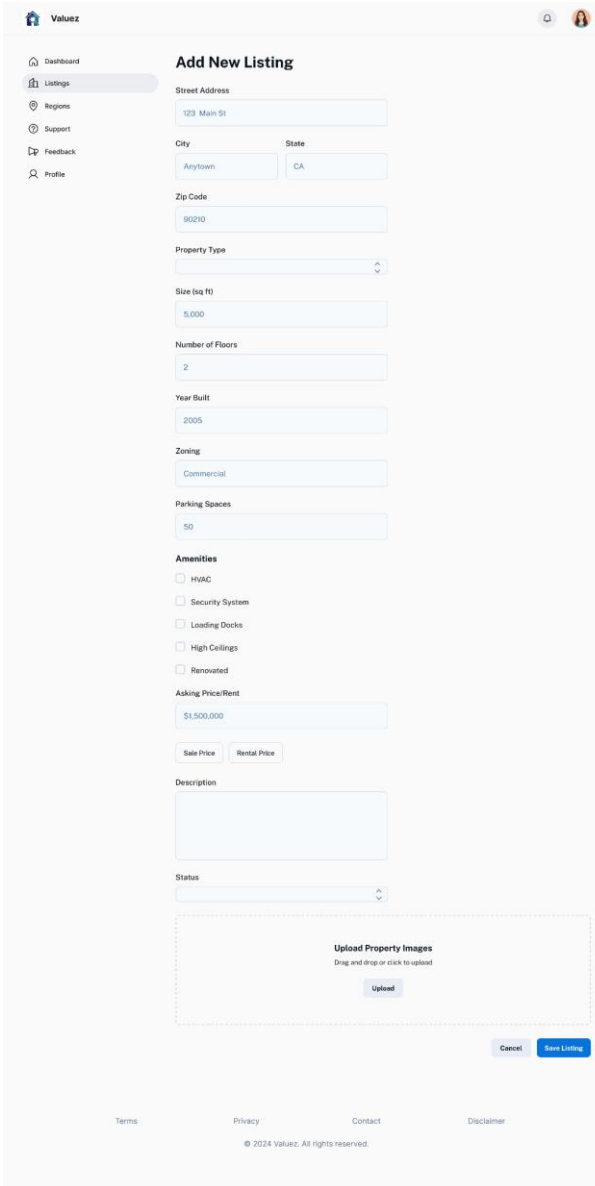
Manage Regions



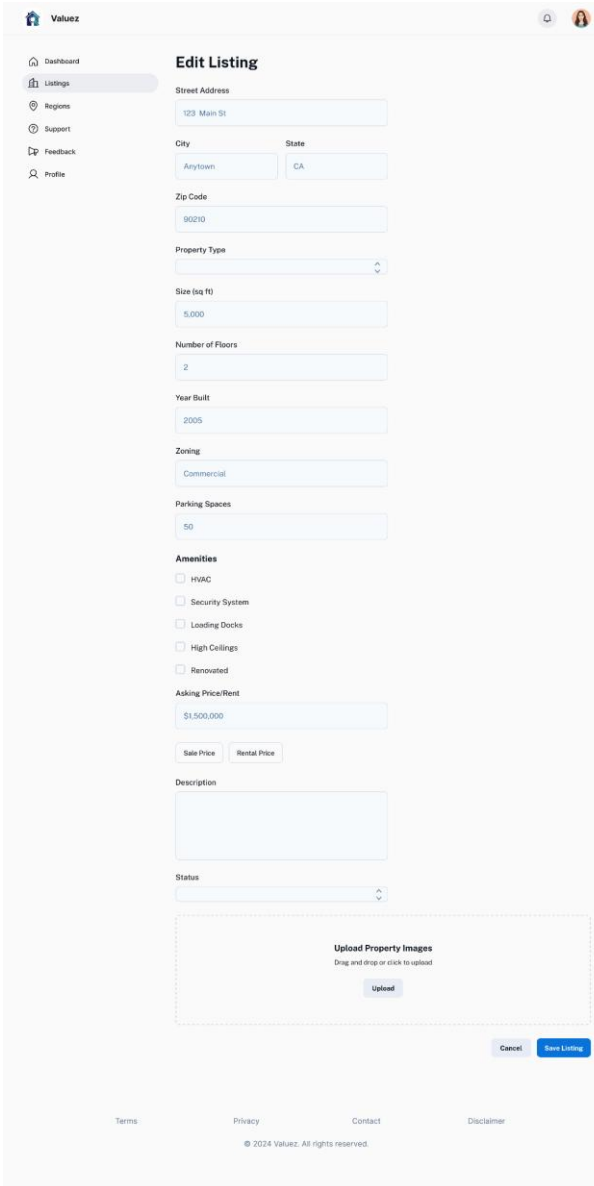
Manage Listings



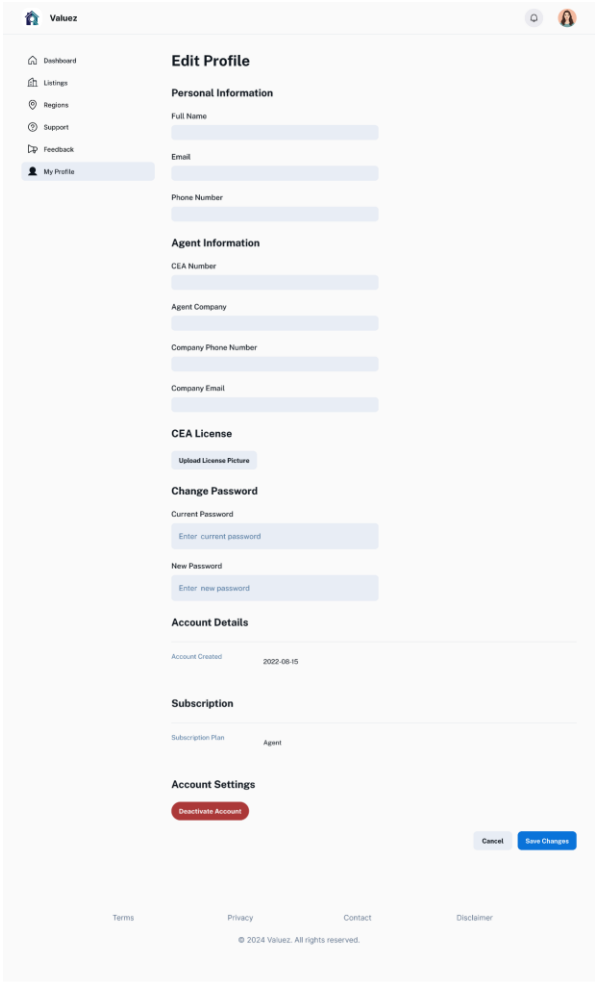
Manage Listings - Add Listings



Manage Listings - Edit Listings

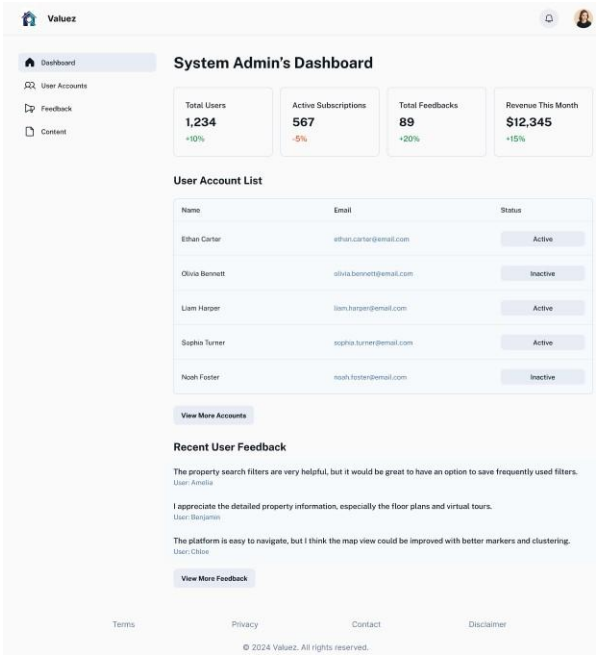


User Profile

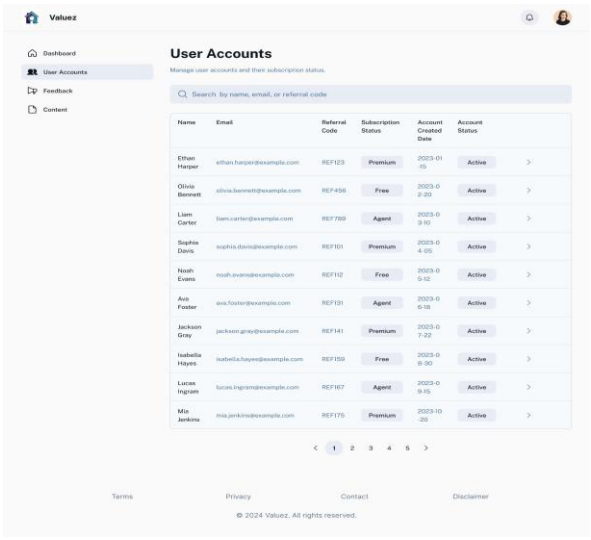


16.4.9 System Admin

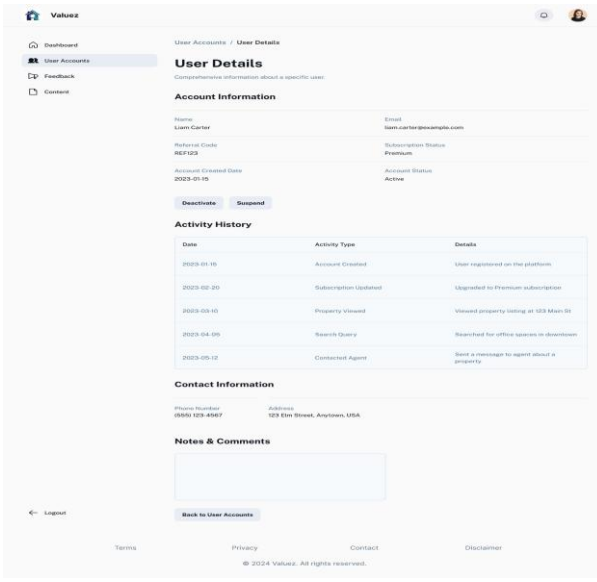
Dashboard



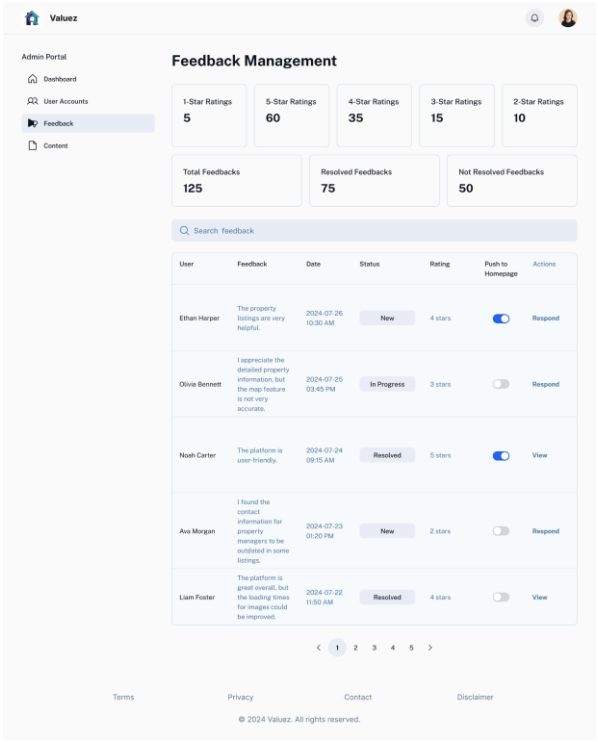
User Accounts



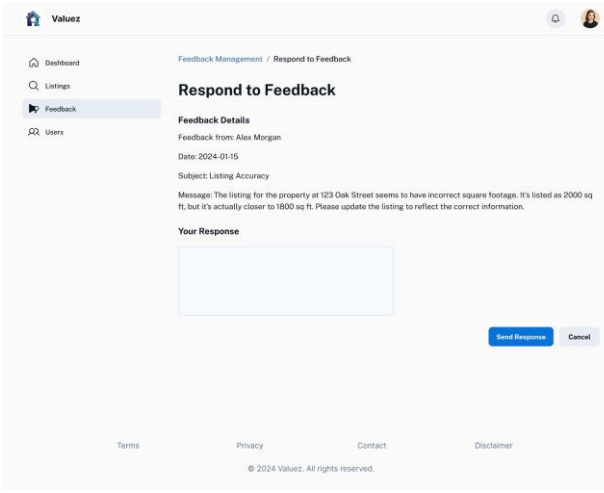
User Accounts - User Details



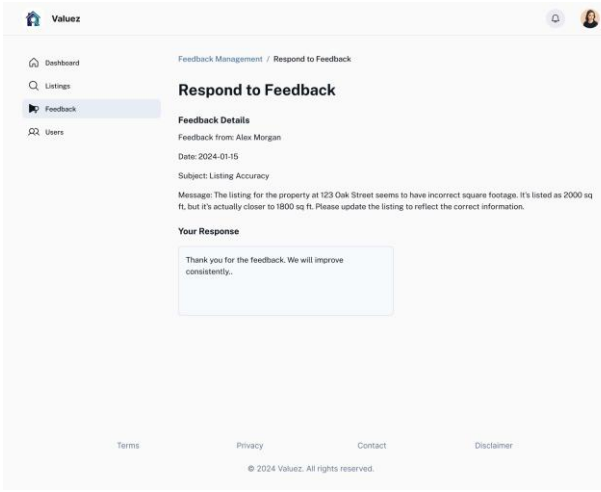
Feedback Management



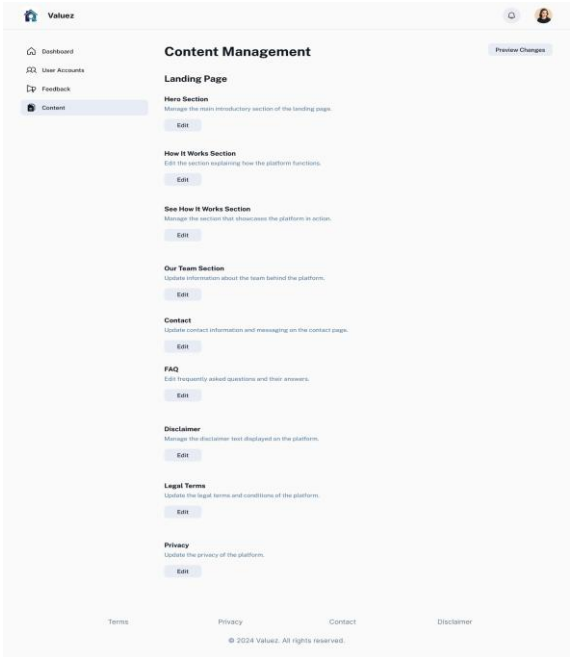
Feedback Management - Respond



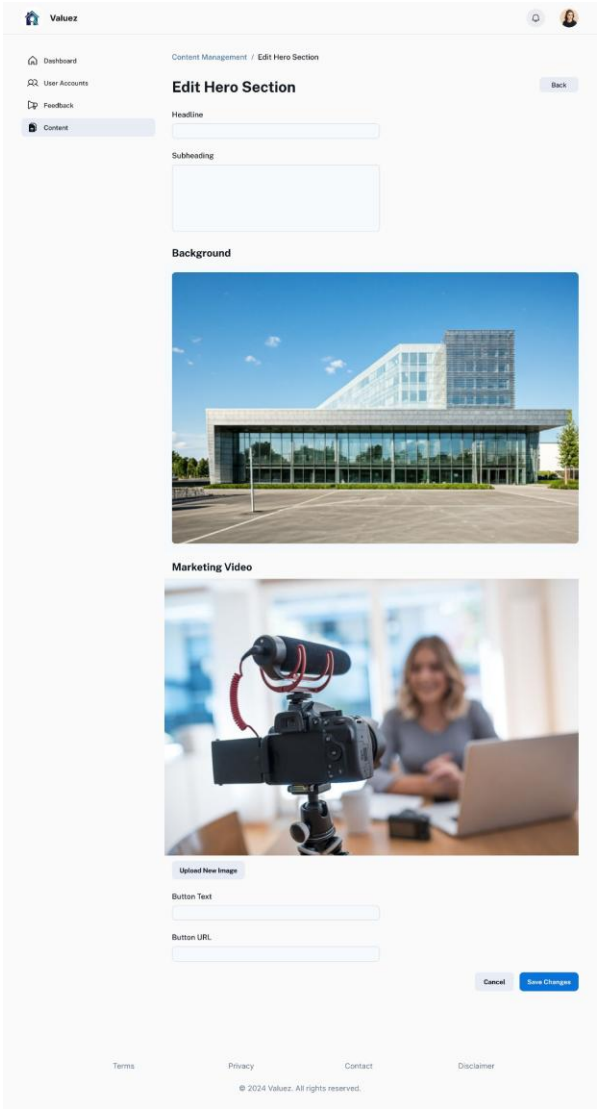
Feedback Management - View



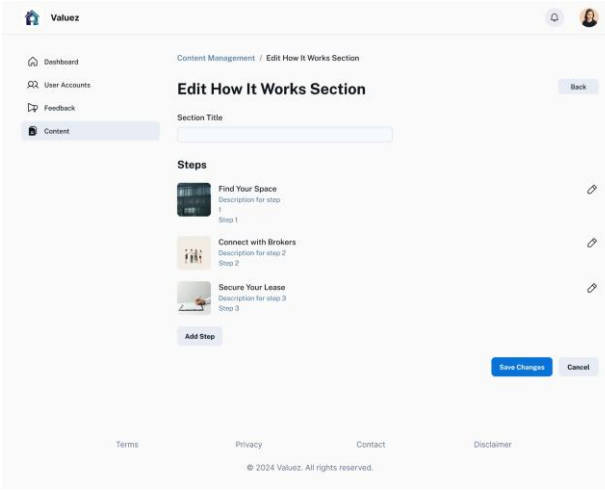
Content Management



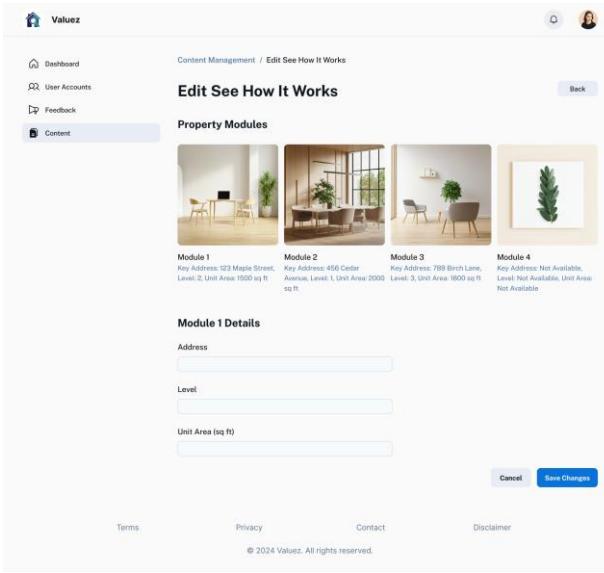
Content Management - Edit Hero Section



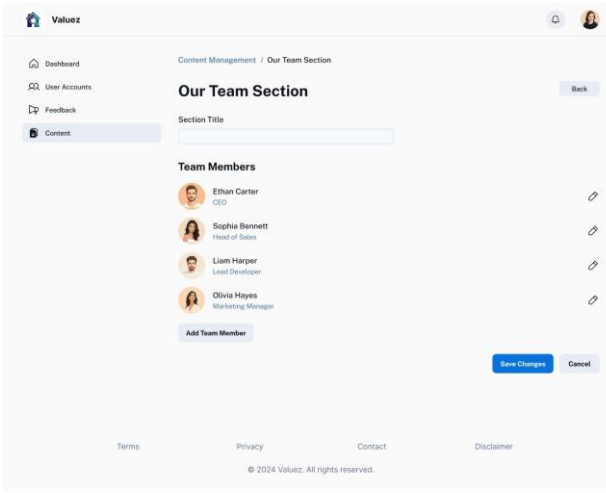
Content Management - Edit How it Works Section



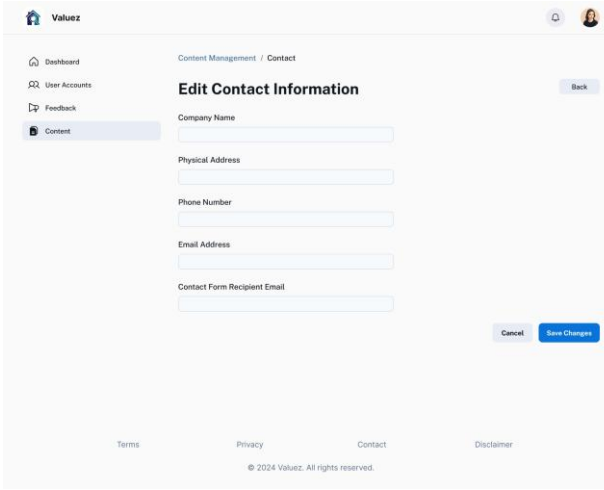
Content Management - Edit See How it Works Section



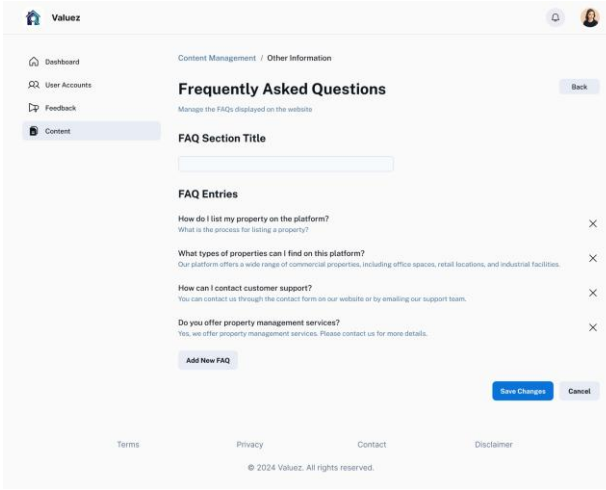
Content Management - Edit Our Team Section



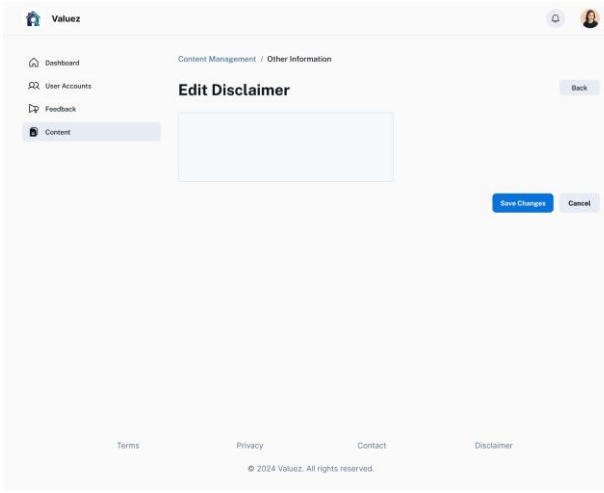
Content Management - Edit Contact Information



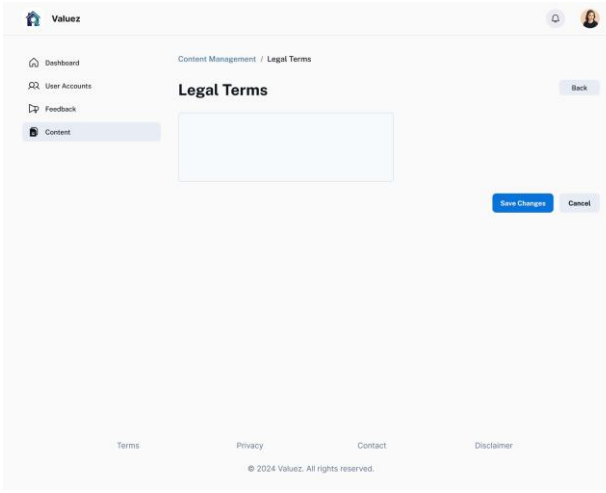
Content Management - Edit FAQ Section



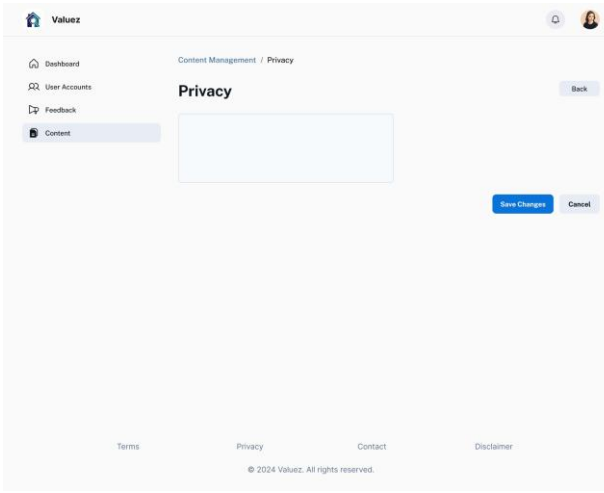
Content Management - Edit Disclaimer



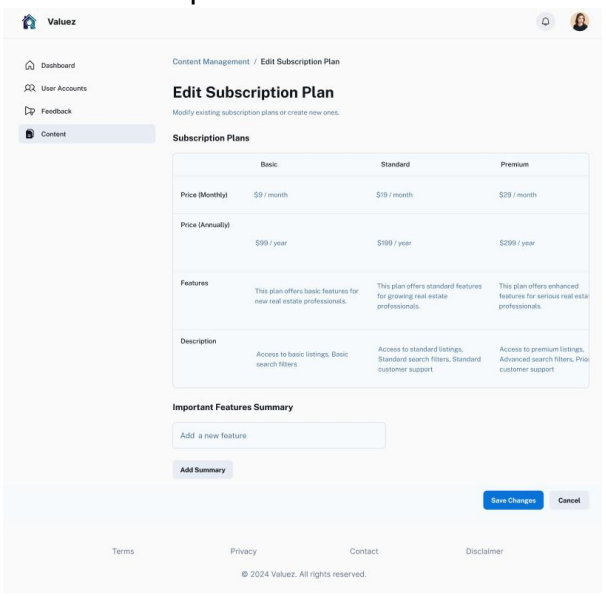
Content Management - Edit Legal Term



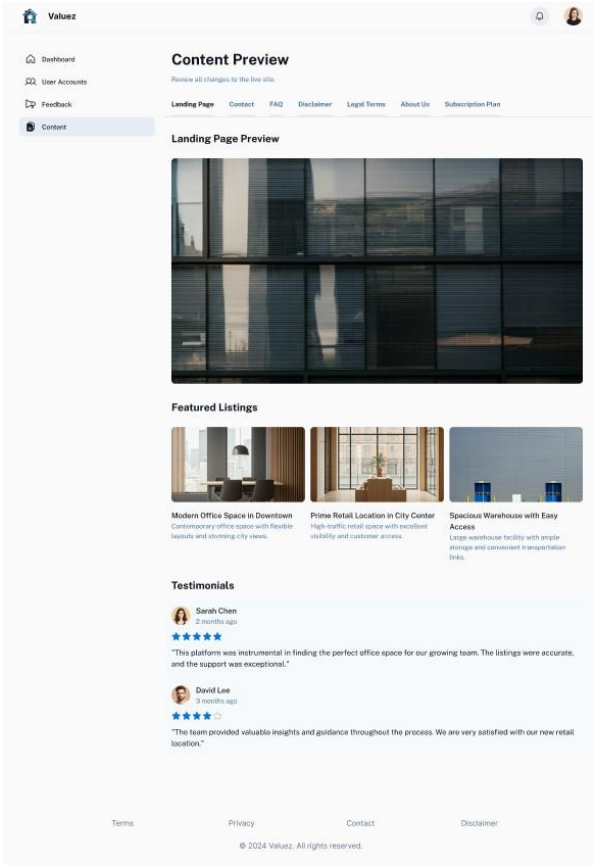
Content Management - Edit Privacy



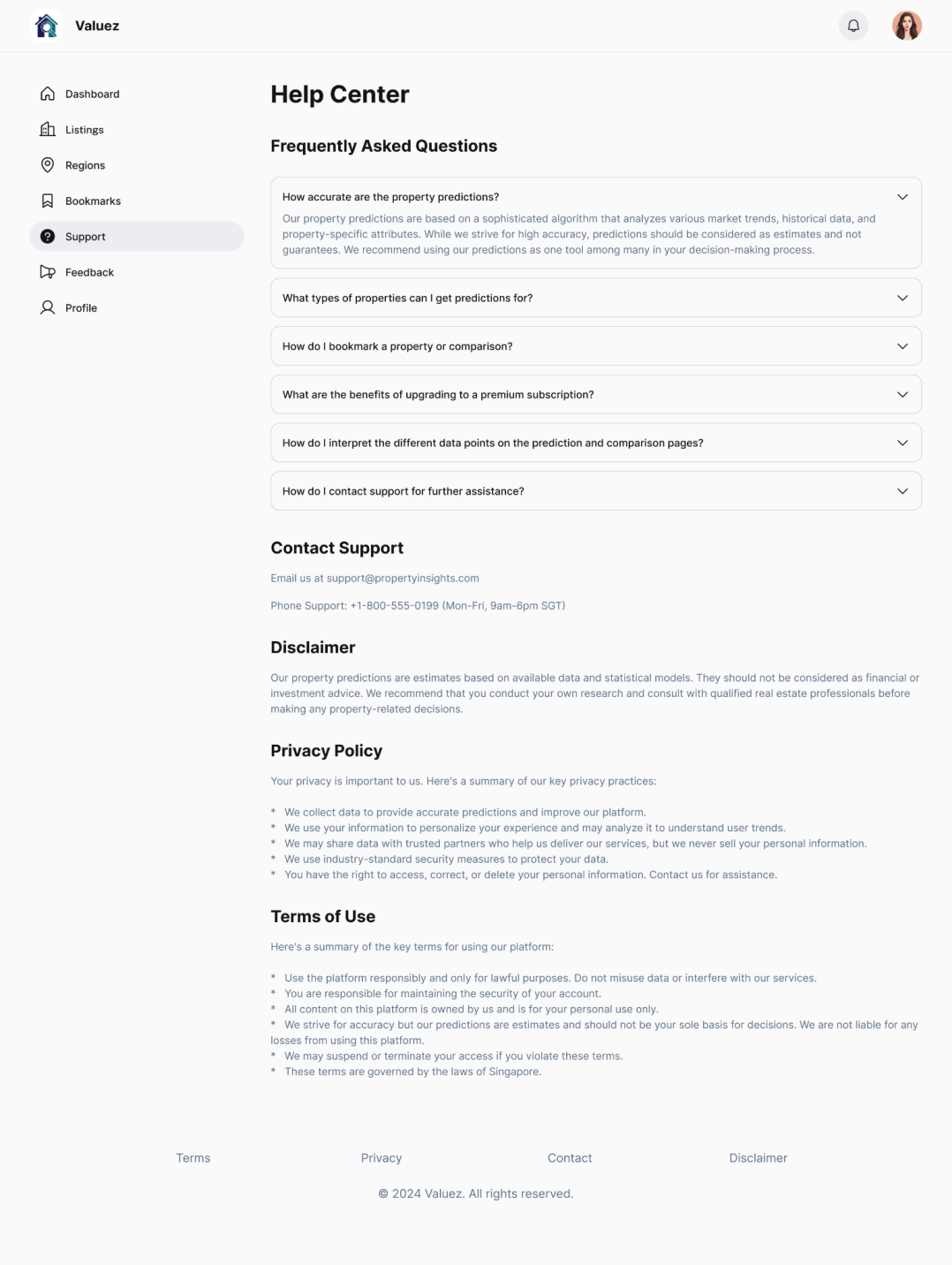
Content Management - Edit Subscription Plan



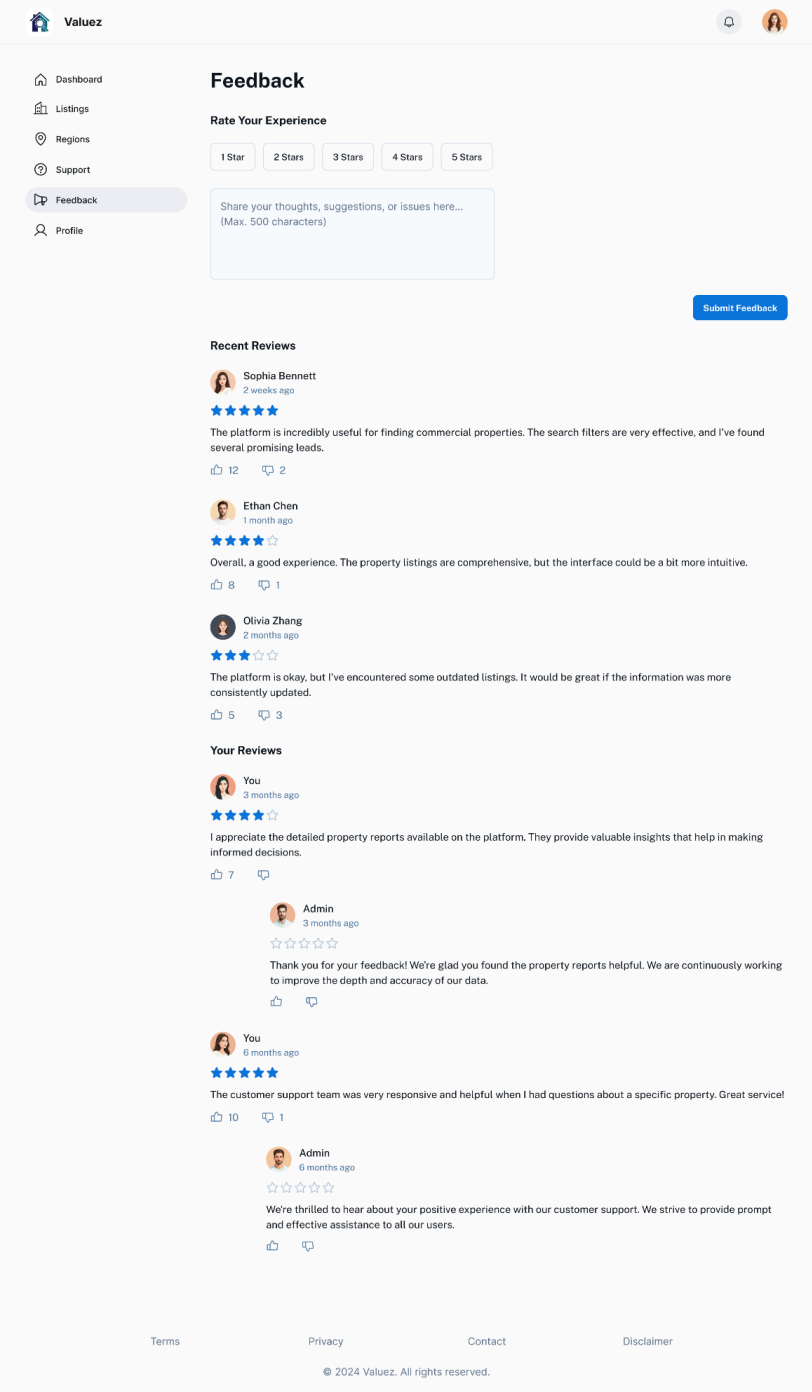
Content Management - Preview Changes



16.4.10 Support



16.4.11 Feedback



17. Conclusion

This project combines open data, location intelligence, and machine learning to create an interactive website that predicts property prices in Singapore with speed and accuracy. By allowing users to enter their own property details and receive instant, data-driven estimates, it removes the delays and limitations found in many existing tools.

More than just a price predictor, the platform provides historical trends and location-based insights to help buyers, sellers, investors, and agents make informed decisions with confidence. With its clean design, reliable data processing, and focus on user needs, the system offers a modern, transparent, and accessible approach to property valuation that can benefit both the public and the real estate industry.

18. Glossary

API (Application Programming Interface) : A set of rules and tools that allows different software applications to communicate and exchange data.

Backend : The server-side part of the application that processes requests, manages databases, and executes the business logic.

CI/CD (Continuous Integration/Continuous Deployment) : Automated processes for testing, integrating, and deploying code changes to ensure fast and reliable updates.

Cloud Hosting : Hosting services provided over the internet, enabling applications to be accessible from anywhere without managing physical servers.

Database : A structured system used to store, manage, and retrieve data efficiently.

ETL (Extract, Transform, Load) : The process of collecting data from various sources, cleaning and transforming it, and loading it into a database for analysis.

Feature Engineering : The process of selecting, modifying, and creating new data features to improve machine learning model performance.

Frontend : The client-facing part of the application that users interact with, typically displayed through a web browser.

Geospatial Data : Data that includes geographical components, such as coordinates or addresses, used to analyze location-based patterns.

GitHub : A web-based platform for version control and collaboration, allowing teams to track changes in code and manage projects.

Machine Learning : A field of artificial intelligence where algorithms learn patterns from data to make predictions or decisions without explicit programming.

Microsoft SQL Server : A relational database management system (RDBMS) used for storing and managing structured data.

Model Training : The process of feeding data into a machine learning algorithm so it can learn patterns and make accurate predictions.

OneMap SG API : A Singapore government API providing geospatial data such as maps, coordinates, and nearby amenities.

Open Data : Data that is freely available for anyone to access, use, and share without restrictions.

Prediction Model : An algorithm that uses historical data and features to estimate future outcomes, such as property prices.

Price Prediction : The process of estimating a property's market value using historical transaction data, location features, and machine learning models.

Property Attributes : Characteristics of a property, such as type, size, floor level, and age, that influence its value.

React : A JavaScript library used for building dynamic and responsive user interfaces.

Render/Vercel : Cloud deployment platforms that automatically build and host applications when changes are pushed to a connected GitHub repository.

Stakeholder : Any person or group with an interest in the project, such as developers, users, supervisors, and assessors.

URA (Urban Redevelopment Authority) Map API : An API by Singapore's URA that provides access to planning maps, zoning data, and other land-related information.

User Interface (UI) : The visual and interactive part of an application that users interact with directly.

User Experience (UX) : The overall satisfaction and usability that users feel when interacting with an application.

Version Control : A system for managing and tracking changes to code over time, enabling collaboration and rollback when needed.

Web Scraping : The process of extracting information from websites that do not have APIs.

20. Appendix

A. Competitors

PropertyGuru : https://www.propertyguru.com.sg/my-home

SRX : https://www.srx.com.sg/xvalue-pricing

EdgeProp : https://www.edgeprop.sg/analytic/edgefairvalue

Propseller : https://propseller.com/home-value-estimate

Ohmyhome : https://www.homerai.sg

Bluenest: https://www.bluenest.sg/myhomevalue

B. Data Sources

OneMap API Documentation. Available at: https://www.onemap.gov.sg/apidocs/

Google Maps API. Available at: https://developers.google.com/maps

Commercial Property Transactions. Available at: https://www.teeco.sg/commercial

Industrial Property Transactions. Available at: https://www.teeco.sg/industrial

Retail Rental Analysis. Available at: https://www.ura.gov.sg/Corporate/Property/Retail-Rental

Office Rental Analysis. Available at: https://www.ura.gov.sg/Corporate/Property/Office-Rental

Commercial Rental Index. Available at: https://data.gov.sg/dataset/commercial-rental-index

Industrial Rental Index. Available at: https://data.gov.sg/dataset/industrial-rental-index

C. Tools and Technologies

Why We Use React.js in Development. Available at: https://www.techmagic.co/blog/why-we-use-react-js-in-the-development

ReactJS vs. Other Frameworks: What Is Better? Available at: https://www.uplers.com/blog/reactjs-vs-other-framework-what-is-better

Comparing Frontend Frameworks: Angular vs. React vs. Vue.js—Which One Fits Your Project Best? Available at: https://dev.to/okoye_ndidiamaka_5e3b7d30/comparing-frontend-frameworks-angular-vs-react-vs-vuejs-which-one-fits-your-project-best-5epm

React Official Documentation. Available at: https://react.dev/learn

Flask vs. Django: Which Python Framework Is Better for Your Web Development? Available at: https://www.stxnext.com/blog/flask-vs-django-comparison

Flask vs FastAPI: An In-Depth Framework Comparison. Available at: https://betterstack.com/community/guides/scaling-python/flask-vs-fastapi

Flask vs. Node.js: Features, Performance, and Scalability Compared. Available at: https://artoonsolutions.com/flask-vs-nodejs

PostgreSQL Official Documentation. Available at: https://www.postgresql.org/docs

What is PostgreSQL? Key Features, Benefits, and Real-World Uses. Available at: https://www.percona.com/blog/what-is-postgresql-used-for

An Introduction to PostGIS. Available at: https://postgis.net/workshops/postgis-intro

What's the Difference Between MySQL and PostgreSQL? Available at: https://aws.amazon.com/compare/the-difference-between-mysql-vs-postgresql

PostgreSQL vs. MongoDB: Differences, Strengths, and Use Cases. Available at: https://estuary.dev/blog/postgresql-vs-mongodb

PostgreSQL vs. MongoDB vs. Firebase: Which Database Should You Use? Available at: https://www.remoteplatz.com/en/blog/postgresql-vs--mongodb-vs--firebase--which-databas

SQLite Limitations and Use Cases. Available at: https://www.sqlite.org/whentouse.html

What Is Scikit-learn, and How Is It Used in AI? Available at: https://www.dataquest.io/blog/what-is-scikit-learn-and-how-is-it-used-in-ai

XGBoost: The Powerhouse Algorithm Behind ML Competitions. Available at: https://royalresearch.in/xgboost-the-powerhouse-algorithm-behind-ml-competitions

An Introduction to SHAP Values and Machine Learning Interpretability. Available at: https://www.datacamp.com/tutorial/introduction-to-shap-values-machine-learning-interpretability

Introduction to GeoPandas. Available at: https://geopandas.org/en/stable/getting_started/introduction.html

LightGBM vs. XGBoost. Available at: https://www.numberanalytics.com/blog/lightgbm-vs-xgboost-comparison

Render vs. Heroku. Available at: https://northflank.com/blog/render-vs-heroku

Continuous Deployment: Vercel vs. Netlify vs. Render. Available at: https://bejamas.com/compare/netlify-vs-render-vs-vercel

5 Best Free Platforms for Hosting Hobby Web Projects. Available at: https://refine.dev/blog/5-top-free-react-hosting-platforms